VisNotebook

October 26, 2026

1 ggplot2 Data Vis. with R

1.1 Libraries and Init

[104]:
```
library(tidyverse)
library(dplyr)
library(ggplot2)
library(patchwork)

library(ggtext)
library(ggthemes)
library(RColorBrewer)
library(viridis)

library(datasets)
library(help = "datasets")

library(readxl)
```

```
Documentation for package 'datasets'


                Information on package 'datasets'

Description:

Package:       datasets
Version:       4.6.1
Priority:      base
Title:         The R Datasets Package
Author:        R Core Team and contributors worldwide
Maintainer:    R Core Team <do-use-Contact-address@r-project.org>
Contact:       R-help mailing list <r-help@r-project.org>
Description:   Base R datasets.
License:       Part of R 4.6.1
Encoding:      UTF-8
Built:         R 4.6.1; ; 2026-06-24 08:19:02 UTC; windows

Index:
```

AirPassengers	Monthly Airline Passenger Numbers 1949-1960
BJsales	Sales Data with Leading Indicator
BOD	Biochemical Oxygen Demand
CO2	Carbon Dioxide Uptake in Grass Plants
ChickWeight	Weight Versus Age of Chicks on Different Diets
DNase	Elisa assay of DNase
EuStockMarkets	Daily Closing Prices of Major European Stock Indices, 1991-1998
Formaldehyde	Determination of Formaldehyde
HairEyeColor	Hair and Eye Color of Statistics Students
Harman23.cor	Harman Example 2.3
Harman74.cor	Harman Example 7.4
Indometh	Pharmacokinetics of Indomethacin
InsectSprays	Effectiveness of Insect Sprays
JohnsonJohnson	Quarterly Earnings per Johnson & Johnson Share
LakeHuron	Level of Lake Huron 1875-1972
LifeCycleSavings	Intercountry Life-Cycle Savings Data
Loblolly	Growth of Loblolly Pine Trees
Nile	Flow of the River Nile
Orange	Growth of Orange Trees
OrchardSprays	Potency of Orchard Sprays
PlantGrowth	Results from an Experiment on Plant Growth
Puromycin	Reaction Velocity of an Enzymatic Reaction
Theoph	Pharmacokinetics of Theophylline
Titanic	Survival of passengers on the Titanic
ToothGrowth	The Effect of Vitamin C on Tooth Growth in Guinea Pigs
UCBAdmissions	Student Admissions at UC Berkeley
UKDriverDeaths	Road Casualties in Great Britain 1969-84
UKLungDeaths	Monthly Deaths from Lung Diseases in the UK
UKgas	UK Quarterly Gas Consumption
USAccDeaths	Accidental Deaths in the US 1973-1978
USArrests	Violent Crime Rates by US State
USJudgeRatings	Lawyers' Ratings of State Judges in the US Superior Court
USPersonalExpenditure	Personal Expenditure Data
VADeaths	Death Rates in Virginia (1940)
WWWusage	Internet Usage per Minute
WorldPhones	The World's Telephones
ability.cov	Ability and Intelligence Tests
airmiles	Passenger Miles on Commercial US Airlines, 1937-1960
airquality	New York Air Quality Measurements
anscombe	Anscombe's Quartet of 'Identical' Simple Linear Regressions
attenu	The Joyner-Boore Attenuation Data
attitude	The Chatterjee-Price Attitude Data

austres	Quarterly Time Series of the Number of Australian Residents
beavers	Body Temperature Series of Two Beavers
cars	Speed and Stopping Distances of Cars
chickwts	Chicken Weights by Feed Type
co2	Mauna Loa Atmospheric CO2 Concentration
crimtab	Student's 3000 Criminals Data
datasets-package	The R Datasets Package
discoveries	Yearly Numbers of Important Discoveries
esoph	Smoking, Alcohol and (O)esophageal Cancer
euro	Conversion Rates of Euro Currencies
eurodist	Distances Between European Cities and Between US Cities
faithful	Old Faithful Geyser Data
freeny	Freeny's Revenue Data
gait	Hip and Knee Angle while Walking
infert	Infertility after Spontaneous and Induced Abortion
iris	Edgar Anderson's Iris Data
islands	Areas of the World's Major Landmasses
lh	Luteinizing Hormone in Blood Samples
longley	Longley's Economic Regression Data
lynx	Annual Canadian Lynx trappings 1821-1934
morley	Michelson Speed of Light Data
mtcars	Motor Trend Car Road Tests
nhtemp	Average Yearly Temperatures in New Haven
nottem	Average Monthly Temperatures at Nottingham, 1920-1939
npk	Classical N, P, K Factorial Experiment
occupationalStatus	Occupational Status of Fathers and their Sons
penguins	Measurements of Penguins near Palmer Station, Antarctica
precip	Annual Precipitation in Selected US Cities
presidents	Quarterly Approval Ratings of US Presidents
pressure	Vapor Pressure of Mercury as a Function of Temperature
quakes	Locations of Earthquakes off Fiji
randu	Random Numbers from Congruential Generator RANDU
rivers	Lengths of Major North American Rivers
rock	Measurements on Petroleum Rock Samples
sleep	Student's Sleep Data
stackloss	Brownlee's Stack Loss Plant Data
state	US State Facts and Figures
sunspot.month	Monthly Sunspot Data, from 1749 to "Present"
sunspot.year	Yearly Sunspot Data, 1700-1988
sunspots	Monthly Sunspot Numbers, 1749-1983
swiss	Swiss Fertility and Socioeconomic Indicators

```
                        (1888) Data
treering                Yearly Tree-Ring Data, -6000-1979
trees                   Diameter, Height and Volume for Black Cherry
                        Trees
uspop                   Populations Recorded by the US Census
volcano                 Topographic Information on Auckland's Maunga
                        Whau Volcano
warpbreaks              The Number of Breaks in Yarn during Weaving
women                   Average Heights and Weights for American Women
```

install.packages('ggtext') <- BAD MANNERS, when installing packages do it in console rather than script

1.2 Tasks

==========

[1] first visualisation, point geometry, lines and smoothers, geom_bar and geom_column [5] histograms, box-and-whisker diagrams, backgrounds, axes, using scales (e.g. scale_x_continuous) [10] Key/legend, Annotations, Titles, Themes (brewer & viridis), Faceting, secondary axis -

==========

1.2.1 [1] First Visualisation

first, let's start off and install our first database

[4]:
```
library(tidyverse) #installing the library
college <- read_csv('http://672258.youcanlearnit.net/college.csv') #installing␣
 ↪the database
college <- college |>
  mutate(state=as.factor(state), region=as.factor(region),
         highest_degree=as.factor(highest_degree),
         control=as.factor(control), gender=as.factor(gender),
         loan_default_rate=as.numeric(loan_default_rate))
summary(college)
```

```
Rows: 1269 Columns: 17
  Column specification


Delimiter: ","
chr (8): name, city, state, region, highest_degree, control, gender,
loan_de…
dbl (9): id, admission_rate, sat_avg, undergrads, tuition,
faculty_salary_av…

 Use `spec()` to retrieve the full column specification for this
data.
 Specify the column types or set `show_col_types = FALSE` to quiet
this message.
Warning message:
```

```
"There was 1 warning in `mutate()`.
 In argument: `loan_default_rate =
as.numeric(loan_default_rate)`.
Caused by warning:
! NAs introduced by coercion"
```

```
       id                name             city          state
 Min.   :100654   Length   :1269   Length   :1269   PA     :101
 1st Qu.:153250   N.unique :1260   N.unique : 834   NY     : 84
 Median :186283   N.blank  :   0   N.blank  :   0   CA     : 71
 Mean   :186988   Min.nchar:   8   Min.nchar:   3   TX     : 63
 3rd Qu.:215284   Max.nchar:  65   Max.nchar:  23   OH     : 52
 Max.   :484905                                     IL     : 47
                                                    (Other):851
       region      highest_degree    control      gender     admission_rate
 Midwest  :353   Associate:  20   Private:763   CoEd :1237   Min.   :0.0509
 Northeast:299   Bachelor : 200   Public :506   Men  :   4   1st Qu.:0.5339
 South    :459   Graduate :1049                 Women:  28   Median :0.6687
 West     :158                                               Mean   :0.6501
                                                             3rd Qu.:0.7859
                                                             Max.   :1.0000

    sat_avg       undergrads       tuition      faculty_salary_avg
 Min.   : 720   Min.   :   47   Min.   : 2732   Min.   : 1451
 1st Qu.: 973   1st Qu.: 1296   1st Qu.: 8970   1st Qu.: 6191
 Median :1040   Median : 2556   Median :20000   Median : 7272
 Mean   :1060   Mean   : 5629   Mean   :21025   Mean   : 7656
 3rd Qu.:1120   3rd Qu.: 6715   3rd Qu.:30364   3rd Qu.: 8671
 Max.   :1545   Max.   :52280   Max.   :51008   Max.   :20650

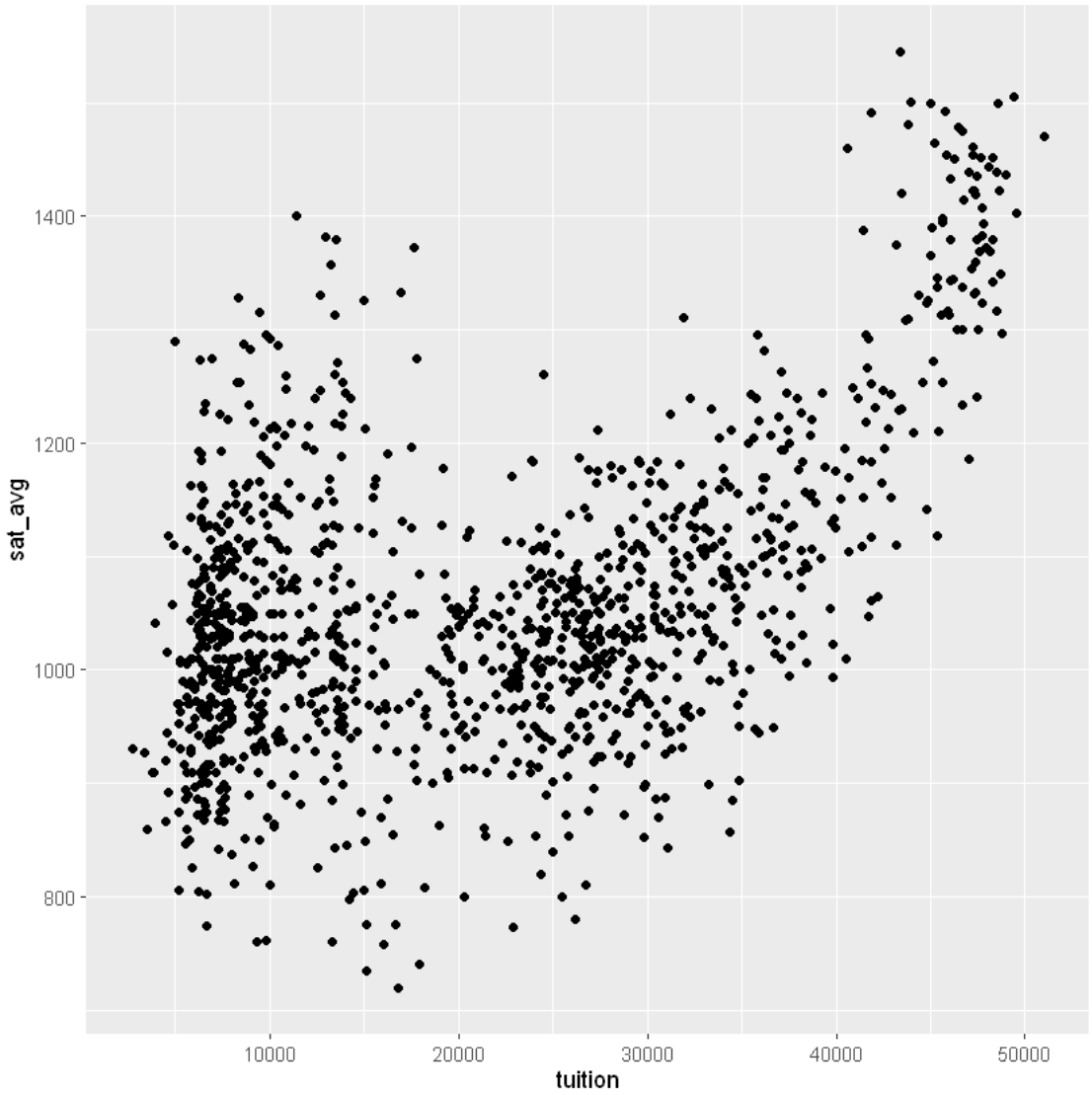
 loan_default_rate  median_debt         lon               lat
 Min.   :0.00000   Min.   : 6056   Min.   :-157.92   Min.   :19.71
 1st Qu.:0.03500   1st Qu.:21250   1st Qu.: -94.17   1st Qu.:35.22
 Median :0.05500   Median :24589   Median : -84.89   Median :39.74
 Mean   :0.06558   Mean   :23483   Mean   : -88.29   Mean   :38.61
 3rd Qu.:0.08300   3rd Qu.:27000   3rd Qu.: -78.63   3rd Qu.:41.81
 Max.   :0.33400   Max.   :41000   Max.   : -68.59   Max.   :61.22
 NAs    :2
```

ggplot, or the grammar of graphics, expands on qplot() and basic R graphic logic, viewing each element of visualisation as a piece of "grammar".with no added arguments, a plot looks something like this:

[7]:
```
# ggplot() called alone makes a blank visualisation => using the 'grammar of␣
 ↪graphics', we build upon this
ggplot()
```

let's try add some values (we will discuss more next section)

[6]:
```
ggplot(data=college) +
  geom_point(mapping=aes(x=tuition, y=sat_avg))
#ggplot [data, type, axis, aesthetics]
```



1.2.2 [2] Point Geometry

the basics of any graphing, is that each point has a coordinate for x and y.

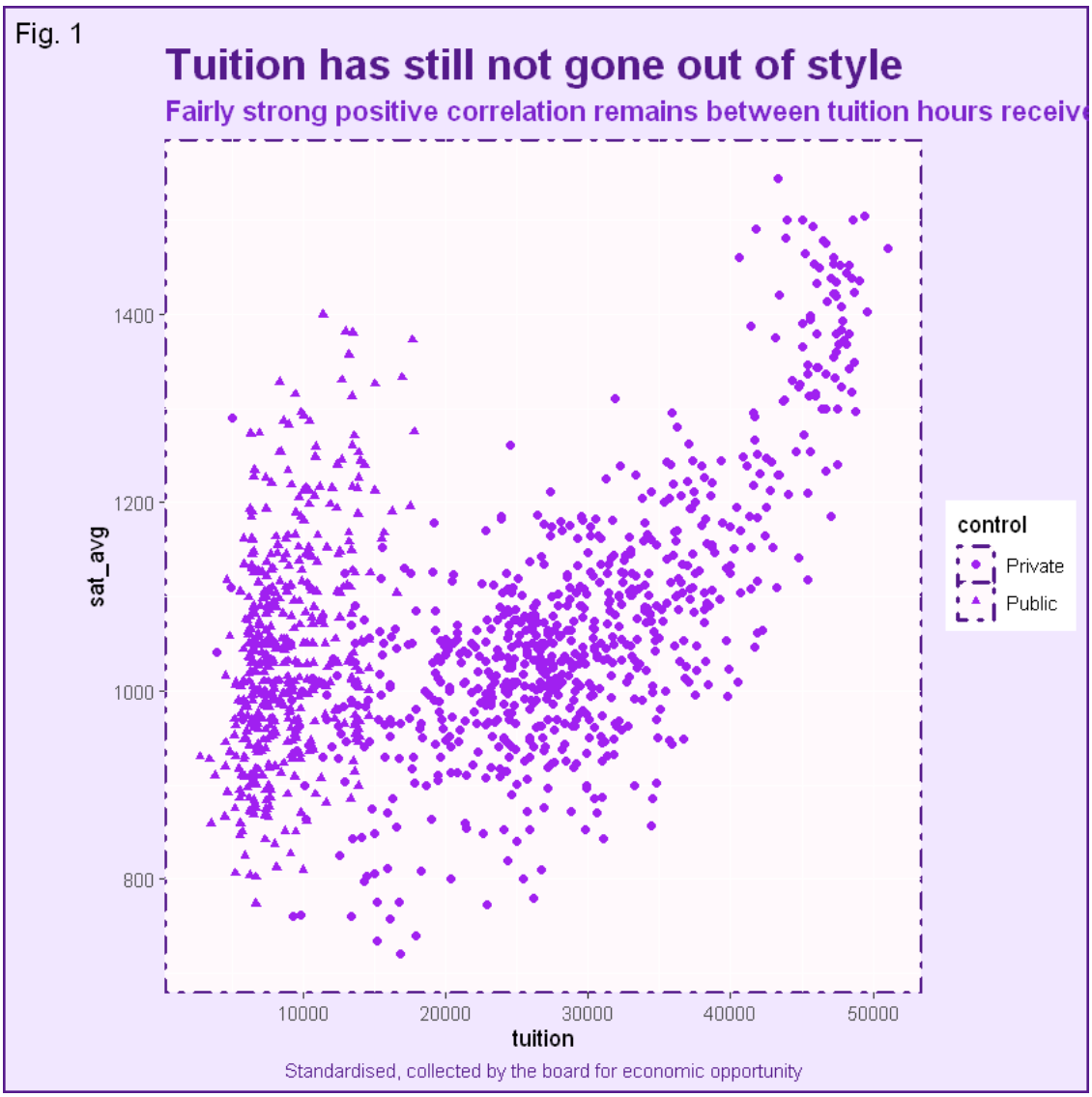
a fairly advanced example of this is seen below - however the core of ggplot remains the same (as above)

[10]:
```
#basics of any graph - each point has a coordinate for x and y

scat2 <- ggplot(data=college) +
  geom_point(mapping=aes(x=tuition,␣
 ↪y=sat_avg,shape=control),colour='purple')+#shape can be used if you have␣
 ↪factor variables
```

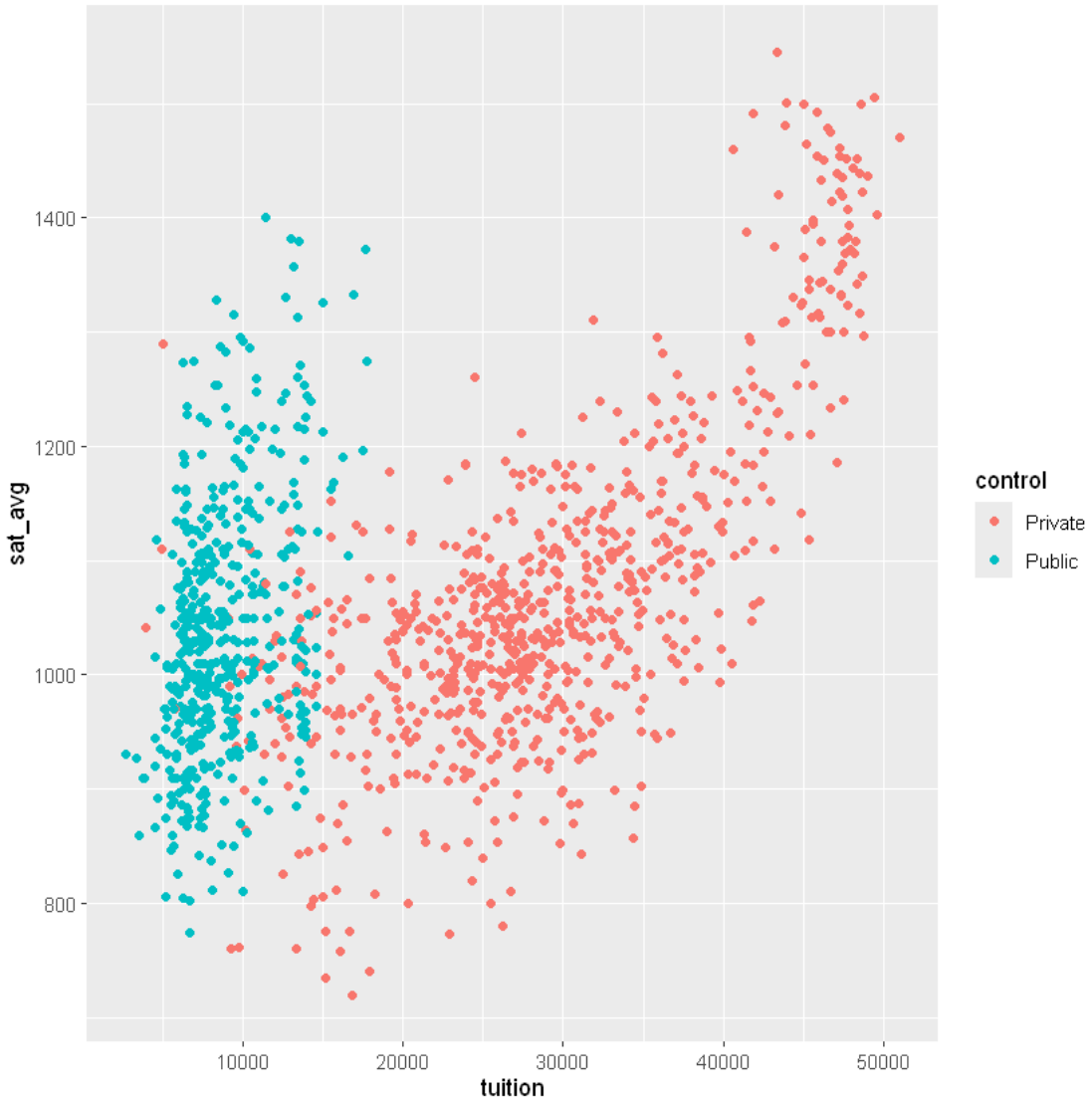
```
  labs(title = "Tuition has still not gone out of style",
       subtitle = "Fairly strong positive correlation remains between tuition␣
↪hours received and SAT Scores",
       caption = "Standardised, collected by the board for economic␣
↪opportunity",
       tag = "Fig. 1")+
 theme(plot.caption = element_text(hjust=0.5,color='purple4'),
       plot.title = element_text(colour='purple4',size=20,face='bold'),
       plot.subtitle= element_text(colour='purple3',size=13,face='bold'))+
 theme(panel.
↪background=element_rect(fill='#fff9fc',linetype=4,linewidth=1,colour='purple4'),
       plot.background=element_rect(fill =␣
↪'#f1e7ff',linewidth=1,colour='purple4'))
```

[13]:
```
print(scat2)
```

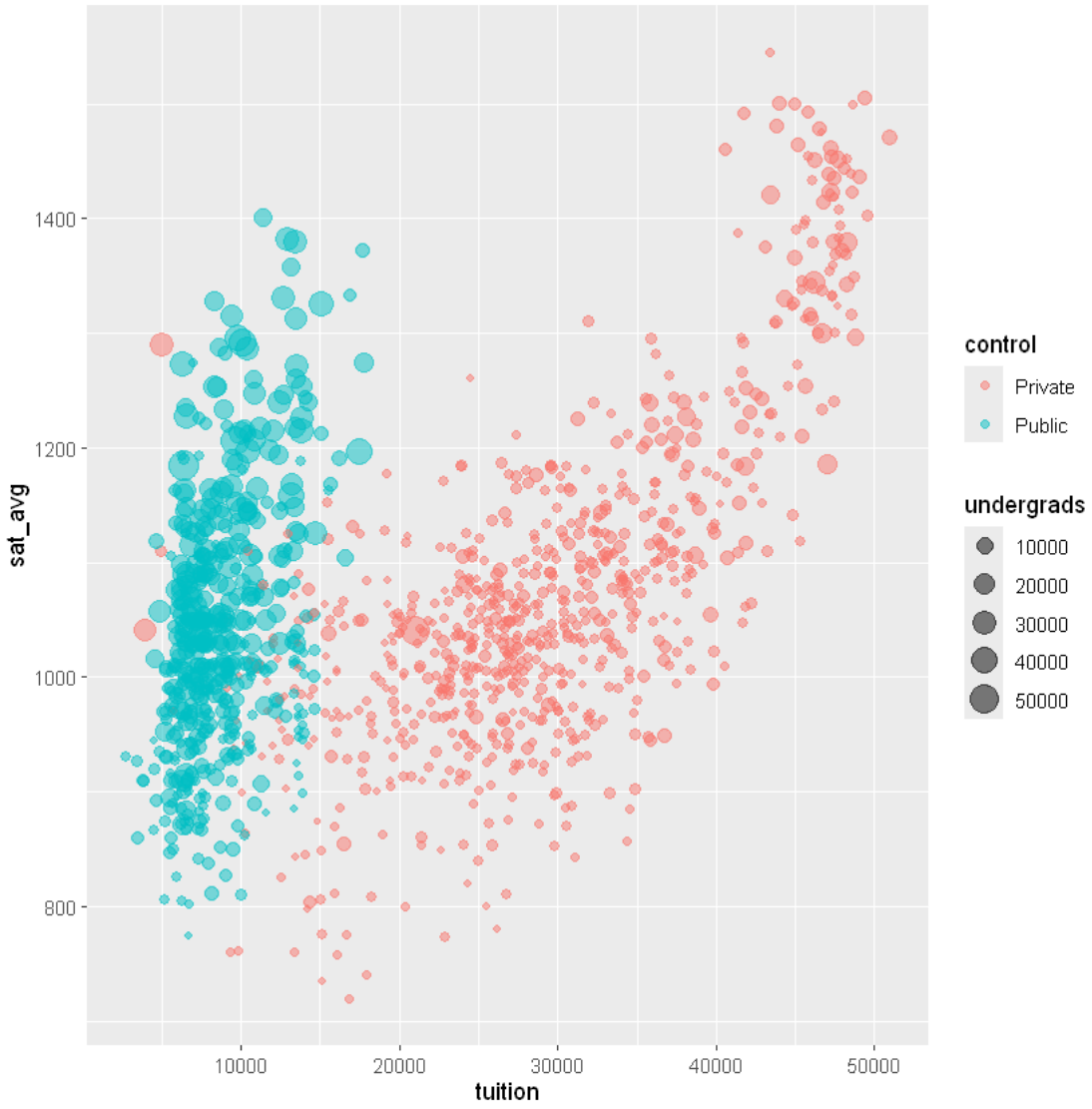


other plots (showing the use of colour, shape and size):

[11]:
```
ggplot(data=college) +
  geom_point(mapping=aes(x=tuition, y=sat_avg,colour=control)) #colour can also␣
 ↪be used if you have factor variables
```



[12]:
```
scat3 <- ggplot(data=college) +
  geom_point(mapping=aes(x=tuition, y=sat_avg,colour=control, size=undergrads),␣
 ↪alpha=50/100) #size self explain; alpha ranges 0 to 1
print(scat3)
```

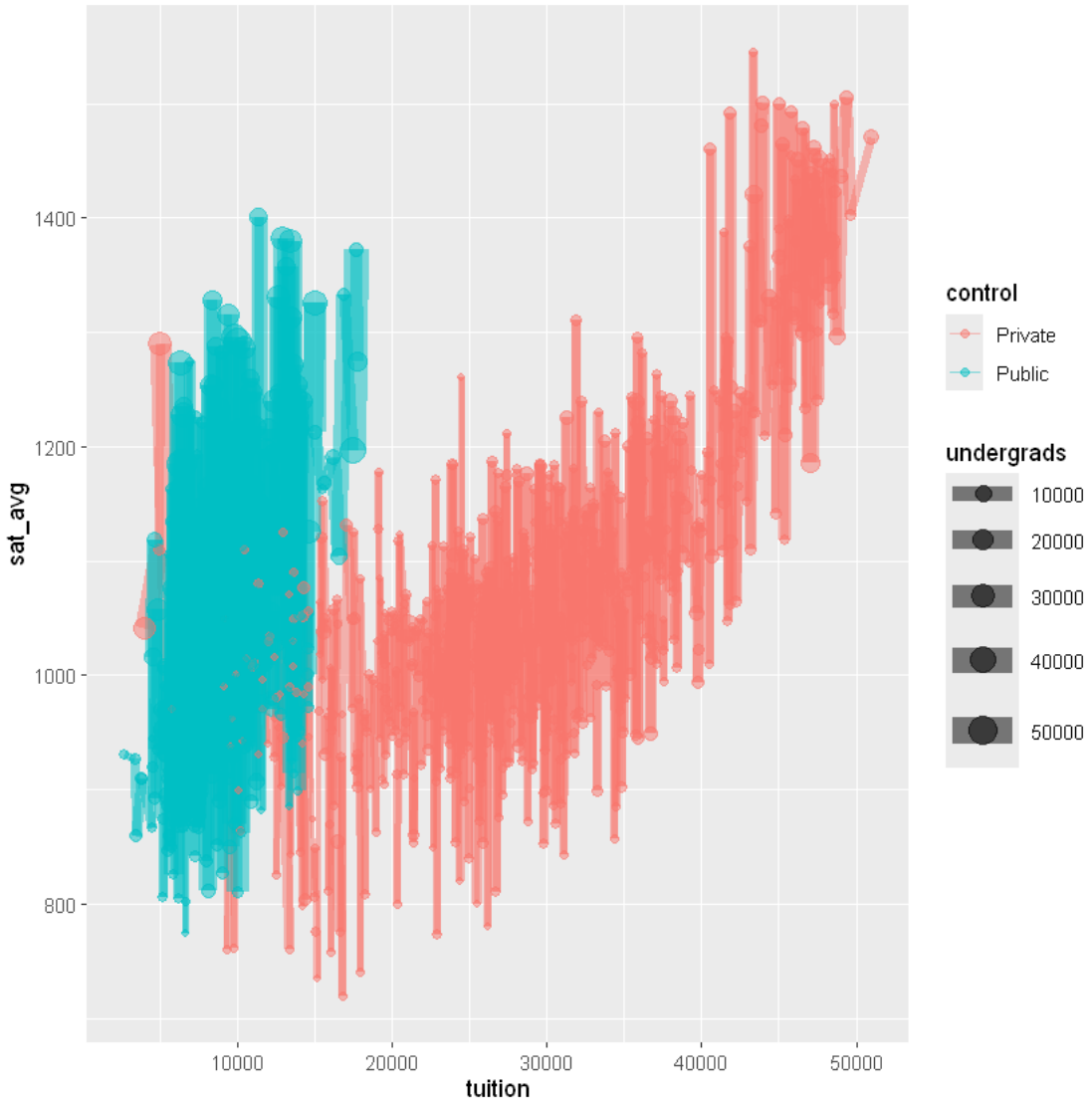


1.2.3 [3] Lines and Smoothers

Lines are effectively the same as point in terms of visualisation, except points are connected by a line

[14]:
```
ggplot(data=college) +
  geom_line(mapping=aes(x=tuition, y=sat_avg,colour=control, size=undergrads),␣
 ↪alpha=50/100)+
  geom_point(mapping=aes(x=tuition, y=sat_avg,colour=control, size=undergrads),␣
 ↪alpha=50/100)
```

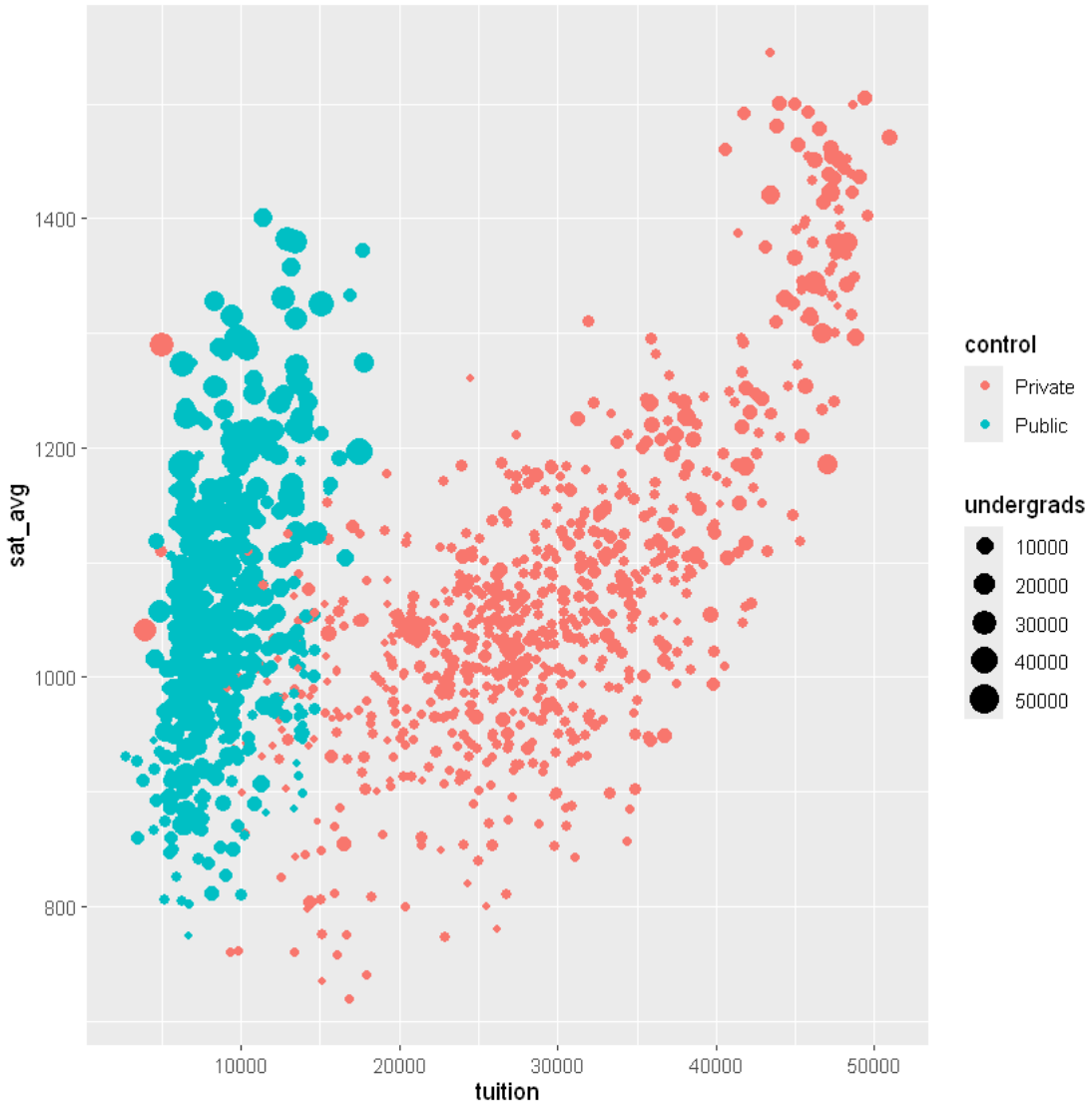
```
Warning message:
"Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 Please use `linewidth` instead."
```



NB: you can run the mapping/grammar on the visualisation level or the plot level, on the case of only one visualisation this is effectively the same

[15]:
```
ggplot(data=college,mapping=aes(x=tuition, y=sat_avg,colour=control,␣
↪size=undergrads),alpha=0.5) +
 geom_point()
```

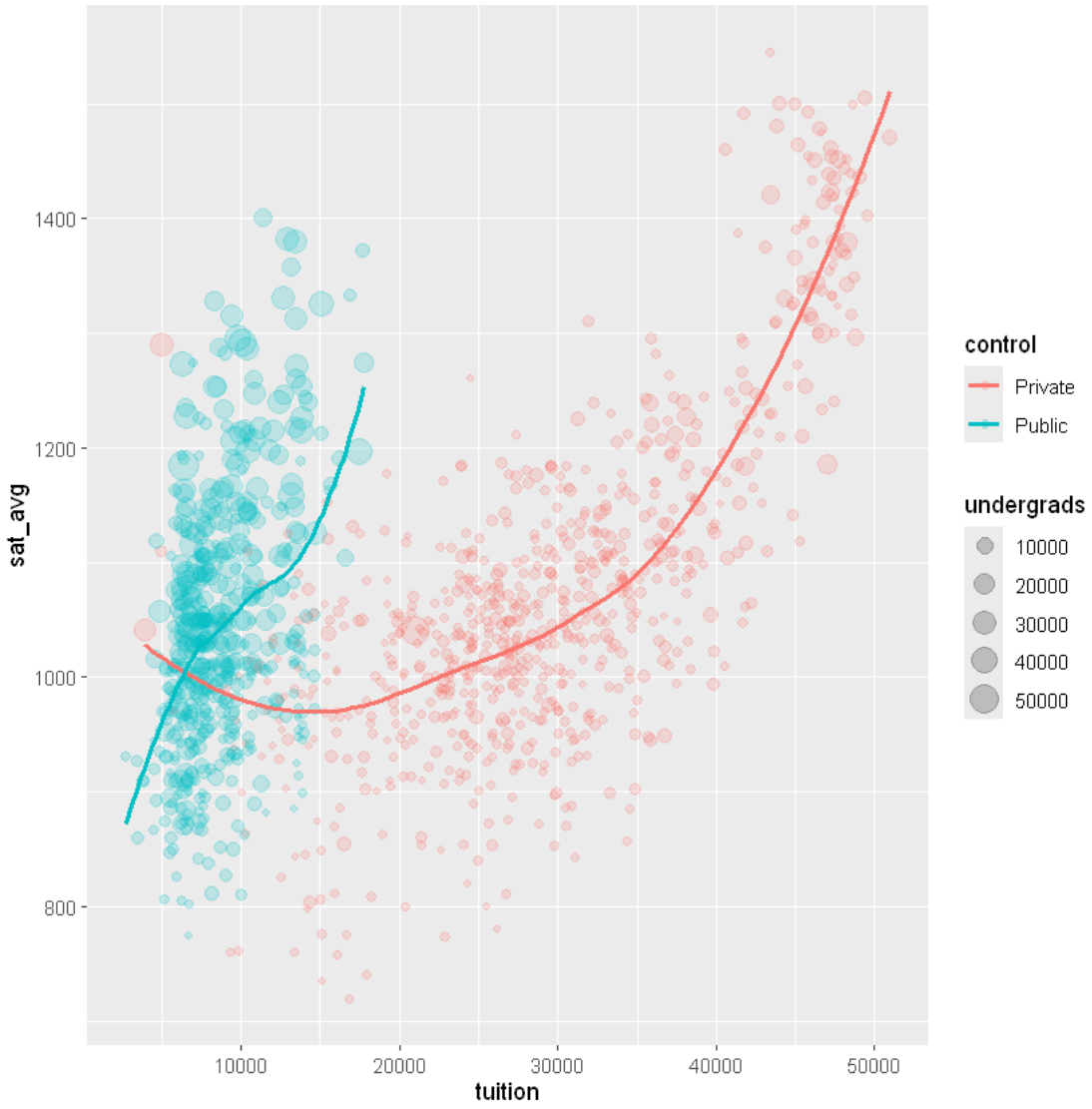
```
Warning message in fortify(data, …):
"Arguments in `…` must be used.
 Problematic argument:
• alpha = 0.5
 Did you misspell an argument name?"
```



Geom Smooth:the smooth() line is also very useful - imagine this as line of best fit. the default value is LOESS, but can be changed to OLS

[17]:
```
ggplot(data=college,mapping=aes(x=tuition, y=sat_avg,colour=control)) +
  geom_point(aes(size=undergrads),alpha=0.2) + geom_smooth(se=FALSE)
#smooth fits line to dataset; SE=standard error of this line in fitting to data
```

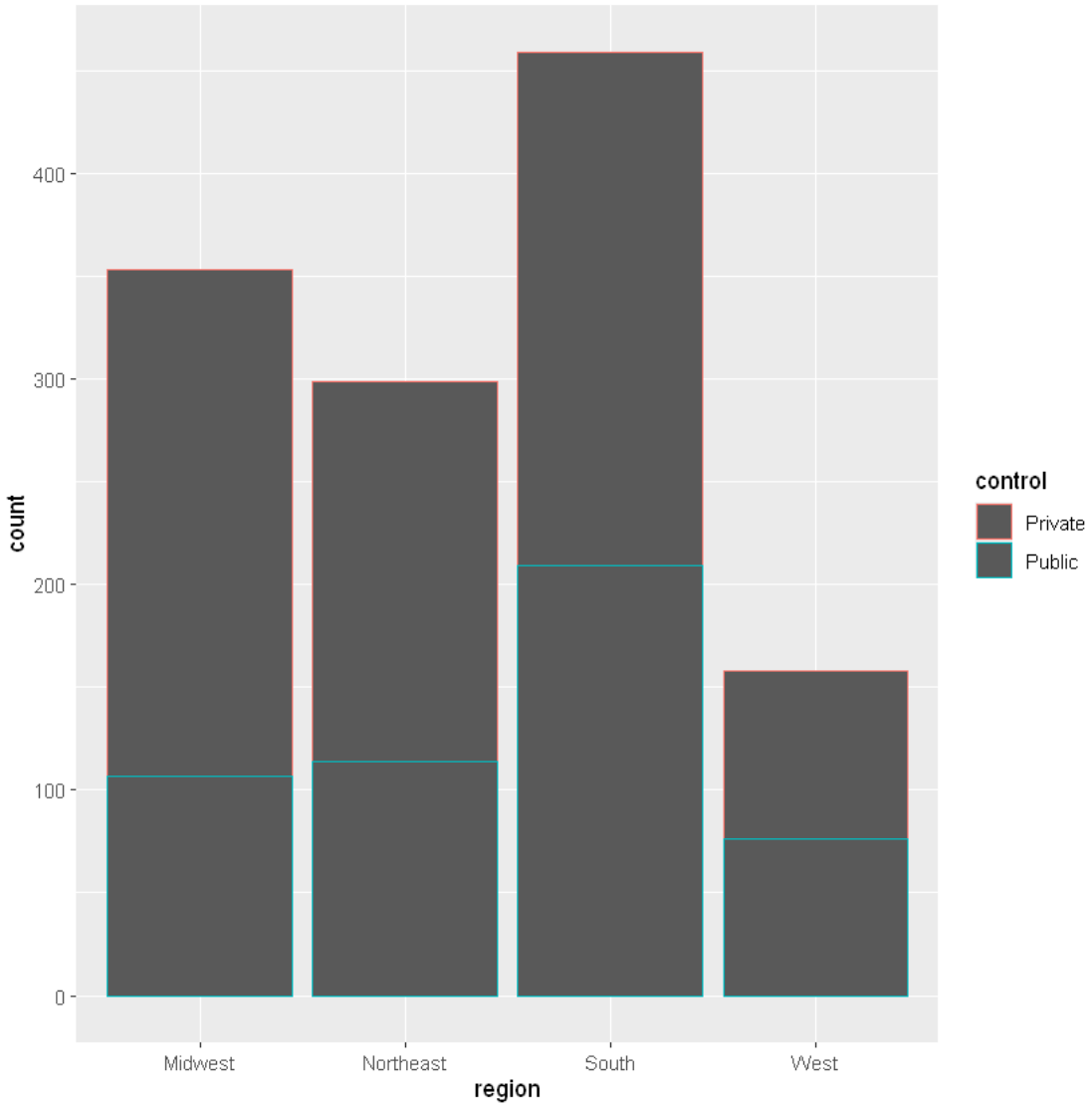
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



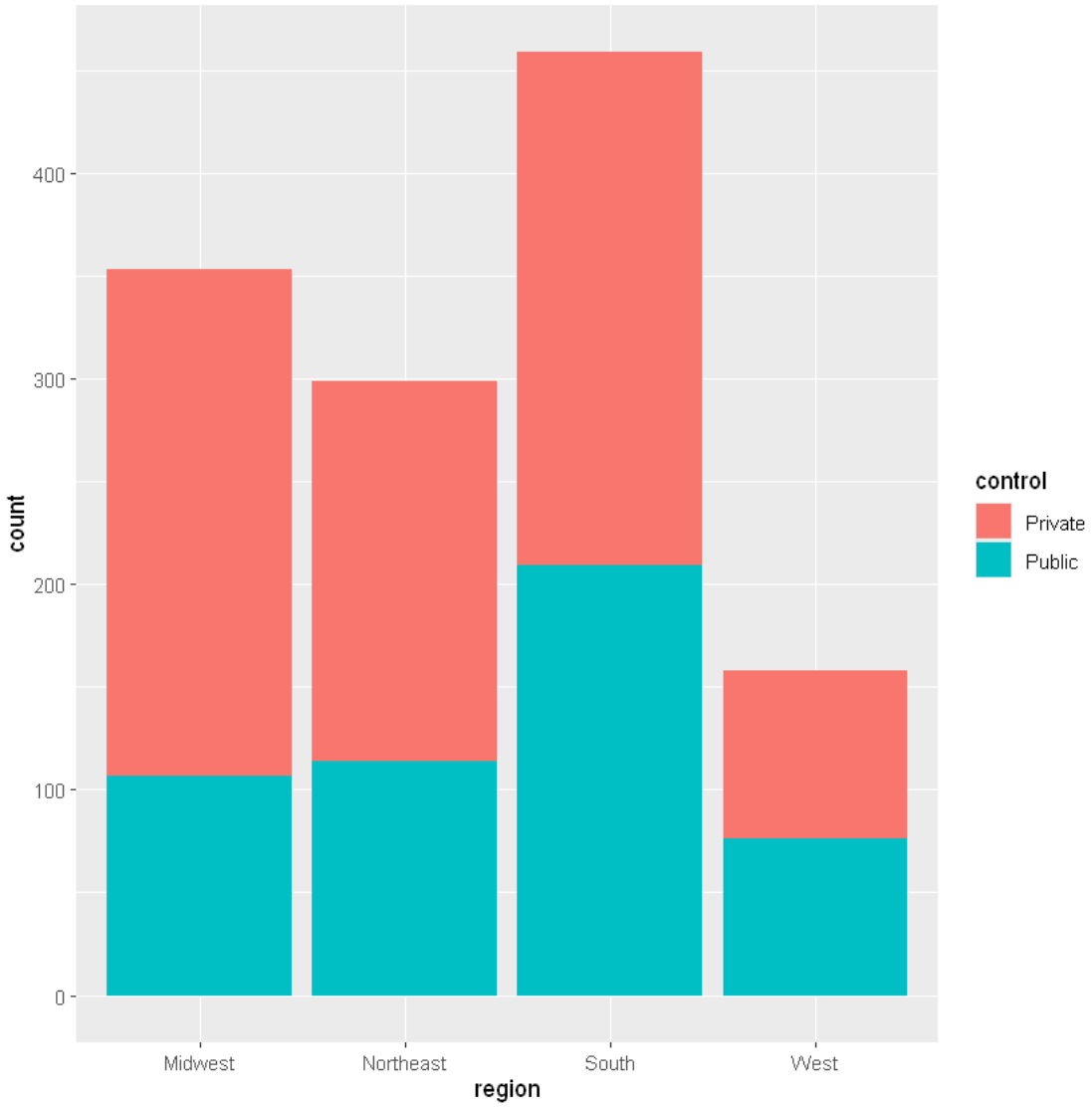
1.2.4 [4] geom_bar and column

geom_bar maps a variable; to the count/number of times it occurs (frequency)

[18]:
```
ggplot(data=college) +
  geom_bar(mapping=aes(x=region,colour=control)) #for bar aesthetic,␣
 ↪colour=outline
```

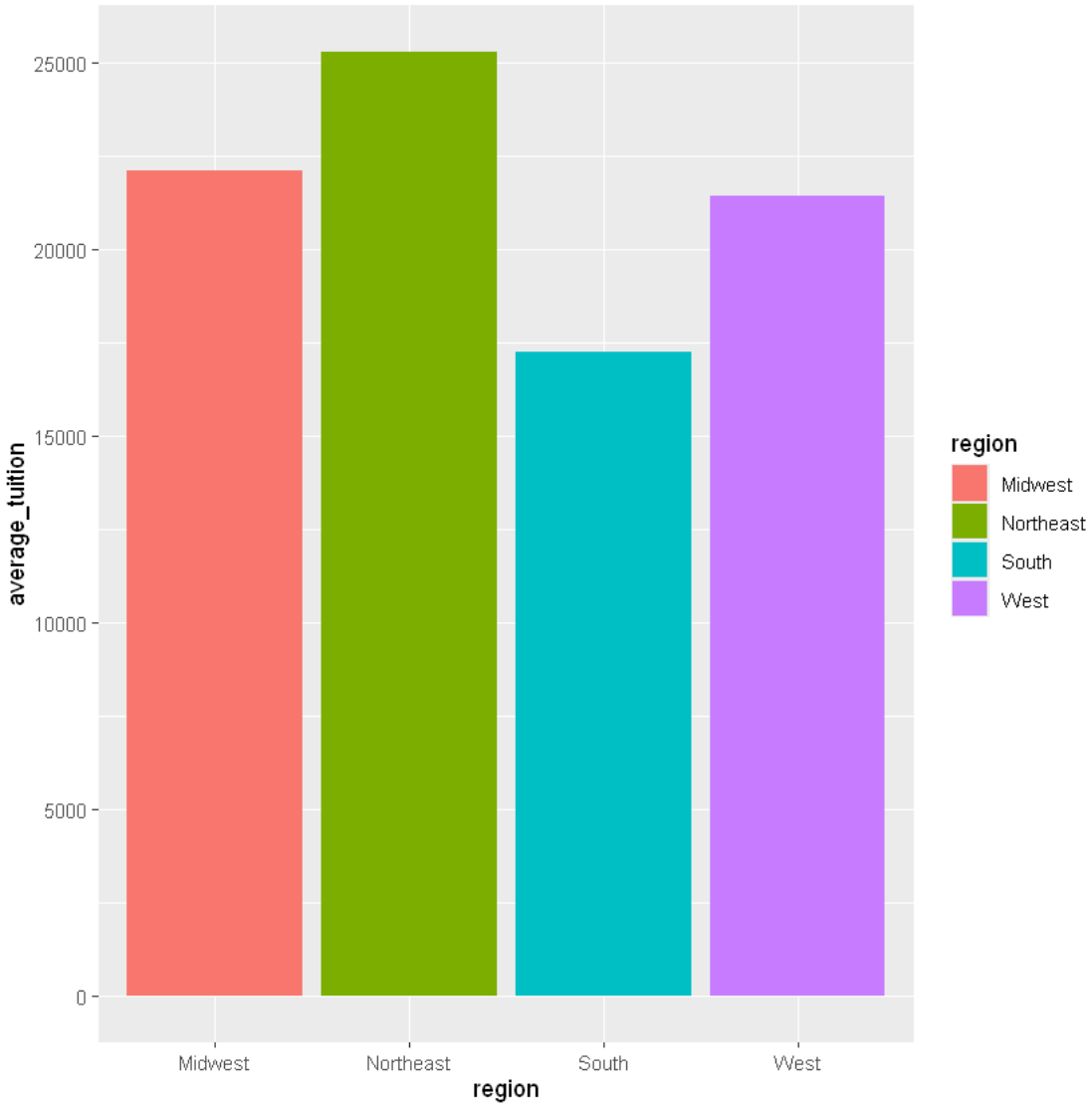


[21]:
```
ggplot(data=college) +
  geom_bar(mapping=aes(x=region,fill=control)) #for bar aesthetic, fill=bar␣
↪colouring in (can be patterned or used to make stacked bars)
```



geom_col() is literally the same; except you manually set y. we will quickly use dplyr to extract a statistic (average tuition); then map that

[22]:
```
college %>%
  group_by(region) %>%
  summarize(average_tuition=mean(tuition)) %>%
  ggplot() + #don't put in college we use piping and save code just take the␣
 ↪new college and pipe it as the input
  geom_col(mapping=aes(x=region, y=average_tuition, fill=region))
```



Quick recap on group_by() and R's answer to pivot tables:

[51]:
```
#?group_by() #this is effectively saying =SUMIF() and forming a new table
```

[25]:
```
# instead of...
# Name    | Region | Sales
# Mary    | North  | 30
# Phil    | North  | 30
# Cara    | South  | 70
# Elliot  | North  | 15
# William | South  | 12
# Emma    | South  | 50
```

[26]:
```
# test <- dataframe %>%
#   group_by(Region) %>%
#   summarise(Total=sum(Sales))
#
# we get
#
# Region | TotalSales
# North  | 30+30+15 = 75
# South  | 70+12+50 = 132
```

(be warned its just grouping, e.g. for max/min searches through all the north/south values) so in effect; group_by() groups the table so R treats them the same (adding to the "column" field; then the summarise() makes the table completely)

[29]:
```
m <- "adding another column to the group(by) simply does that; add another␣
 ↪column"
# test <- dataframe %>%
#   group_by(Region,Season) %>%
#   summarise(Total=sum(Sales),.groups="drop") -> R; its ok to stop grouping now
#
```

If u wanted to combine and have one of the variables be wide; just separately do a pivot_wider() operation

[30]:
```
# pivot_wider(dataframe,names_from=Season, values_from=Total)
```

Doing this means the original total column has been spread effectively into the four seasonsHence you have to make a new column for total just do mutate(dataframe,Total=spring+summer+autumn+winter)

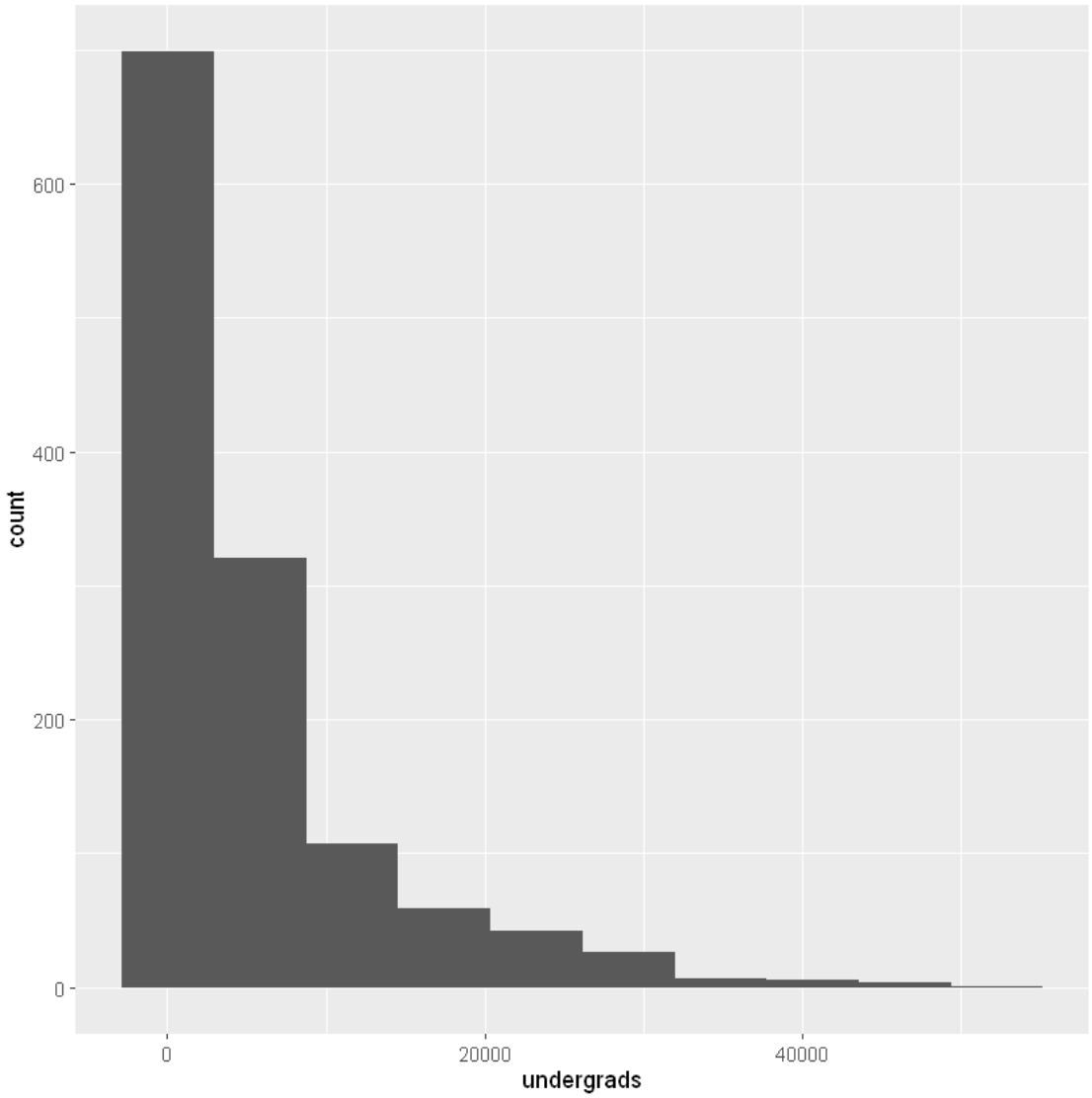
1.2.5 [5] Histograms

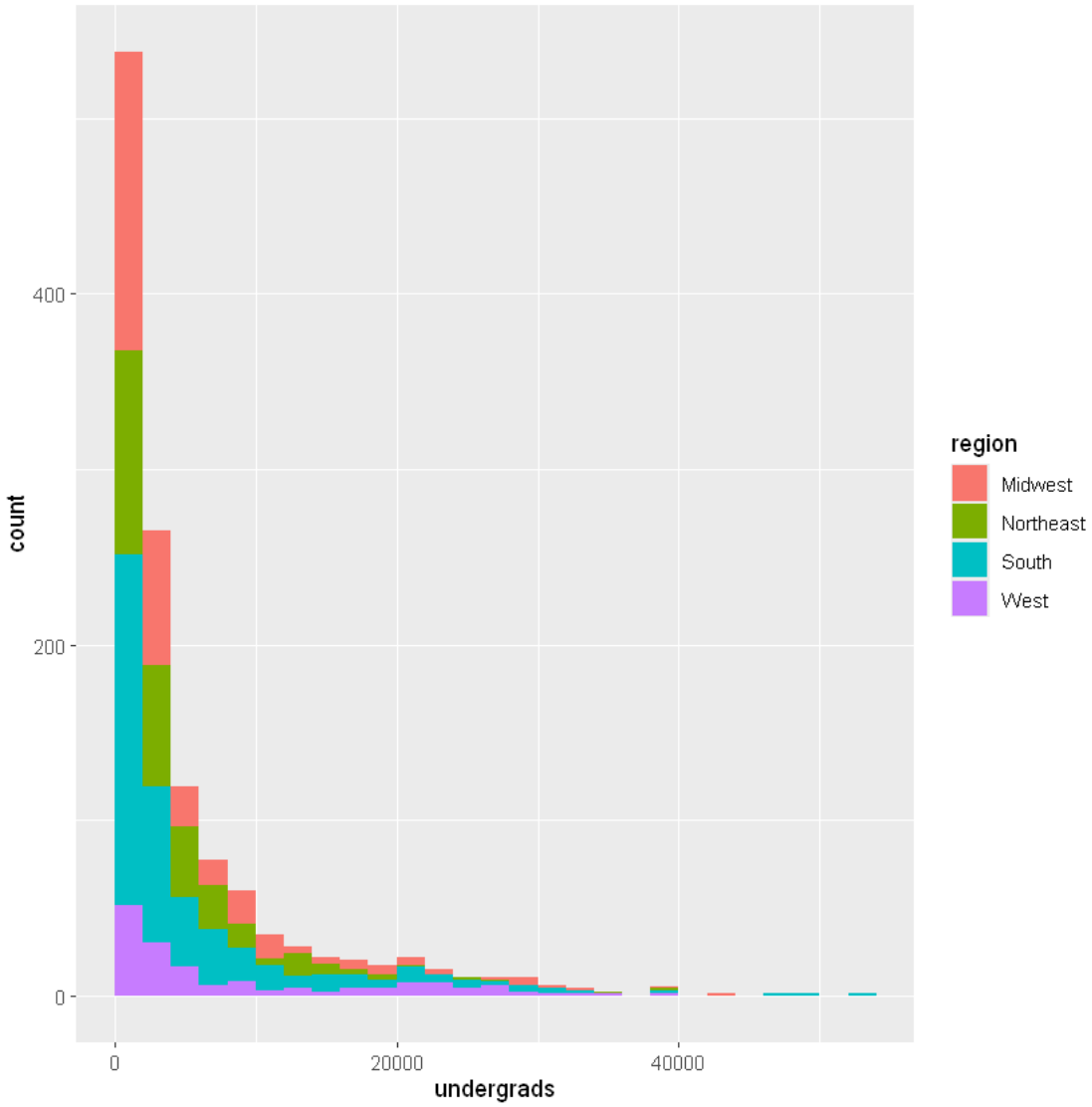
Say you have a data value like (12,1) and (100000,1); the chart can be very wide and spaced out. 'Histograms helps with this; by grouping data into "bins"'

[31]:
```
college %>%
  ggplot() + geom_histogram(mapping=aes(x=undergrads),origin=0,bins=10)


ggplot(data=college) +␣
 ↪geom_histogram(mapping=aes(x=undergrads,fill=region),boundary=0,binwidth=2000)
```

```
Warning message in geom_histogram(mapping = aes(x = undergrads), origin = 0,
bins = 10):
"Ignoring unknown parameters: `origin`"
```

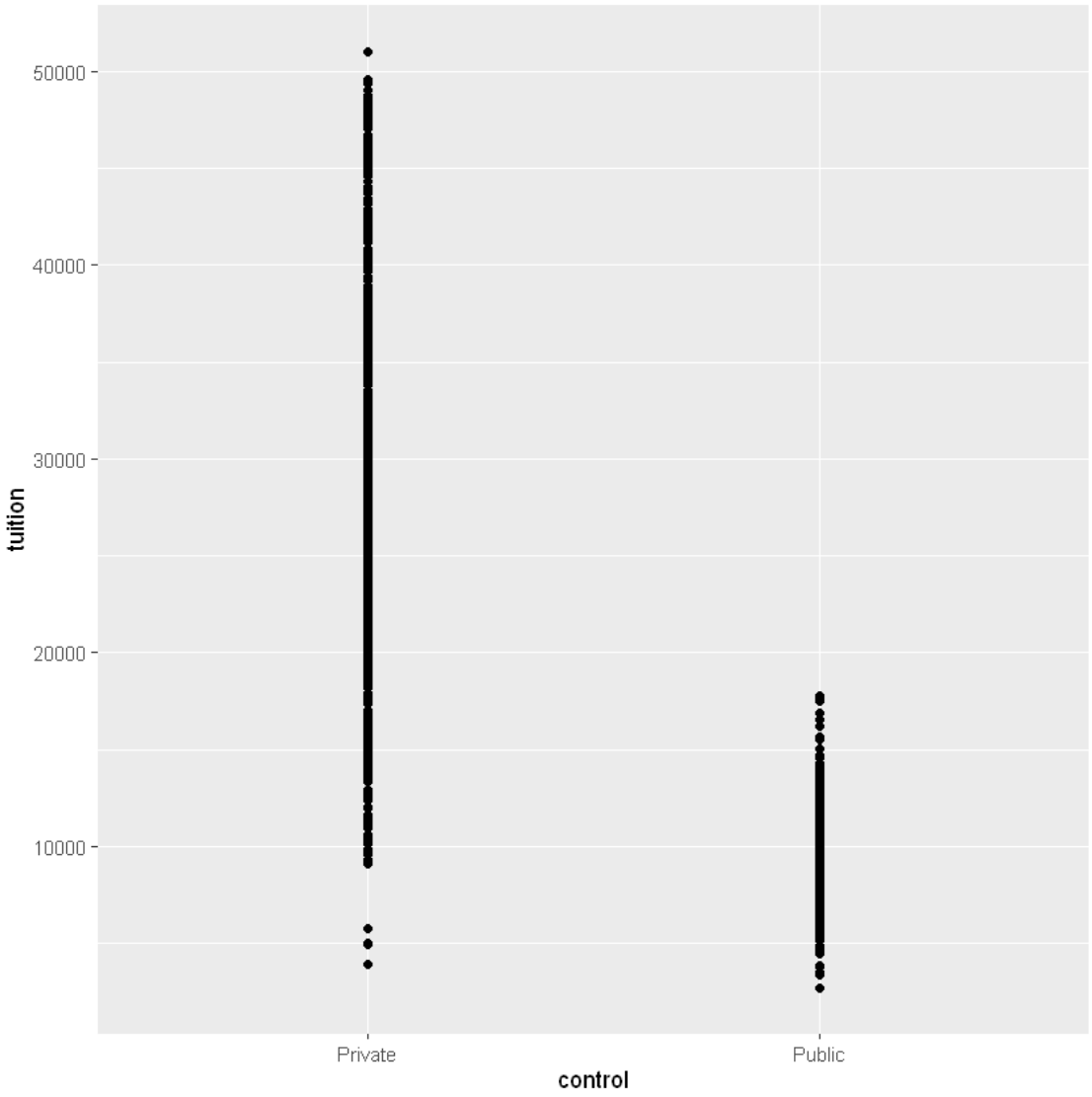

600
400
200
0
count
0
20000
40000
undergrads



1.2.6 [6] Box-and-Whisker Diagrams

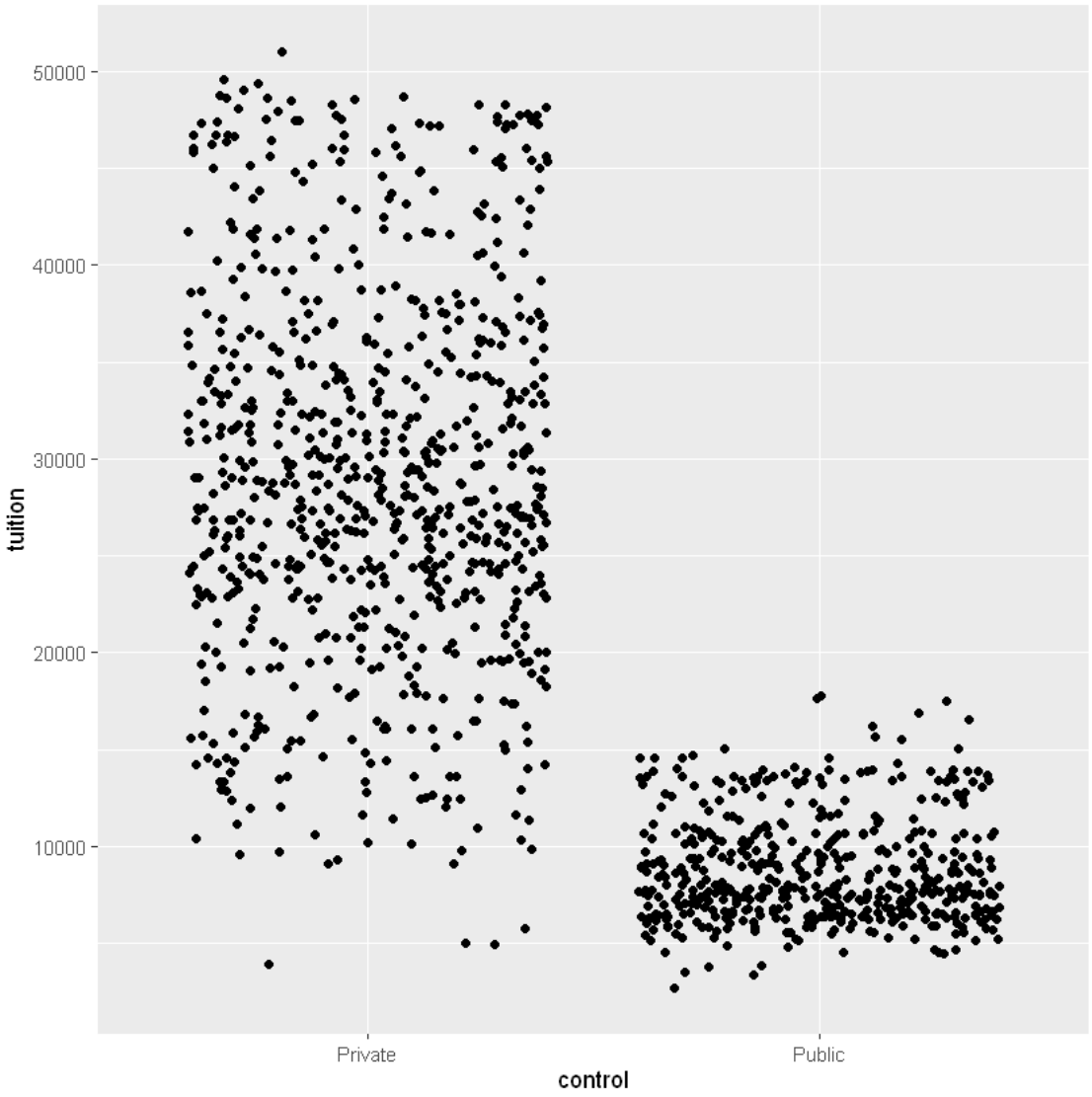
Let's try plot private against public schooling, and see how this affects tuition.

[32]:
```
college %>%
  ggplot() + geom_point(mapping=aes(x=control, y=tuition)) #not very useful...
```



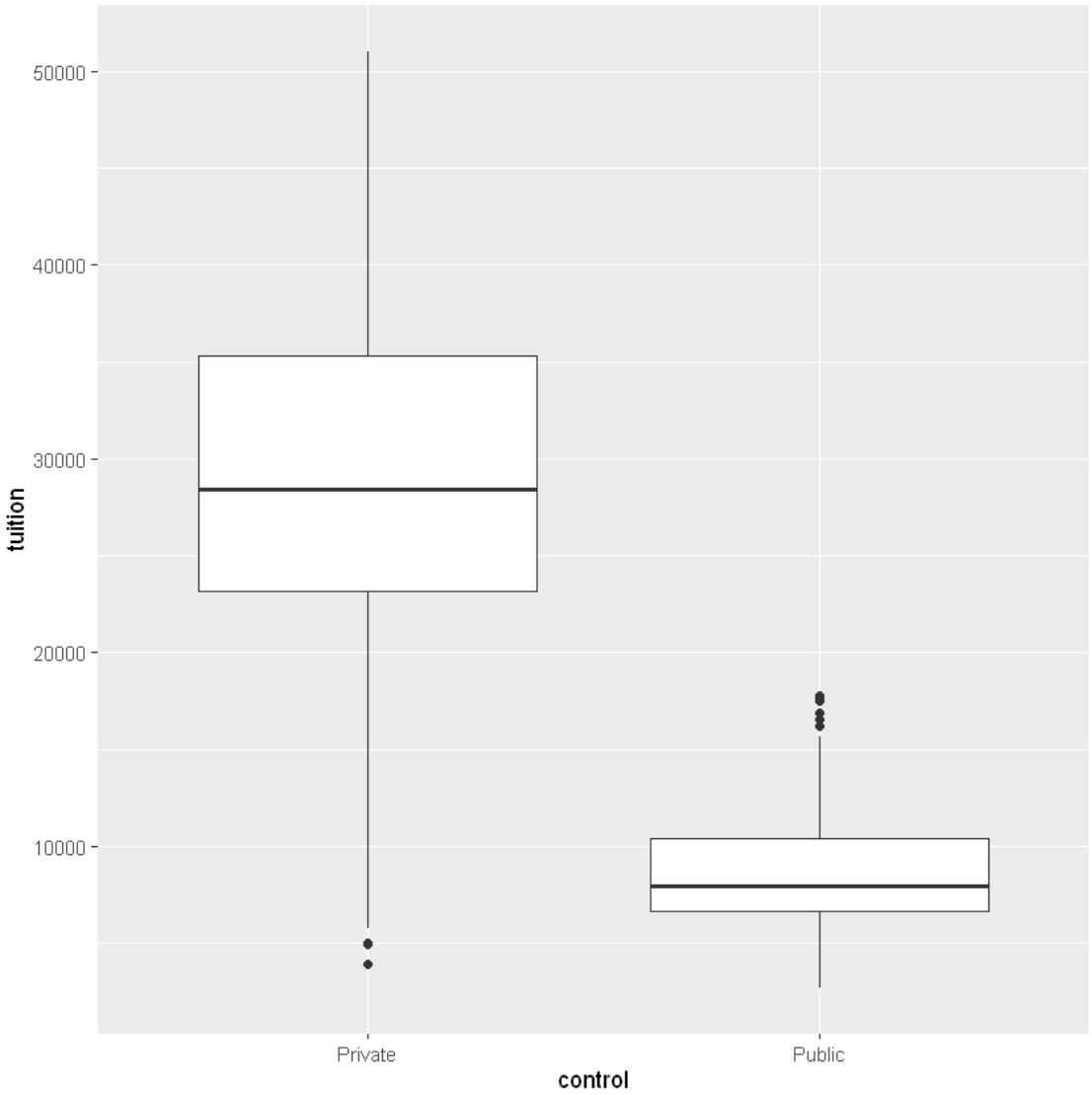
jitter plots slightly avert the problem, but still difficult to read

[33]:
```
college %>%
  ggplot() + geom_jitter(mapping=aes(x=control, y=tuition))
#similar; slightly averts the problem
```



the best and most commonplace action here is to present the data using a boxplot (box-and-whiskers) diagram

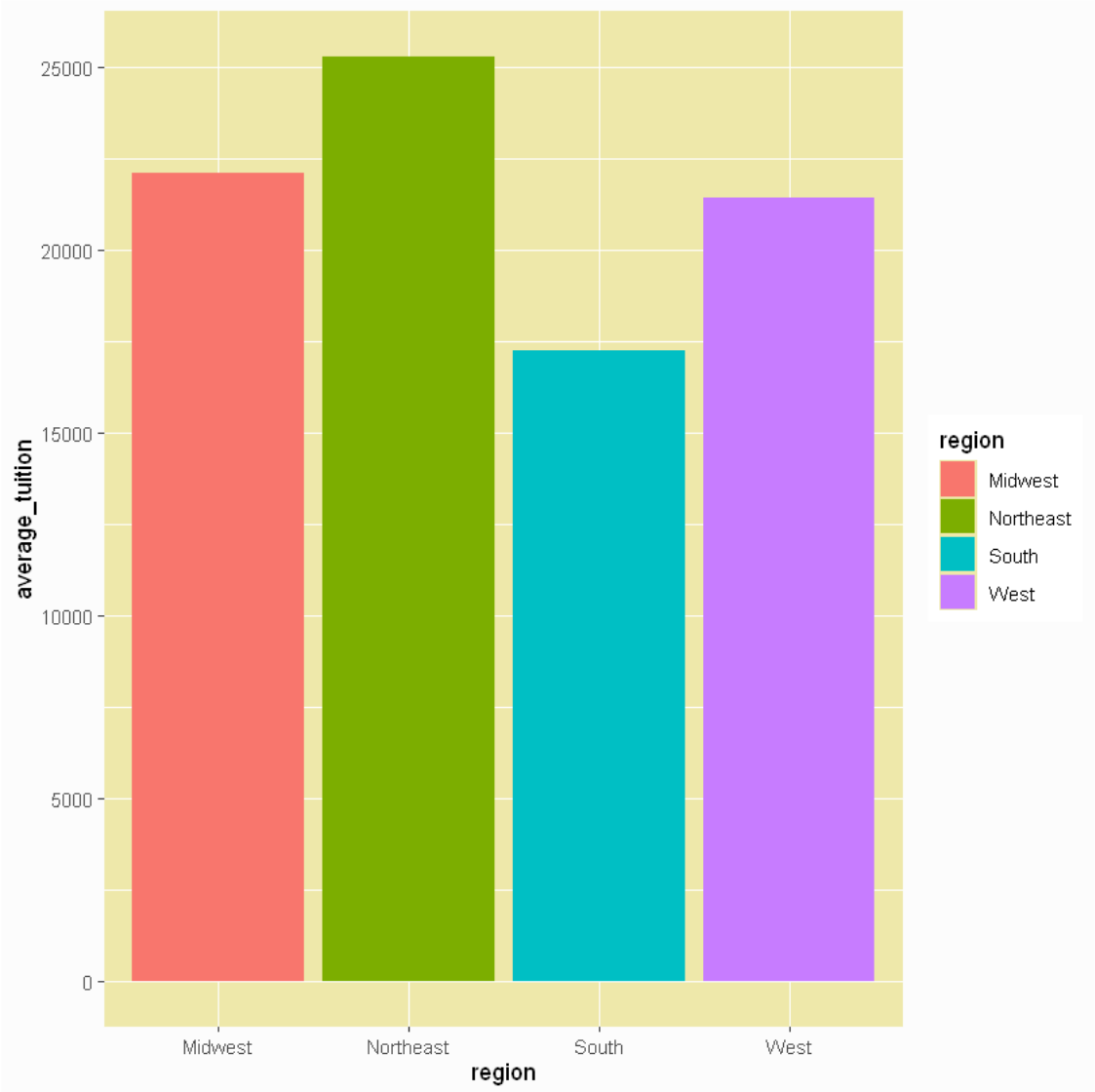
[34]:
```
college %>%
  ggplot() + geom_boxplot(mapping=aes(x=control, y=tuition))
```



1.2.7 [7] Backgrounds

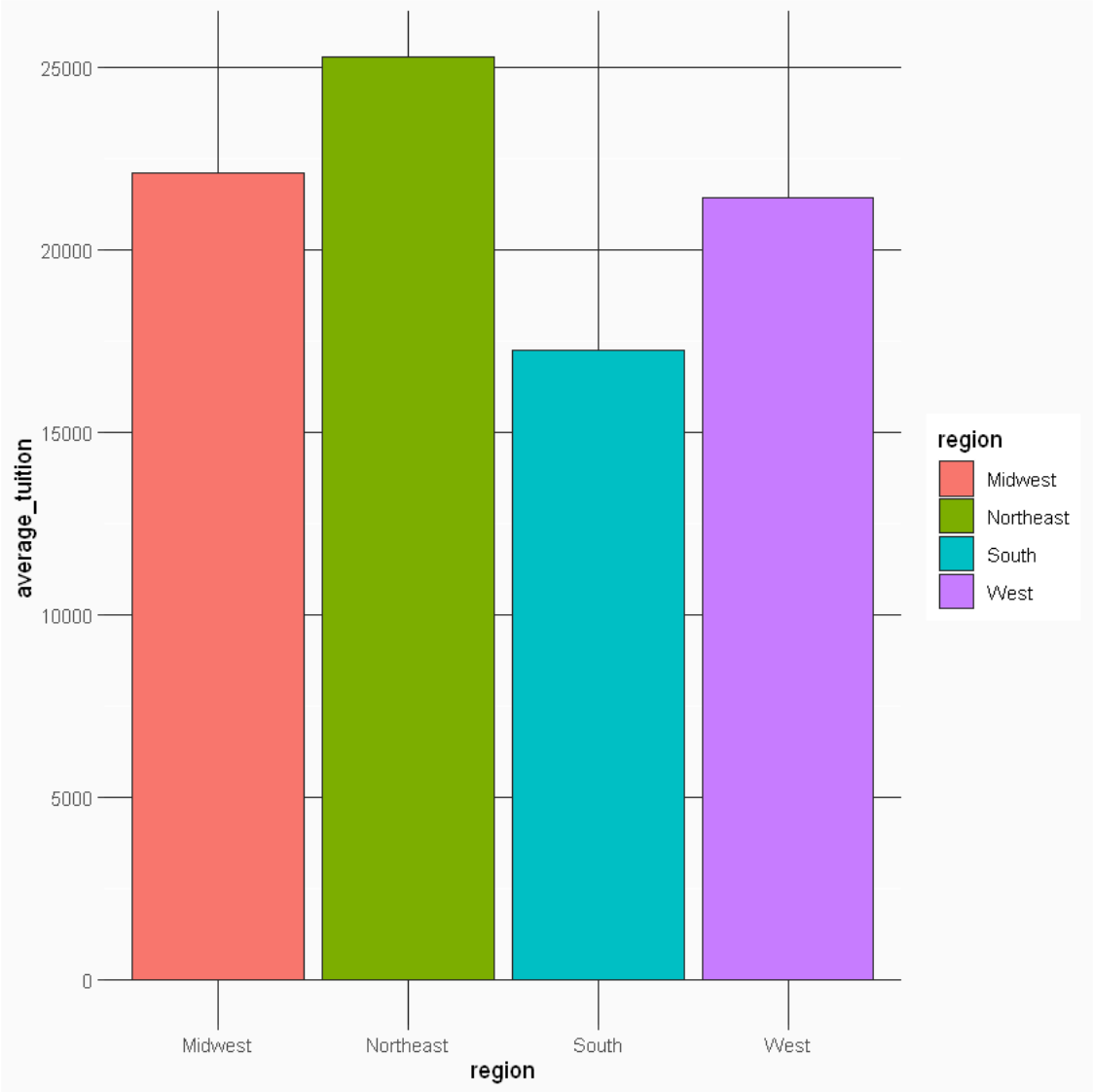
This section covers some R visualisations and graphics, please view below (more on themes later):*to summarise, use HEX codes for colours (or R presets)*

[38]:
```
Plot1 <- college %>%
  group_by(region) %>%
  summarize(average_tuition=mean(tuition)) %>%
  ggplot() +
  geom_col(mapping=aes(x=region, y=average_tuition, fill=region)) +
  theme(plot.background=element_rect(fill='grey99')) +
  theme(panel.background=element_rect(fill='palegoldenrod'))
print(Plot1)
```



plot that changes grid line colours:

[39]:
```
Plot2 <- college %>%
  group_by(region) %>%
  summarize(average_tuition=mean(tuition)) %>%
  ggplot() +
  geom_col(mapping=aes(x=region,␣
 ↪y=average_tuition,fill=region),colour='grey15') +
  theme(plot.background=element_rect(fill='grey98')) +
  theme(panel.background=element_rect(fill='grey98')) +
  theme(panel.grid.major=element_line(colour='grey15')) #changes grid line␣
 ↪colours
print(Plot2)
```



[40]:
```
colours() #all the colours in R
```

1. 'white' 2. 'aliceblue' 3. 'antiquewhite' 4. 'antiquewhite1' 5. 'antiquewhite2' 6. 'antiquewhite3' 7. 'antiquewhite4' 8. 'aquamarine' 9. 'aquamarine1' 10. 'aquamarine2' 11. 'aquamarine3' 12. 'aquamarine4' 13. 'azure' 14. 'azure1' 15. 'azure2' 16. 'azure3' 17. 'azure4' 18. 'beige' 19. 'bisque' 20. 'bisque1' 21. 'bisque2' 22. 'bisque3' 23. 'bisque4' 24. 'black' 25. 'blanchedalmond' 26. 'blue' 27. 'blue1' 28. 'blue2' 29. 'blue3' 30. 'blue4' 31. 'blueviolet' 32. 'brown' 33. 'brown1' 34. 'brown2' 35. 'brown3' 36. 'brown4' 37. 'burlywood' 38. 'burlywood1' 39. 'burlywood2' 40. 'burlywood3' 41. 'burlywood4' 42. 'cadetblue' 43. 'cadetblue1' 44. 'cadetblue2' 45. 'cadetblue3' 46. 'cadetblue4' 47. 'chartreuse' 48. 'chartreuse1' 49. 'chartreuse2' 50. 'chartreuse3' 51. 'chartreuse4' 52. 'chocolate' 53. 'chocolate1' 54. 'chocolate2' 55. 'chocolate3' 56. 'chocolate4' 57. 'coral' 58. 'coral1' 59. 'coral2' 60. 'coral3' 61. 'coral4' 62. 'cornflowerblue' 63. 'cornsilk' 64. 'cornsilk1' 65. 'cornsilk2' 66. 'cornsilk3' 67. 'cornsilk4' 68. 'cyan' 69. 'cyan1' 70. 'cyan2' 71. 'cyan3' 72. 'cyan4' 73. 'darkblue'

74. 'darkcyan' 75. 'darkgoldenrod' 76. 'darkgoldenrod1' 77. 'darkgoldenrod2' 78. 'darkgoldenrod3' 79. 'darkgoldenrod4' 80. 'darkgray' 81. 'darkgreen' 82. 'darkgrey' 83. 'darkkhaki' 84. 'darkmagenta' 85. 'darkolivegreen' 86. 'darkolivegreen1' 87. 'darkolivegreen2' 88. 'darkolivegreen3' 89. 'darkolivegreen4' 90. 'darkorange' 91. 'darkorange1' 92. 'darkorange2' 93. 'darkorange3' 94. 'darkorange4' 95. 'darkorchid' 96. 'darkorchid1' 97. 'darkorchid2' 98. 'darkorchid3' 99. 'darkorchid4' 100. 'darkred' 101. 'darksalmon' 102. 'darkseagreen' 103. 'darkseagreen1' 104. 'darkseagreen2' 105. 'darkseagreen3' 106. 'darkseagreen4' 107. 'darkslateblue' 108. 'darkslategray' 109. 'darkslategray1' 110. 'darkslategray2' 111. 'darkslategray3' 112. 'darkslategray4' 113. 'darkslategrey' 114. 'darkturquoise' 115. 'darkviolet' 116. 'deeppink' 117. 'deeppink1' 118. 'deeppink2' 119. 'deeppink3' 120. 'deeppink4' 121. 'deepskyblue' 122. 'deepskyblue1' 123. 'deepskyblue2' 124. 'deepskyblue3' 125. 'deepskyblue4' 126. 'dimgray' 127. 'dimgrey' 128. 'dodgerblue' 129. 'dodgerblue1' 130. 'dodgerblue2' 131. 'dodgerblue3' 132. 'dodgerblue4' 133. 'firebrick' 134. 'firebrick1' 135. 'firebrick2' 136. 'firebrick3' 137. 'firebrick4' 138. 'floralwhite' 139. 'forestgreen' 140. 'gainsboro' 141. 'ghostwhite' 142. 'gold' 143. 'gold1' 144. 'gold2' 145. 'gold3' 146. 'gold4' 147. 'goldenrod' 148. 'goldenrod1' 149. 'goldenrod2' 150. 'goldenrod3' 151. 'goldenrod4' 152. 'gray' 153. 'gray0' 154. 'gray1' 155. 'gray2' 156. 'gray3' 157. 'gray4' 158. 'gray5' 159. 'gray6' 160. 'gray7' 161. 'gray8' 162. 'gray9' 163. 'gray10' 164. 'gray11' 165. 'gray12' 166. 'gray13' 167. 'gray14' 168. 'gray15' 169. 'gray16' 170. 'gray17' 171. 'gray18' 172. 'gray19' 173. 'gray20' 174. 'gray21' 175. 'gray22' 176. 'gray23' 177. 'gray24' 178. 'gray25' 179. 'gray26' 180. 'gray27' 181. 'gray28' 182. 'gray29' 183. 'gray30' 184. 'gray31' 185. 'gray32' 186. 'gray33' 187. 'gray34' 188. 'gray35' 189. 'gray36' 190. 'gray37' 191. 'gray38' 192. 'gray39' 193. 'gray40' 194. 'gray41' 195. 'gray42' 196. 'gray43' 197. 'gray44' 198. 'gray45' 199. 'gray46' 200. 'gray47' 201. 202. 'maroon3' 203. 'maroon4' 204. 'mediumaquamarine' 205. 'mediumblue' 206. 'mediumorchid' 207. 'mediumorchid1' 208. 'mediumorchid2' 209. 'mediumorchid3' 210. 'mediumorchid4' 211. 'mediumpurple' 212. 'mediumpurple1' 213. 'mediumpurple2' 214. 'mediumpurple3' 215. 'mediumpurple4' 216. 'mediumseagreen' 217. 'mediumslateblue' 218. 'mediumspringgreen' 219. 'mediumturquoise' 220. 'mediumvioletred' 221. 'midnightblue' 222. 'mintcream' 223. 'mistyrose' 224. 'mistyrose1' 225. 'mistyrose2' 226. 'mistyrose3' 227. 'mistyrose4' 228. 'moccasin' 229. 'navajowhite' 230. 'navajowhite1' 231. 'navajowhite2' 232. 'navajowhite3' 233. 'navajowhite4' 234. 'navy' 235. 'navyblue' 236. 'oldlace' 237. 'olivedrab' 238. 'olivedrab1' 239. 'olivedrab2' 240. 'olivedrab3' 241. 'olivedrab4' 242. 'orange' 243. 'orange1' 244. 'orange2' 245. 'orange3' 246. 'orange4' 247. 'orangered' 248. 'orangered1' 249. 'orangered2' 250. 'orangered3' 251. 'orangered4' 252. 'orchid' 253. 'orchid1' 254. 'orchid2' 255. 'orchid3' 256. 'orchid4' 257. 'palegoldenrod' 258. 'palegreen' 259. 'palegreen1' 260. 'palegreen2' 261. 'palegreen3' 262. 'palegreen4' 263. 'paleturquoise' 264. 'paleturquoise1' 265. 'paleturquoise2' 266. 'paleturquoise3' 267. 'paleturquoise4' 268. 'palevioletred' 269. 'palevioletred1' 270. 'palevioletred2' 271. 'palevioletred3' 272. 'palevioletred4' 273. 'papayawhip' 274. 'peachpuff' 275. 'peachpuff1' 276. 'peachpuff2' 277. 'peachpuff3' 278. 'peachpuff4' 279. 'peru' 280. 'pink' 281. 'pink1' 282. 'pink2' 283. 'pink3' 284. 'pink4' 285. 'plum' 286. 'plum1' 287. 'plum2' 288. 'plum3' 289. 'plum4' 290. 'powderblue' 291. 'purple' 292. 'purple1' 293. 'purple2' 294. 'purple3' 295. 'purple4' 296. 'red' 297. 'red1' 298. 'red2' 299. 'red3' 300. 'red4' 301. 'rosybrown' 302. 'rosybrown1' 303. 'rosybrown2' 304. 'rosybrown3' 305. 'rosybrown4' 306. 'royalblue' 307. 'royalblue1' 308. 'royalblue2' 309. 'royalblue3' 310. 'royalblue4' 311. 'saddlebrown' 312. 'salmon' 313. 'salmon1' 314. 'salmon2' 315. 'salmon3' 316. 'salmon4' 317. 'sandybrown' 318. 'seagreen' 319. 'seagreen1' 320. 'seagreen2' 321. 'seagreen3' 322. 'seagreen4' 323. 'seashell' 324. 'seashell1' 325. 'seashell2' 326. 'seashell3' 327. 'seashell4' 328. 'sienna' 329. 'sienna1' 330. 'sienna2' 331. 'sienna3' 332. 'sienna4' 333. 'skyblue' 334. 'skyblue1' 335. 'skyblue2' 336. 'skyblue3' 337. 'skyblue4' 338. 'slateblue' 339. 'slateblue1' 340. 'slateblue2' 341. 'slateblue3' 342. 'slateblue4' 343. 'slategray' 344. 'slategray1' 345. 'slategray2' 346. 'slategray3' 347. 'slategray4' 348. 'slategrey' 349. 'snow' 350. 'snow1' 351. 'snow2' 352. 'snow3' 353. 'snow4' 354. 'spring-

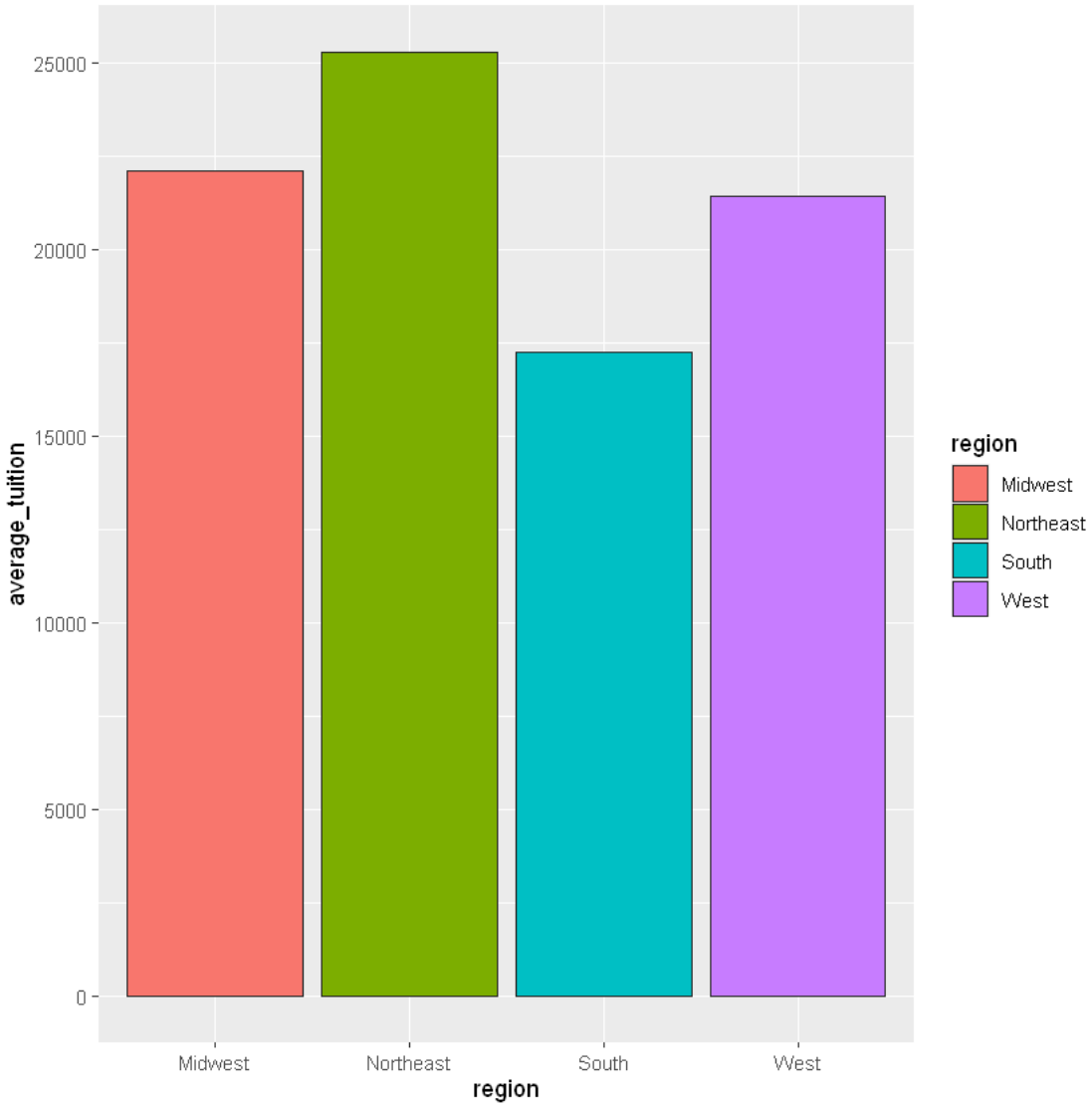
green' 355. 'springgreen1' 356. 'springgreen2' 357. 'springgreen3' 358. 'springgreen4' 359. 'steelblue' 360. 'steelblue1' 361. 'steelblue2' 362. 'steelblue3' 363. 'steelblue4' 364. 'tan' 365. 'tan1' 366. 'tan2' 367. 'tan3' 368. 'tan4' 369. 'thistle' 370. 'thistle1' 371. 'thistle2' 372. 'thistle3' 373. 'thistle4' 374. 'tomato' 375. 'tomato1' 376. 'tomato2' 377. 'tomato3' 378. 'tomato4' 379. 'turquoise' 380. 'turquoise1' 381. 'turquoise2' 382. 'turquoise3' 383. 'turquoise4' 384. 'violet' 385. 'violetred' 386. 'violetred1' 387. 'violetred2' 388. 'violetred3' 389. 'violetred4' 390. 'wheat' 391. 'wheat1' 392. 'wheat2' 393. 'wheat3' 394. 'wheat4' 395. 'whitesmoke' 396. 'yellow' 397. 'yellow1' 398. 'yellow2' 399. 'yellow3' 400. 'yellow4' 401. 'yellowgreen'

[52]:
```
#?theme() #more on R themes
```

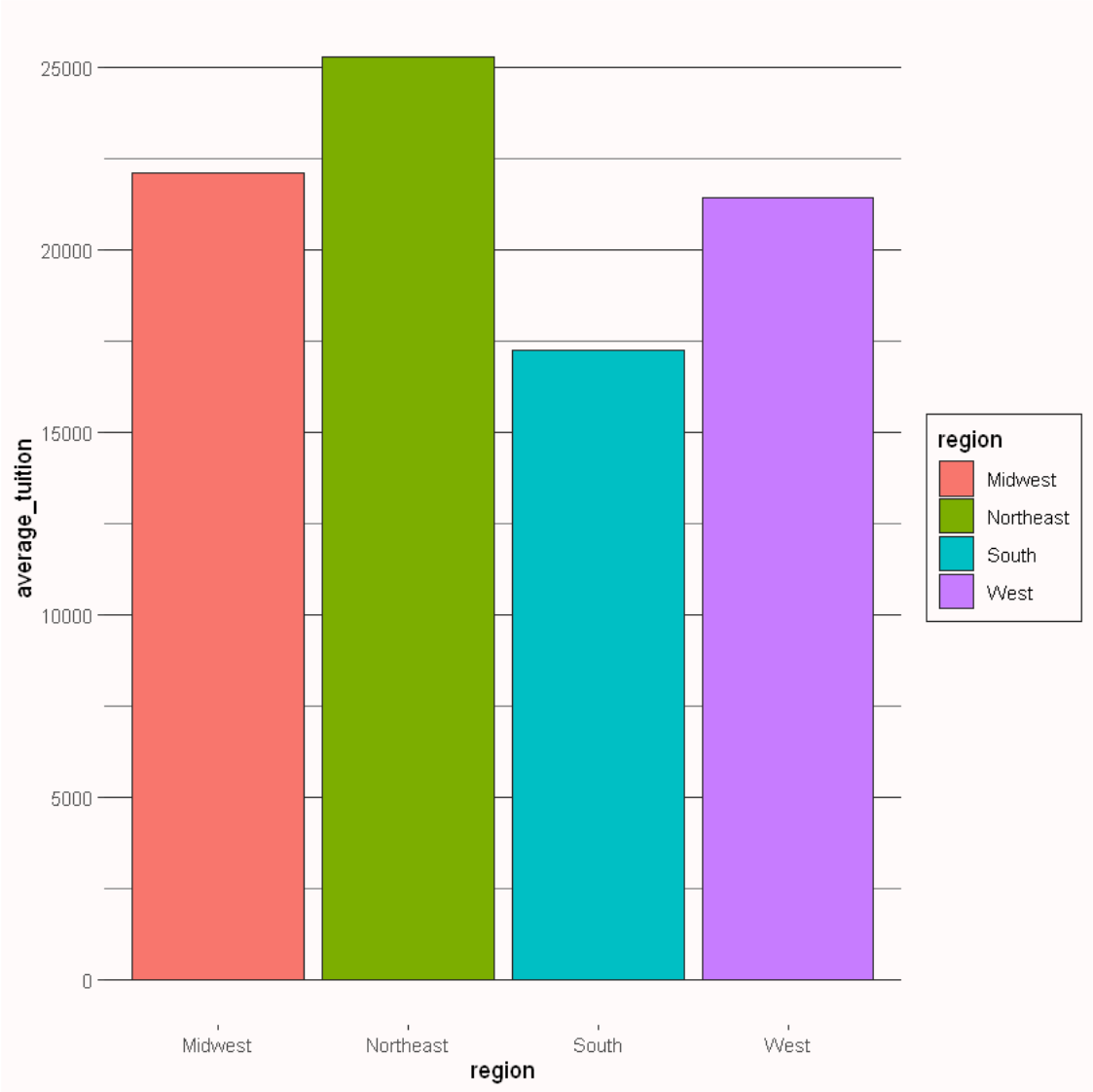
more visualisations:

[42]:
```
Plot3 <- college %>%
  group_by(region) %>%
  summarize(average_tuition=mean(tuition)) %>%
  ggplot() +

  geom_col(mapping=aes(x=region, y=average_tuition,␣
 ↪fill=region),colour='grey15')
print(Plot3)
```



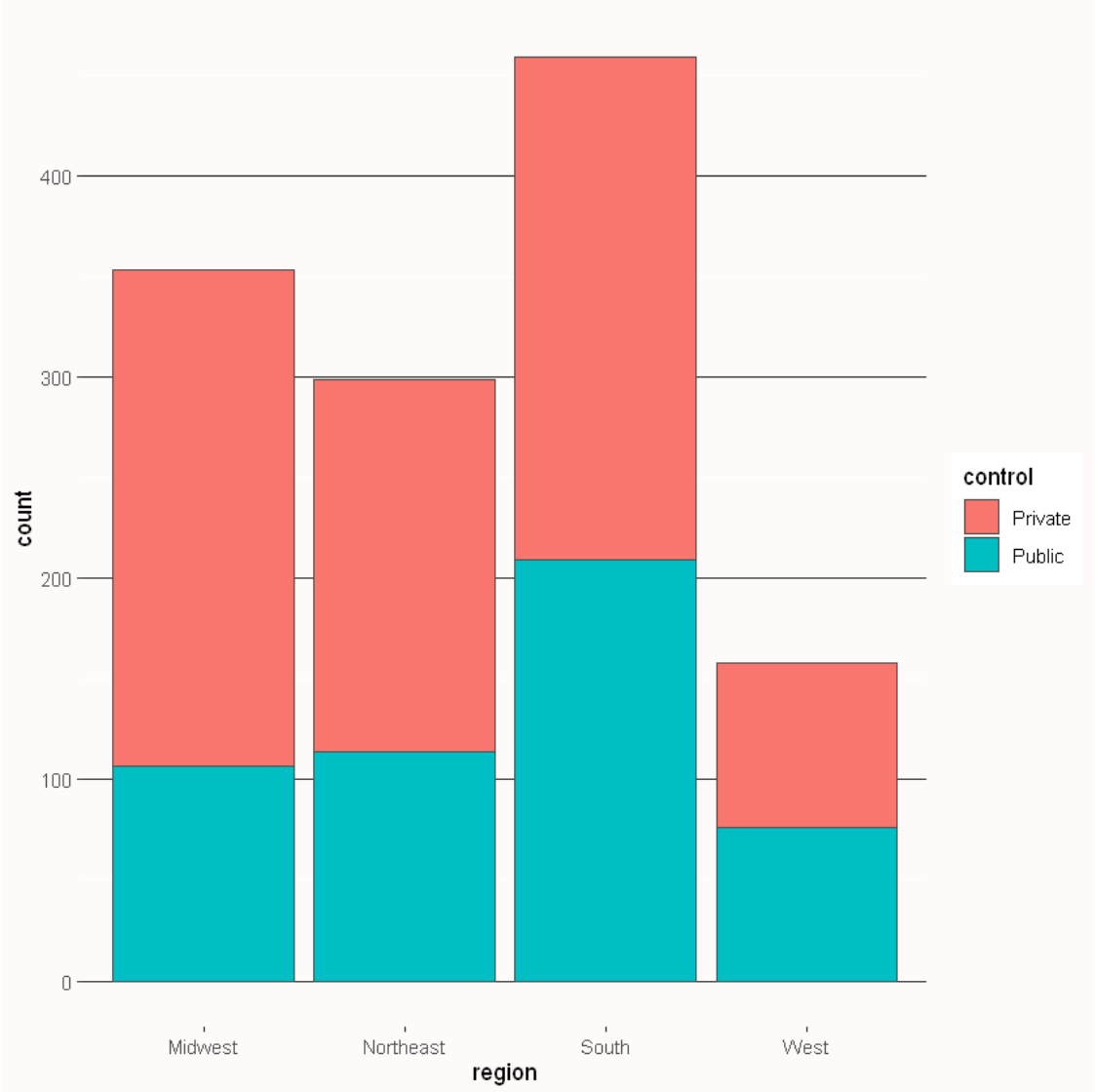
[43]:
```
Plot3 <- Plot3+
  theme(plot.background=element_rect(fill='snow')) +
  theme(panel.background=element_rect(fill='snow')) +
  theme(panel.grid.major.y=element_line(colour='grey15')) + #major grid lines␣
 ↪only for y
  theme(panel.grid.minor.y=element_line(colour='grey50')) +
  theme(panel.grid.major.x=element_blank())+
  theme(legend.background = element_rect(colour='grey15',fill = "snow"),legend.
 ↪key = element_rect(fill = "snow"))
print(Plot3)
```



1.2.8 [8] Axes

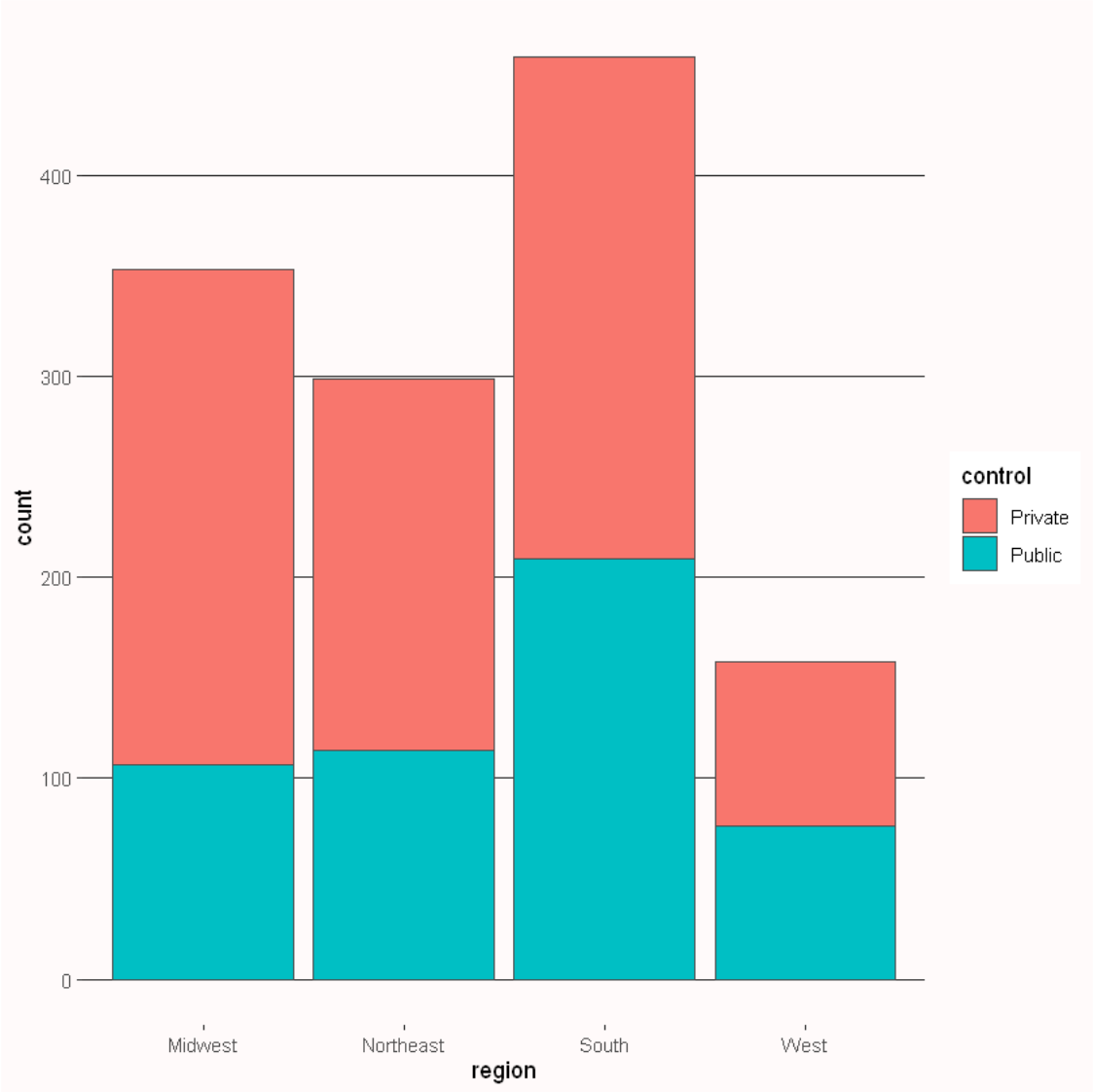
Section summary: showing how you can play around and changes axes on charts

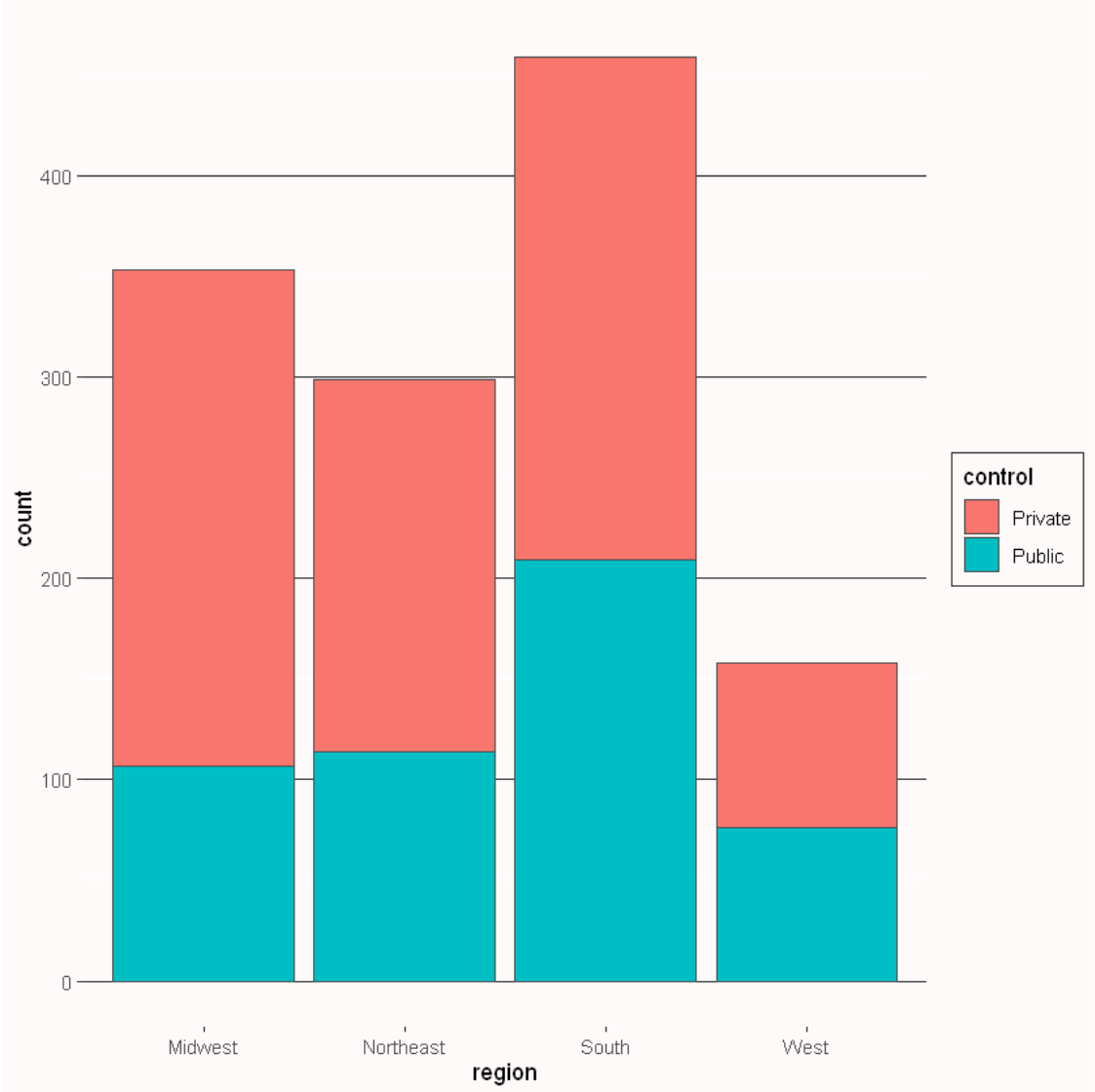
[44]:
```
#Setting up the chart
Plot4 <- ggplot(data=college) +
  geom_bar(mapping=aes(x=region, fill=control),colour='grey30') +
  theme(panel.background=element_rect(fill='snow'),plot.
 ↪background=element_rect(fill='snow')) +
  theme(panel.grid.major.y=element_line(colour='grey15'),panel.grid.major.
 ↪x=element_blank()) #major grid lines only for y
print(Plot4)
```



removing the minor grid panels and changing the backgrounds:

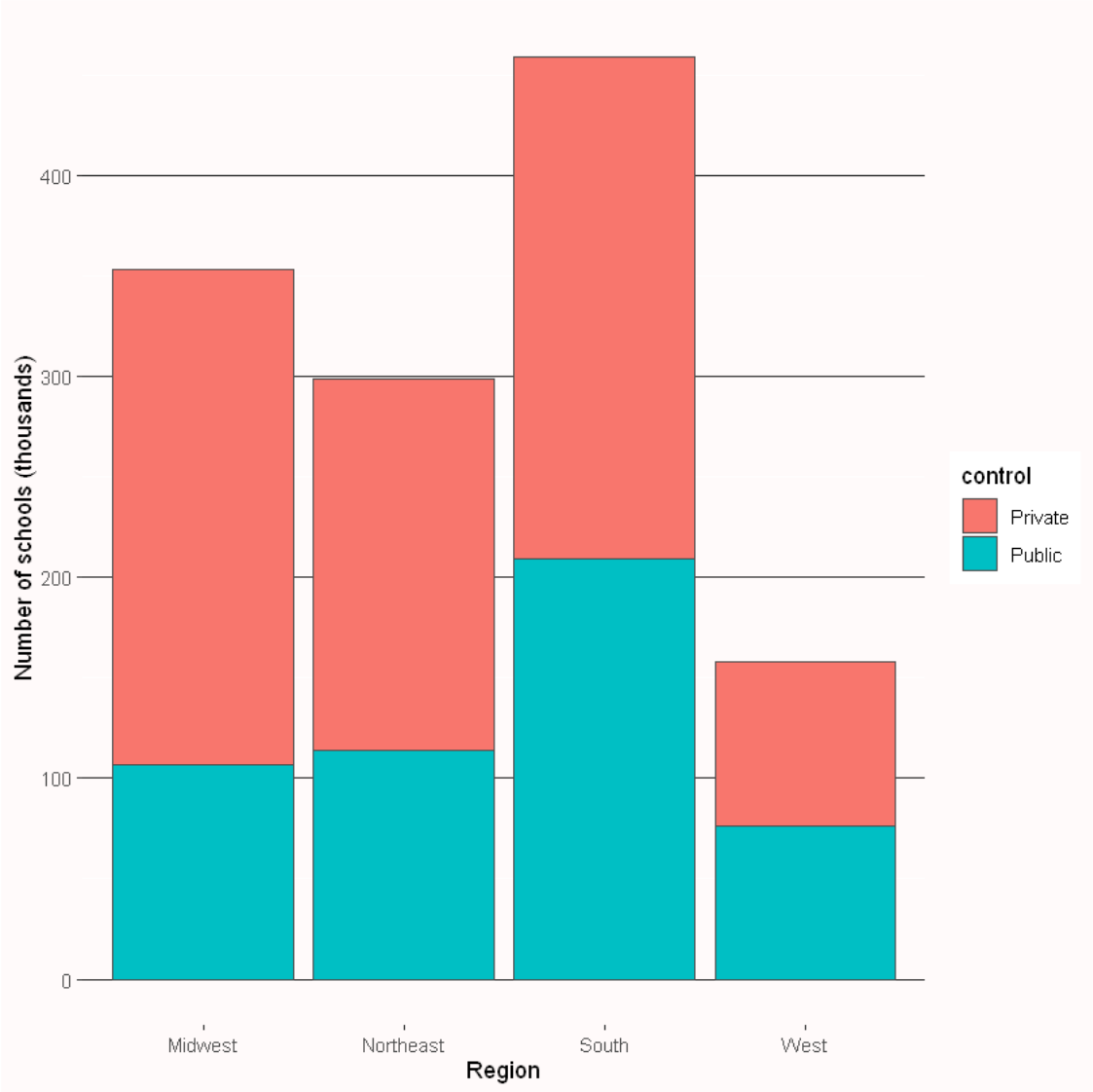
[45]:
```
Plot4 + theme(panel.grid.minor.x=element_blank(),panel.grid.minor.
 ↪y=element_blank())
Plot4 + theme(legend.background = element_rect(colour='grey15',fill =␣
 ↪"snow"),legend.key = element_rect(fill = "#fffcfc"))
```


400
300
200
100
0
count
Midwest
Northeast
South
West
region
control
Private
Public



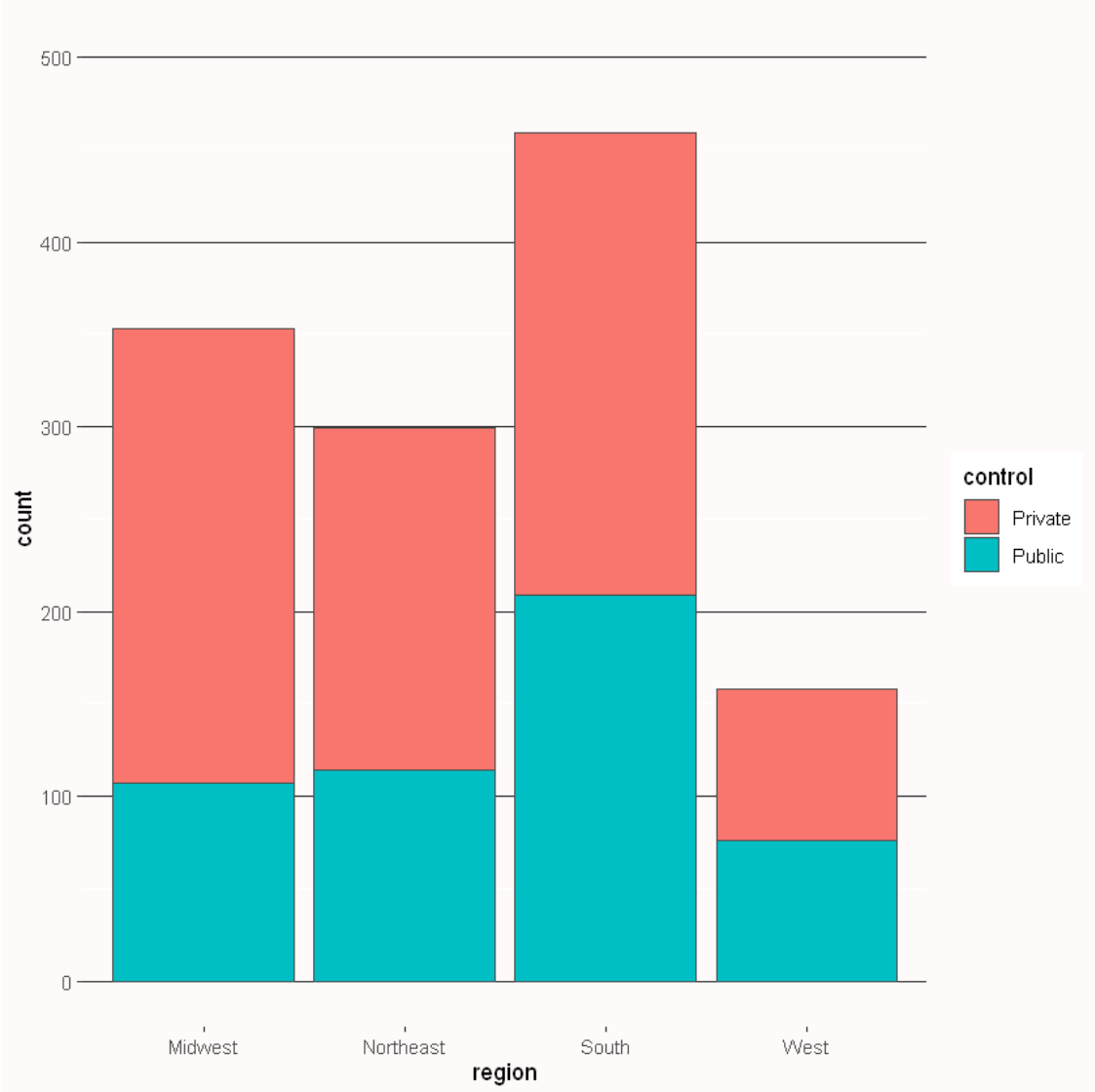
Adding labels:

[46]:
```
#Adding labels
Plot4 + ylab('Number of schools (thousands)') + xlab('Region')
```



Changing the scale of axes:

[47]:
```
#Changing scale
Plot4 + ylim(0,500)
```



[49]:
```
colnames(college) #as a reminder
```

1. 'id' 2. 'name' 3. 'city' 4. 'state' 5. 'region' 6. 'highest_degree' 7. 'control' 8. 'gender' 9. 'admission_rate' 10. 'sat_avg' 11. 'undergrads' 12. 'tuition' 13. 'faculty_salary_avg' 14. 'loan_default_rate' 15. 'median_debt' 16. 'lon' 17. 'lat'

1.2.9 [9] Using scale_ functions in ggplot

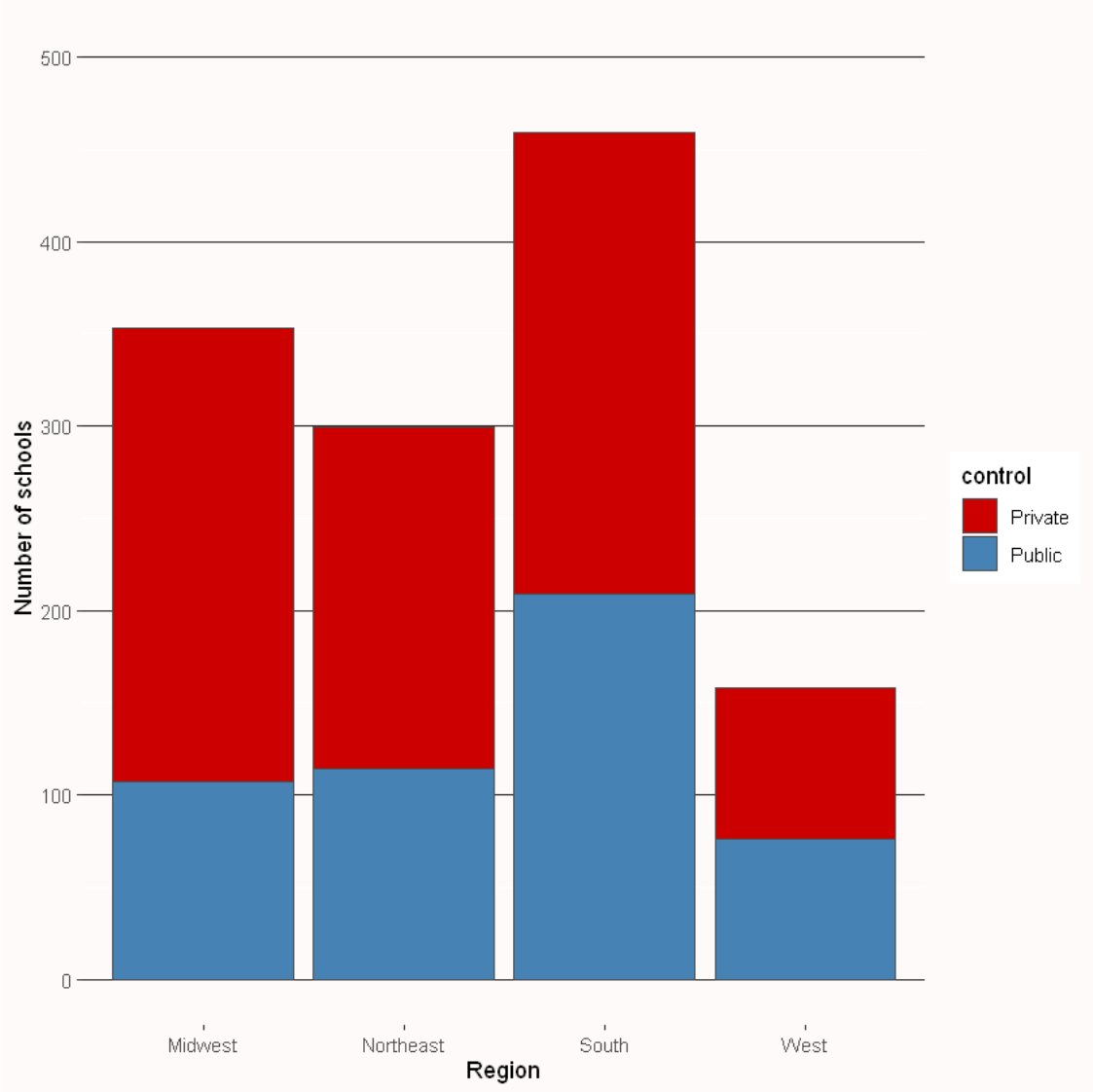
'Scales describe how data is mapped from the dataset to the visualisation.' 'Both discreet and continous scales exist; for discreet and continous variables respectively'

To map a scale, follow the format scale_name_datatype: > *scale_y_discrete* | discrete variable on the y axis > *scale_x_continous* | continous variable on the x axis > *scale_fill_discrete* | discrete variable on the 'fill' axis/scale

Using Plot4 again; we can change label names using scale_

[55]:
```
Plot5 <- Plot4 +
  scale_x_discrete(name='Region')+
  scale_y_continuous(name='Number of schools',limits=c(0,500))+
  scale_fill_manual(values=c('red3','steelblue'))
print(Plot5)

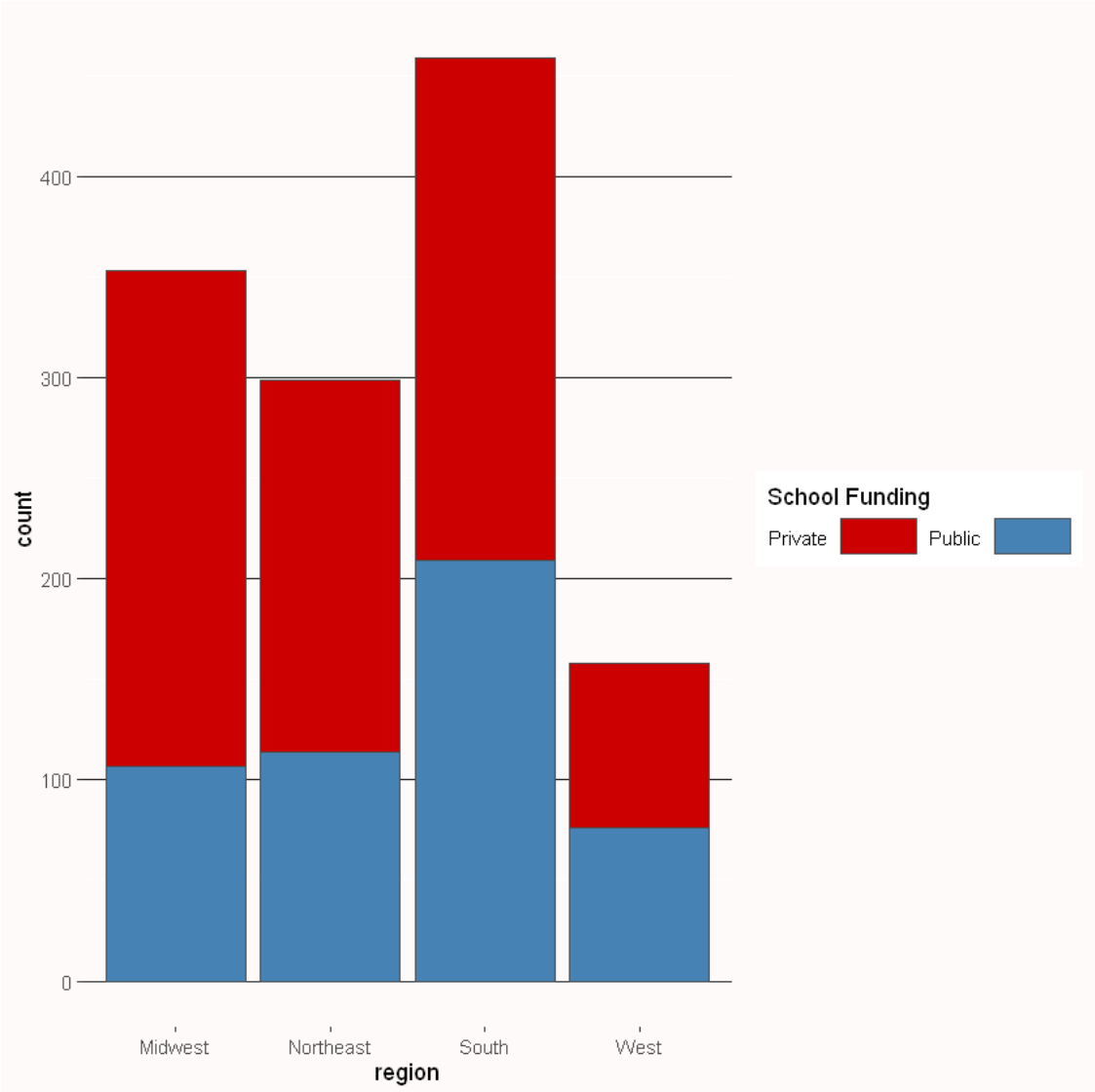
#?scale_x_continuous for more info
```



1.2.10 [10] Key / Legend

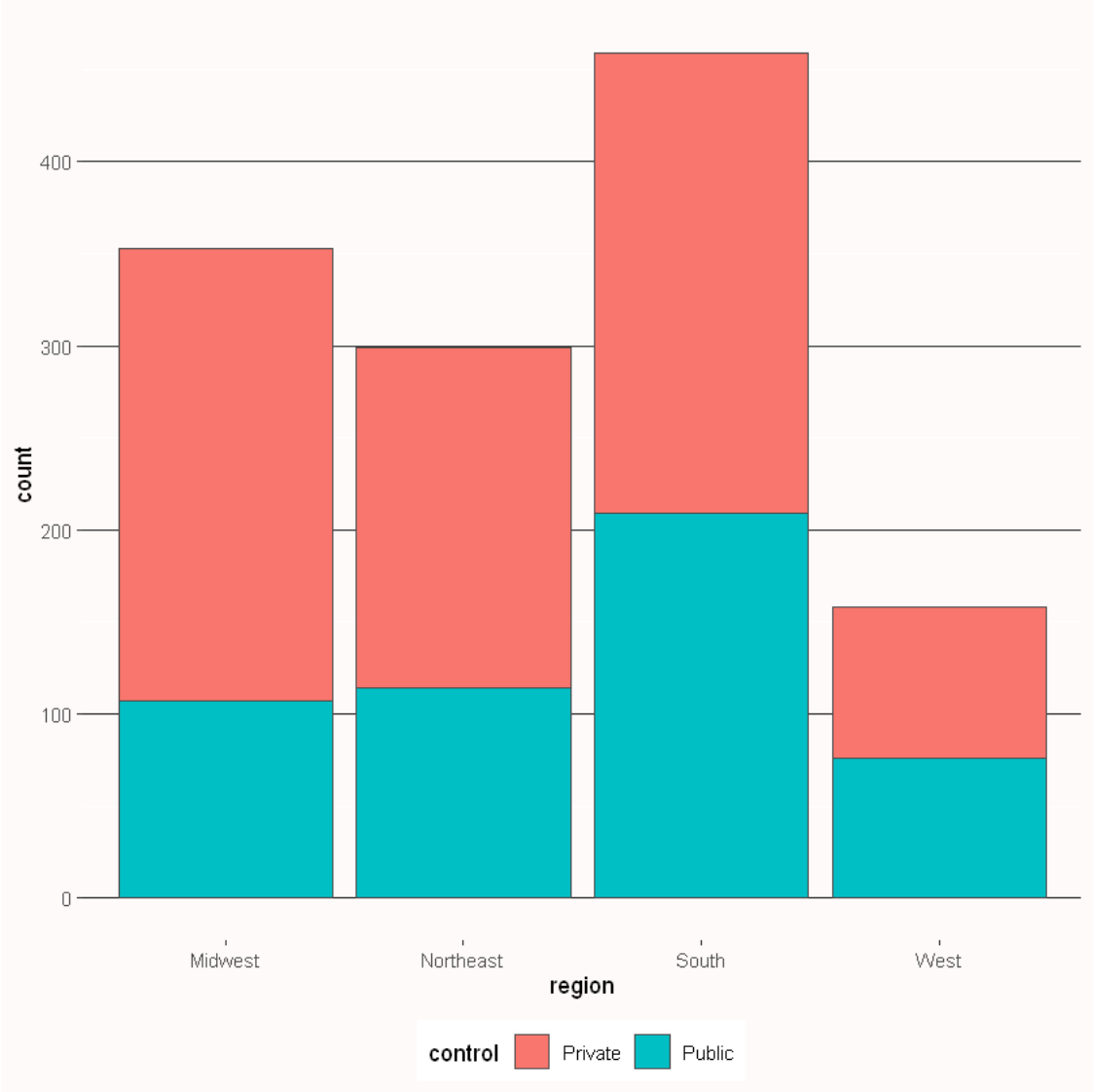
This section covers some visual changes for the key/legend

[57]:
```
Plot4 +
 ␣
 ↪scale_fill_manual(values=c('red3','steelblue'),guide=guide_legend(title='School␣
 ↪Funding',nrow=1,
    label.position='left', keywidth=2.5 )) #changing the legend is to do with␣
 ↪colour....
#.... and hence we edit it on the 'fill'(colour) scale
#this is changing the order of Schoolwidth > Private/Public > The colours(we␣
 ↪changed width with keywidth)
```



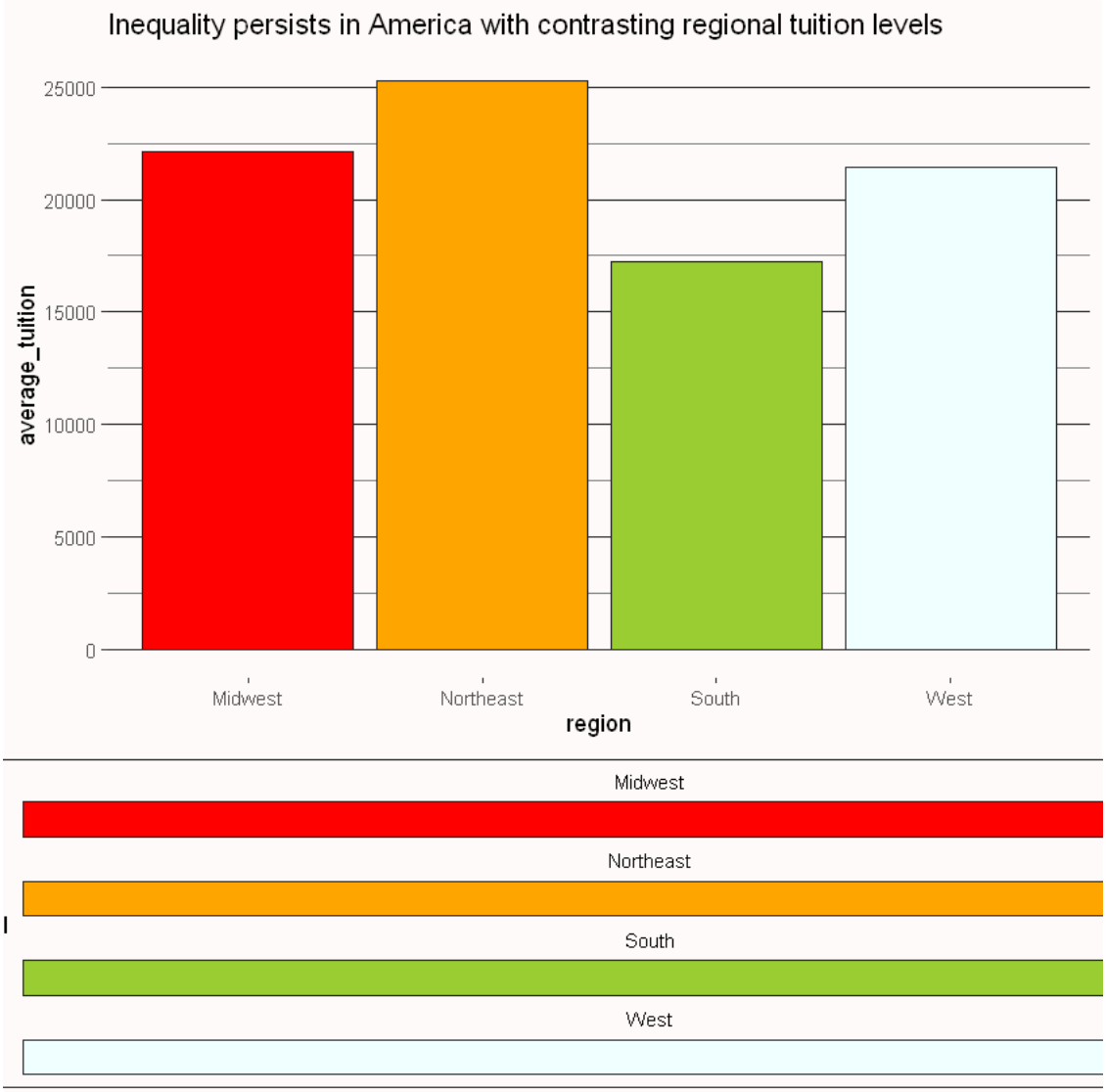
moving legend to the bottom:

[59]:
```
Plot4 + theme(legend.position = 'bottom')
```



One extreme example to demonstrate the point:

[60]:
```
print(Plot3 + theme(legend.position = 'bottom')+
        labs(title='Inequality persists in America with contrasting regional␣
↪tuition levels')+
      ␣
↪scale_fill_manual(values=c('red','orange','yellowgreen','azure'),guide=guide_legend(label.
↪position='top',title='Cardinal',nrow=4,keywidth=40)))
```



1.2.11 [11] Annotations

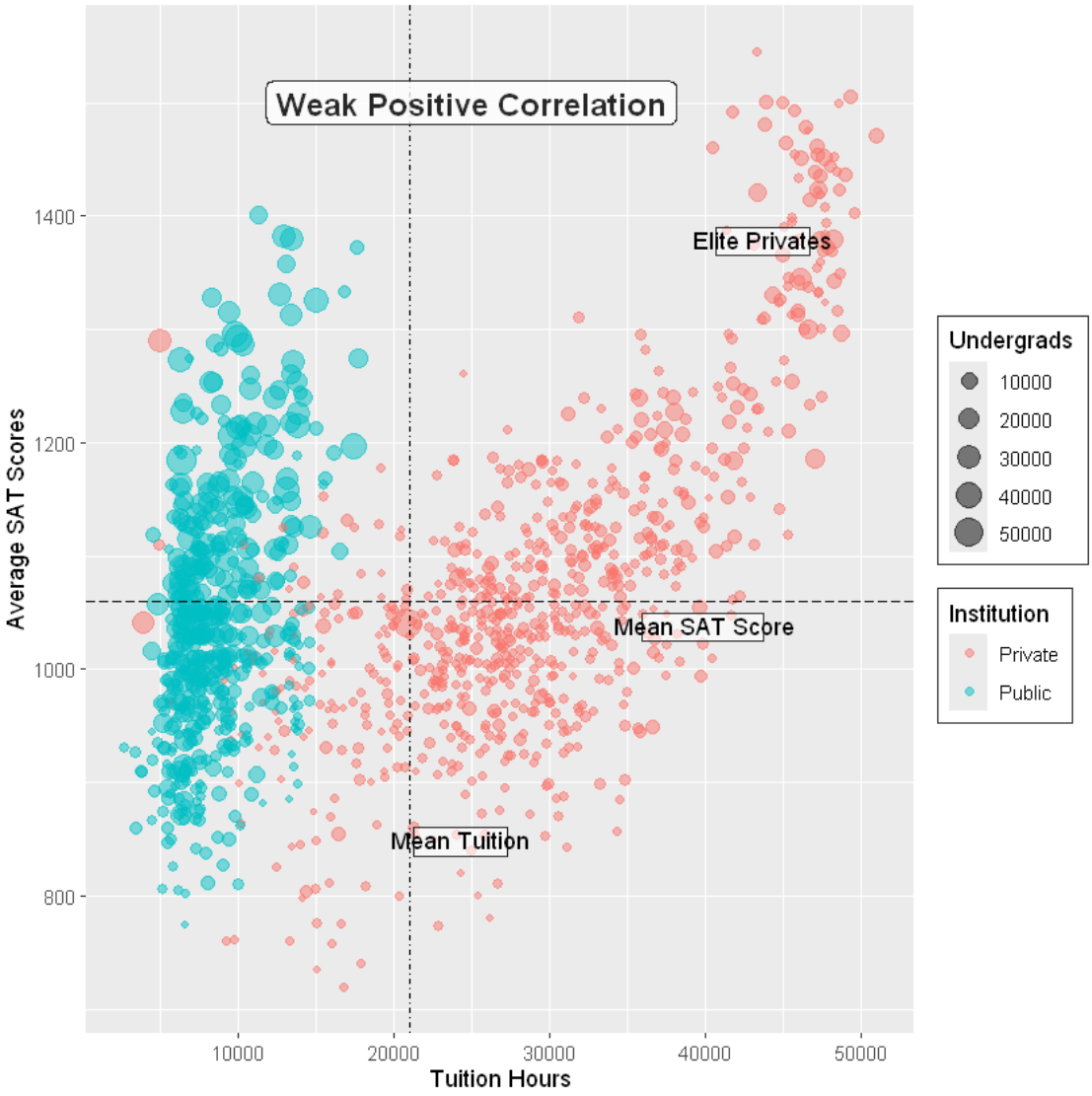
Text annotations (self explanatory) are used often in R. see below my personal attempt as well as R's official labels

[65]:
```
scat3 <- scat3 + theme(legend.background=element_rect(colour='grey15'))
scatter <- scat3 +
      ␣
 ↪annotate('rect',xmin=40750,xmax=46700,ymin=1365,ymax=1390,colour='grey15',fill='white',alph
 ↪6)+
       annotate("text",label='Elite Privates',x=43800,y=1380)+
```

```
       geom_hline(alpha=1,size=0.5,linetype=5,yintercept =␣
↪mean(college$sat_avg))+
      ␣
↪annotate('rect',xmin=36000,xmax=43800,ymin=mean(college$sat_avg)-35,ymax=mean(college$sat_a
↪6)+
       annotate('text',label='Mean SAT␣
↪Score',x=40000,y=mean(college$sat_avg)-20)+

       geom_vline(alpha=1,linetype=4,xintercept = mean(college$tuition))+
      ␣
↪annotate('rect',xmin=mean(college$tuition)+250,xmax=mean(college$tuition)+6300,ymin␣
↪=835 ,ymax =860,colour='grey15',fill='white',alpha=0.6 )+
       annotate('text',label='Mean␣
↪Tuition',x=mean(college$tuition)+3300,y=850)+
       annotate('label',label='Weak Positive␣
↪Correlation',x=25000,y=1500,fill='white',colour='grey15',alpha=0.
↪8,size=5,fontface='bold')+

       scale_x_continuous(name='Tuition Hours')+
       scale_y_continuous(name='Average SAT Scores')+
       scale_colour_discrete(name='Institution')+
       scale_size_continuous(name='Undergrads')
print(scatter)
```



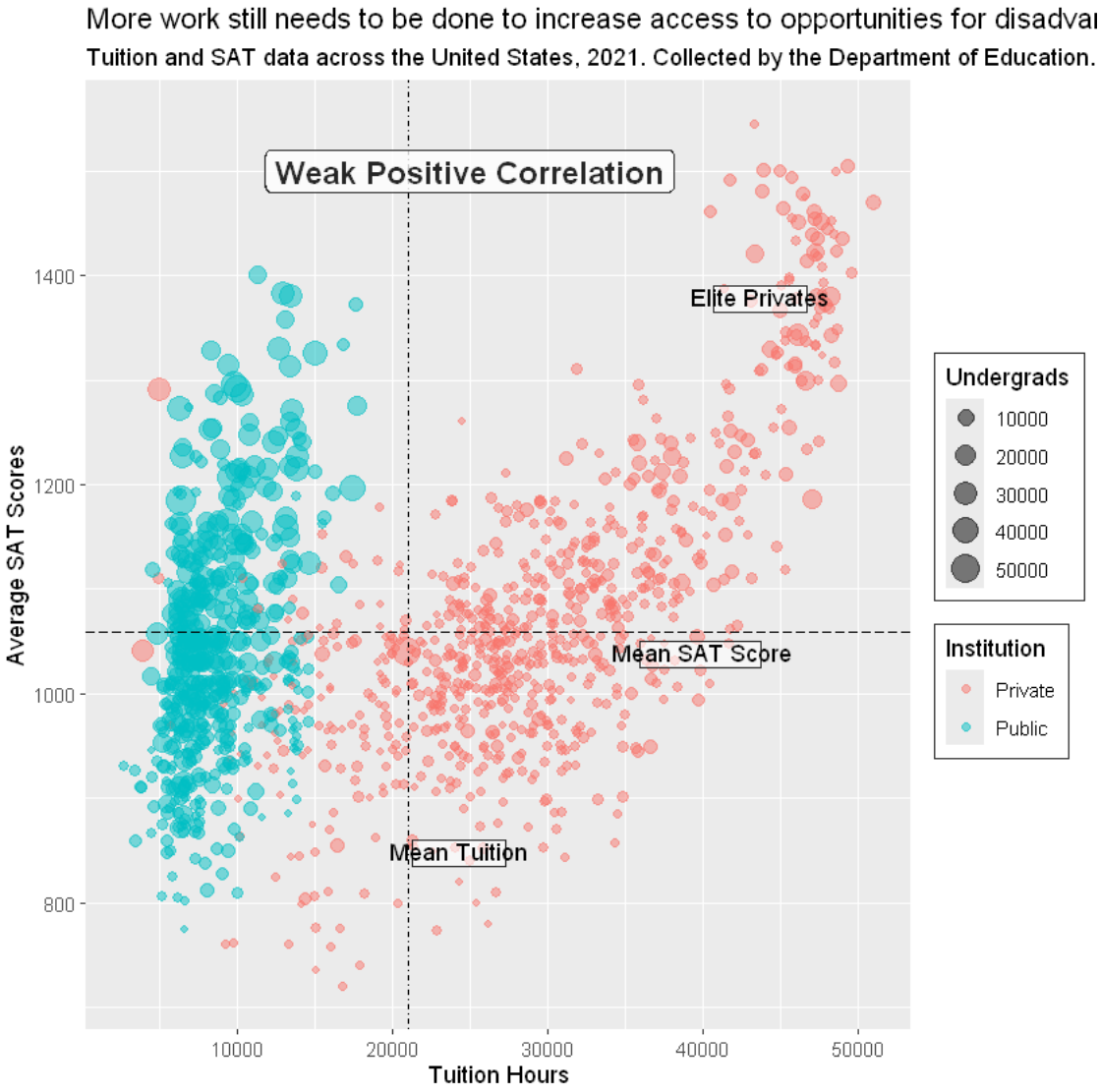
[66]:
```
# ?geom_hline()

# ?annotate()
```

1.2.12 [12] Titles & Subtitles

titles are similarly self explanatory, these can be done through ggplot or base R programming.

[68]:
```
#use lab(title='') in base R or ggtitle() - either works
scatter + ggtitle(label='More work still needs to be done to increase access to␣
 ↪opportunities for disadvantaged communities',
                        subtitle = 'Tuition and SAT data across the United␣
 ↪States, 2021. Collected by the Department of Education.')
```



[69]:
```
windowsFonts() #for customising fonts
```

$serif 'TT Times New Roman'

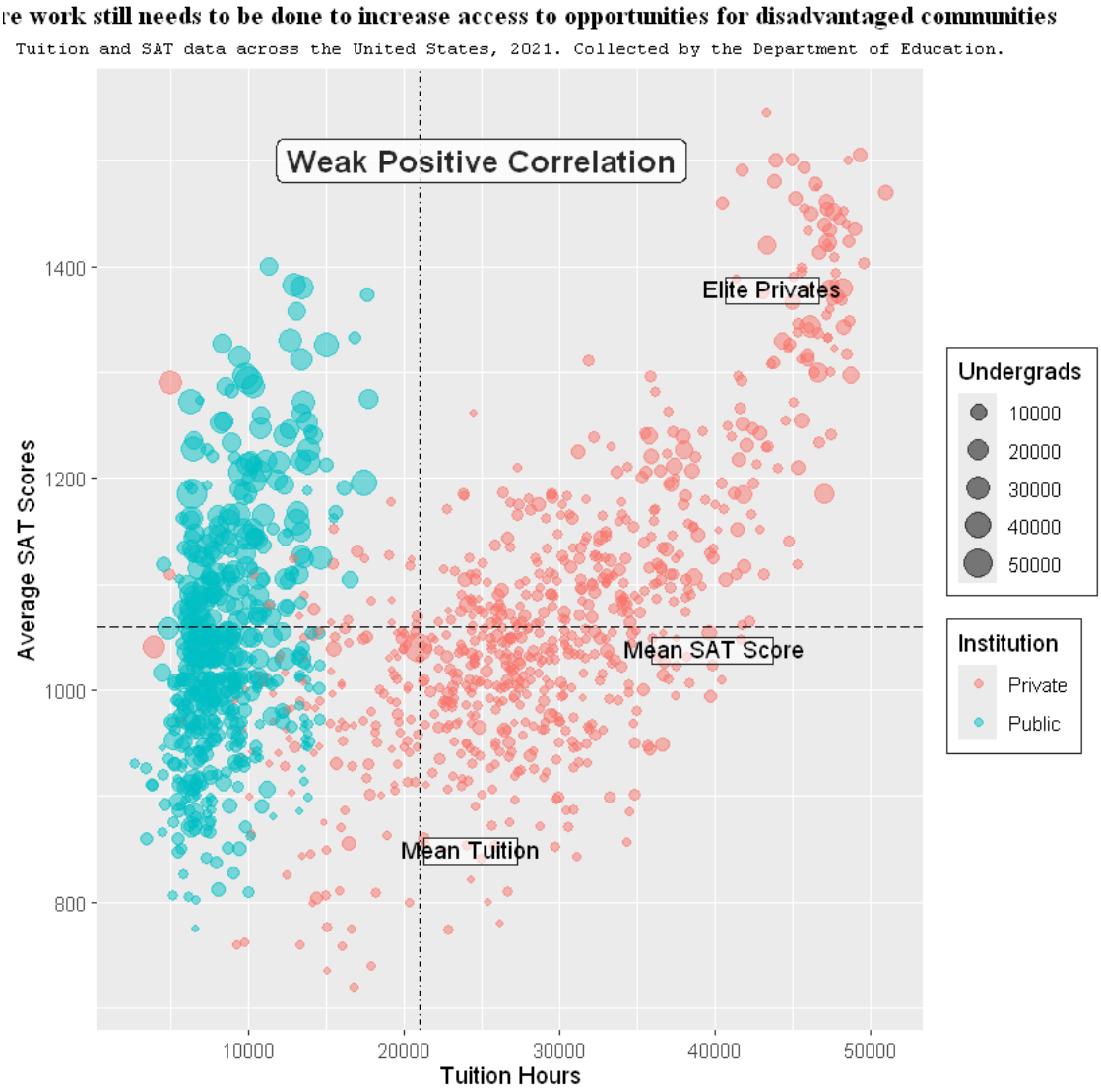
$sans 'TT Arial'

$mono 'TT Courier New'

Customising titles:

[70]:
```
#customising titles
print(scatter +
        ggtitle(label='More work still needs to be done to increase access to␣
 ↪opportunities for disadvantaged communities',
```

```
                subtitle = 'Tuition and SAT data across the United States,␣
↪2021. Collected by the Department of Education.') +
      theme (plot.title = element_text(hjust=0.
↪5,size=12,face='bold',family='serif'))+
      theme(plot.subtitle = element_text(hjust=0.5,size=8,family='mono')))
```

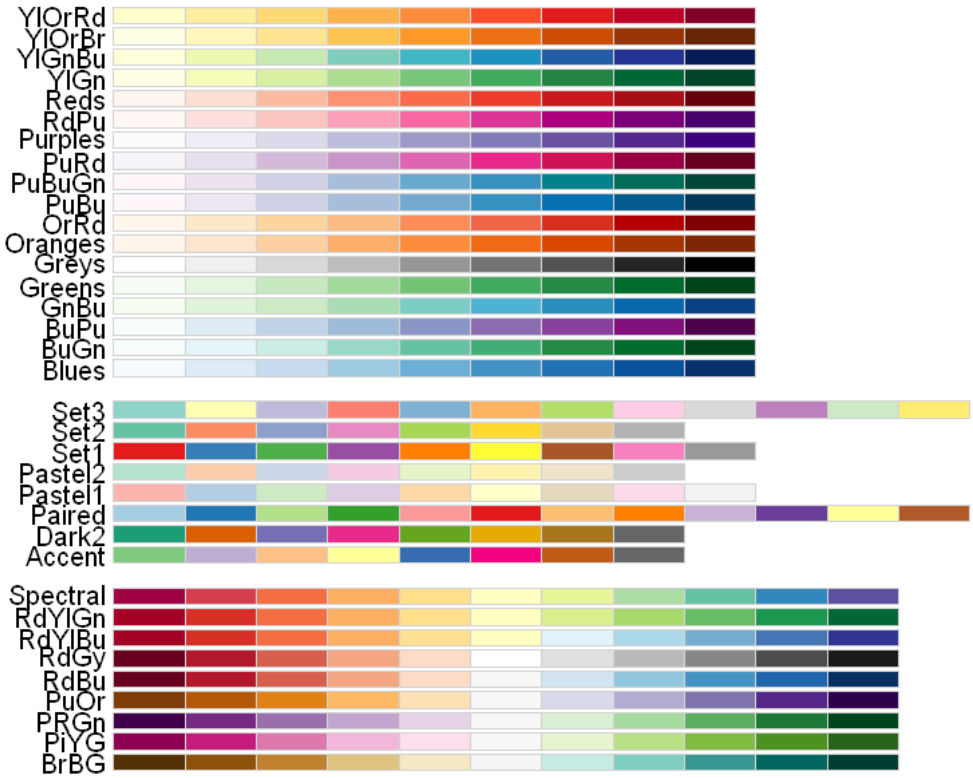


1.2.13 [13] Introducing Themes (brewer + viridis)

As hinted at earlier, themes are a great way to customise charts - with preset backgrounds, colour schemes and so much more.

A glance at the brewer() function colour scheme:

[71]:
```
display.brewer.all()
```



A glance at the Viridis family of colours (with primary aim of raising accessibility for colorblind readers):

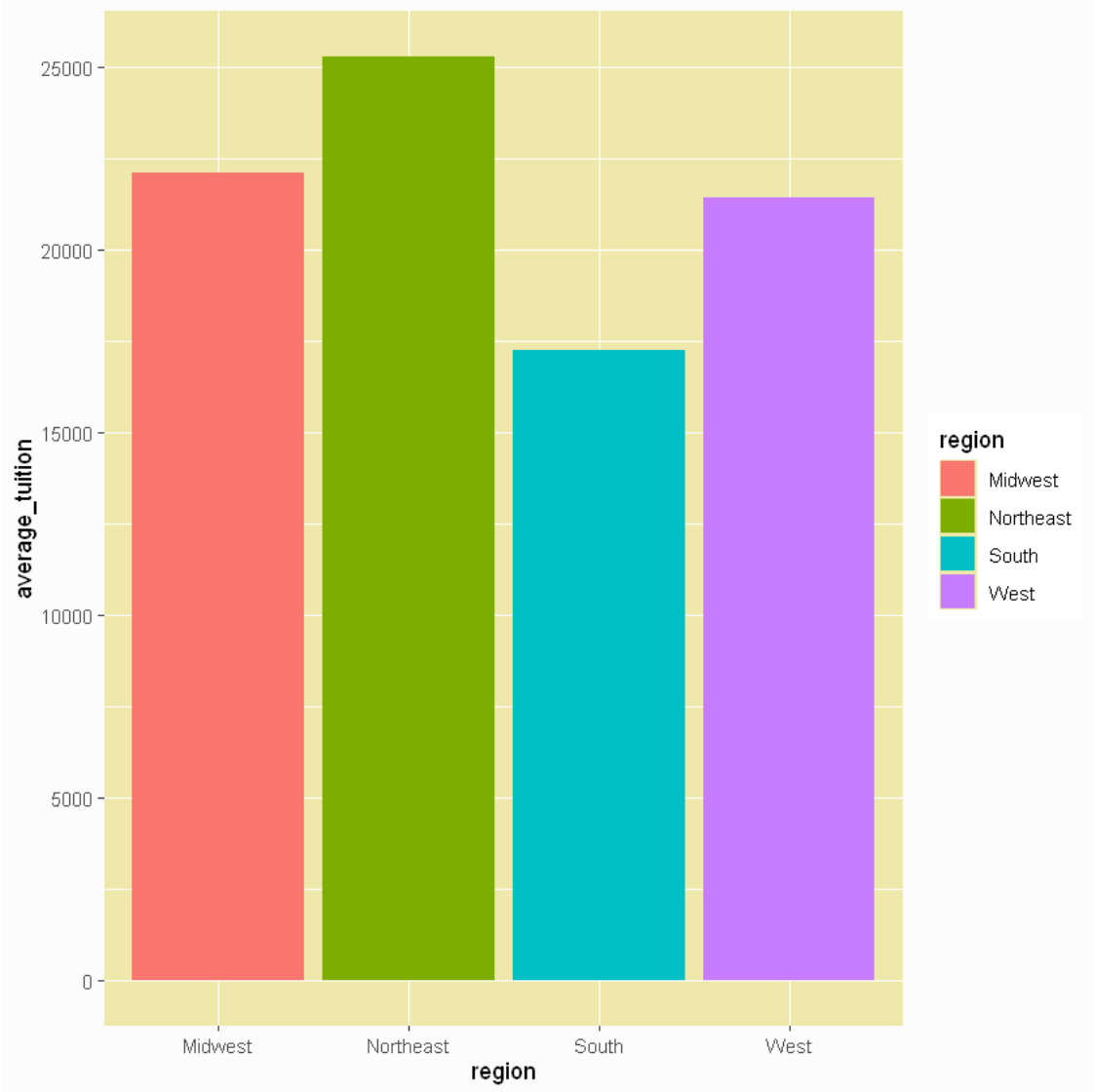
Viridis family >A = magma B = inferno C = plasma D = viridis (default) E = cividis (standard for publications) F = rocket G = mako H = turbo

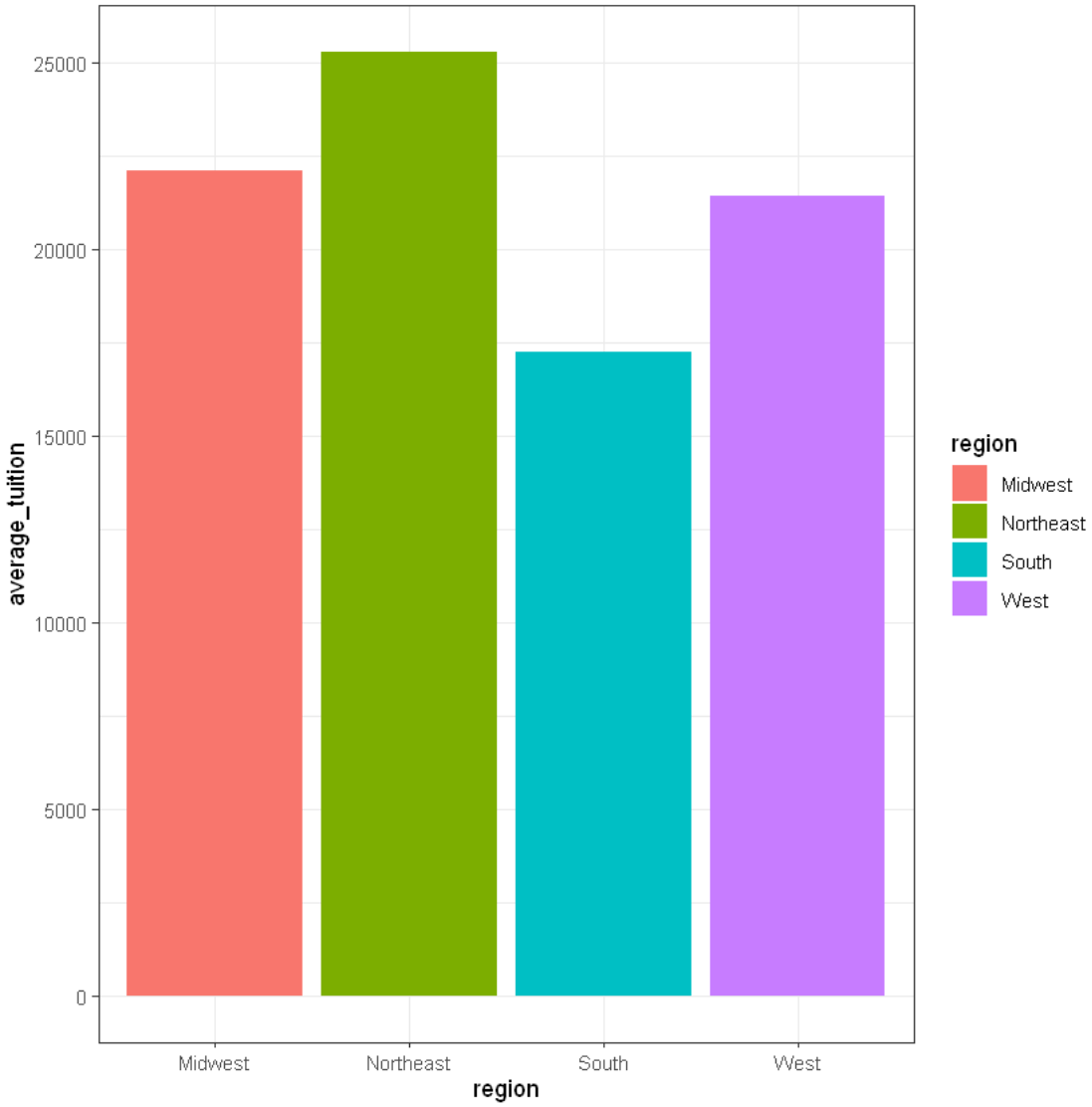
Below is a basic demonstration of some of the customisations:

[75]:
```
#learning set colour sets in R

#download + install ggthemes(); and stick a theme function on the end
Plot1
```

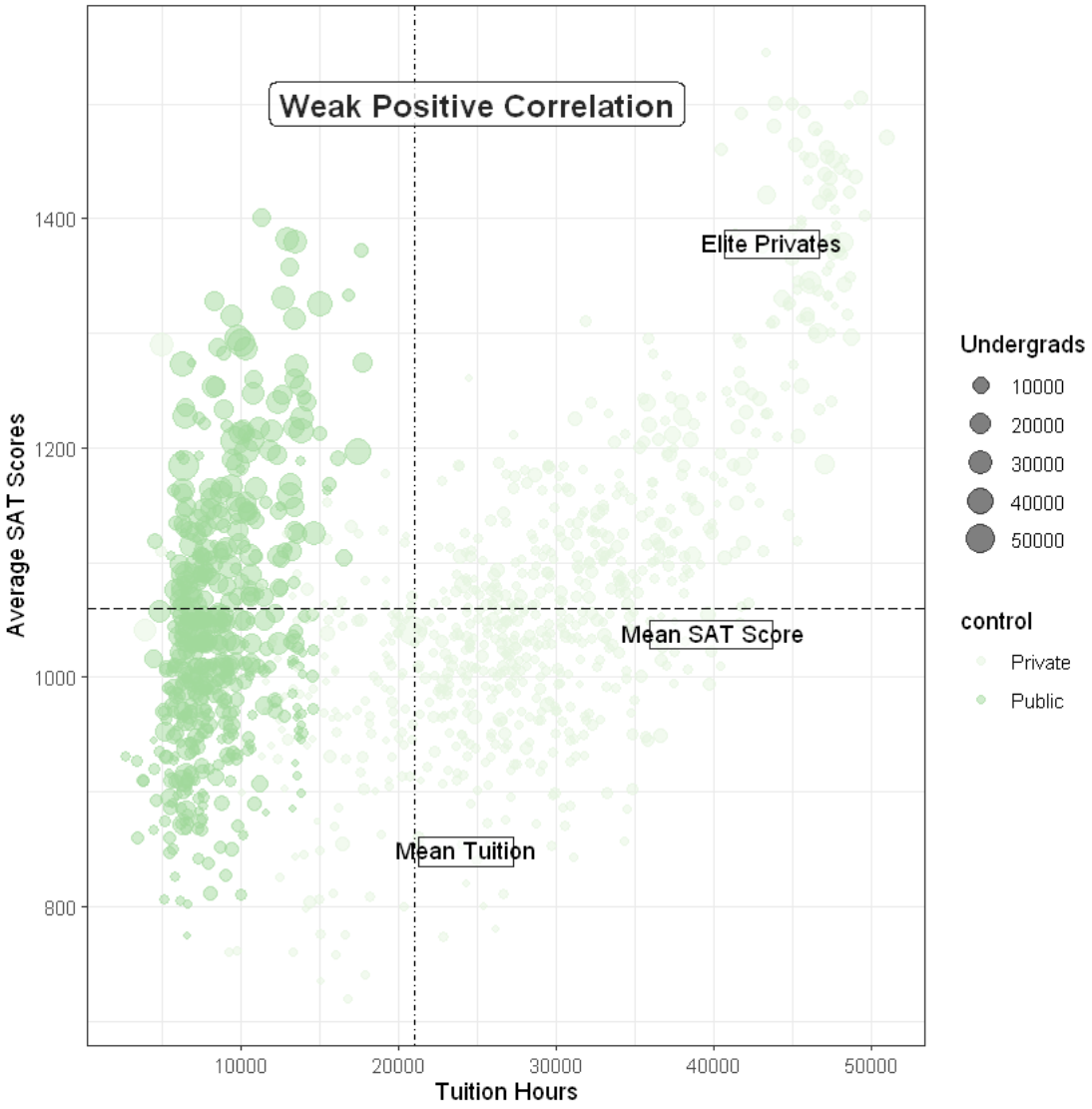
```
Plot1 + theme_bw() #does a preset colour scheming
```



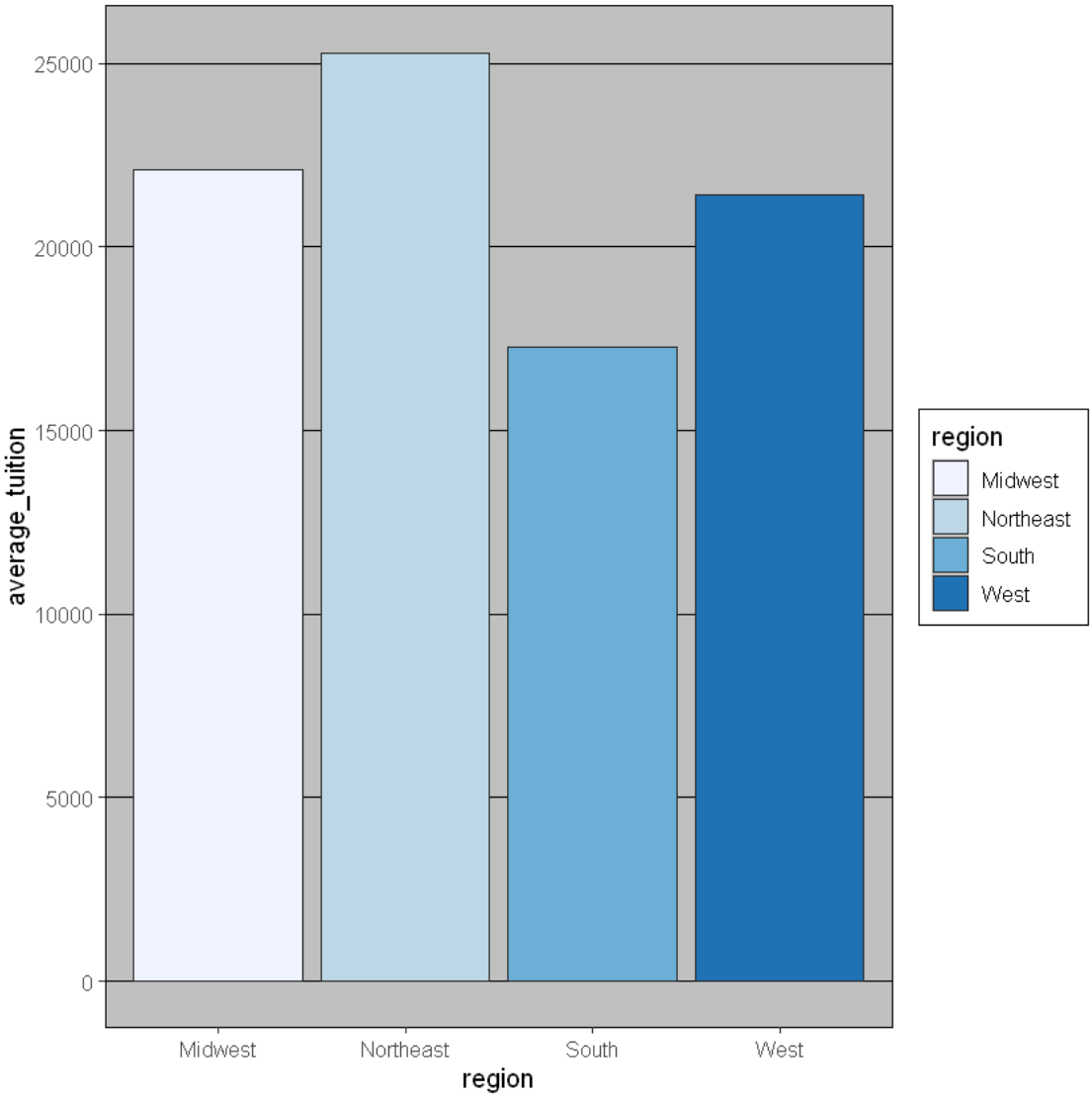


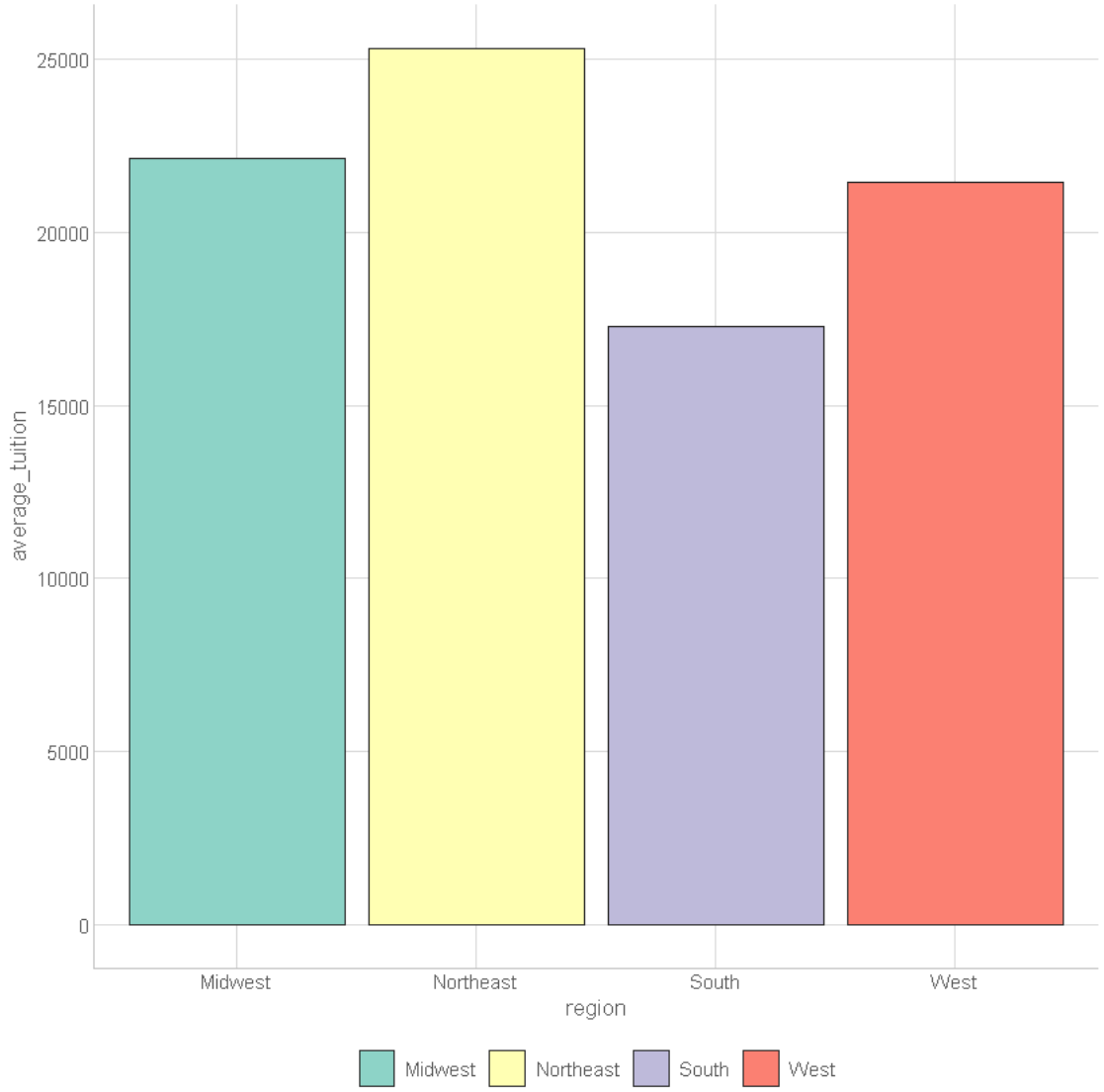
[79]:
```
#add colour_brewer and fill_brewer() instead of manual to use themes
scatter + theme_bw() + scale_colour_brewer(palette='Dark')
scatter + theme_solarized() +scale_colour_brewer(palette='Reds')
```

```
Warning message:
"Unknown palette: "Dark""
Scale for colour is already present.
Adding another scale for colour, which will replace the existing
scale.
```

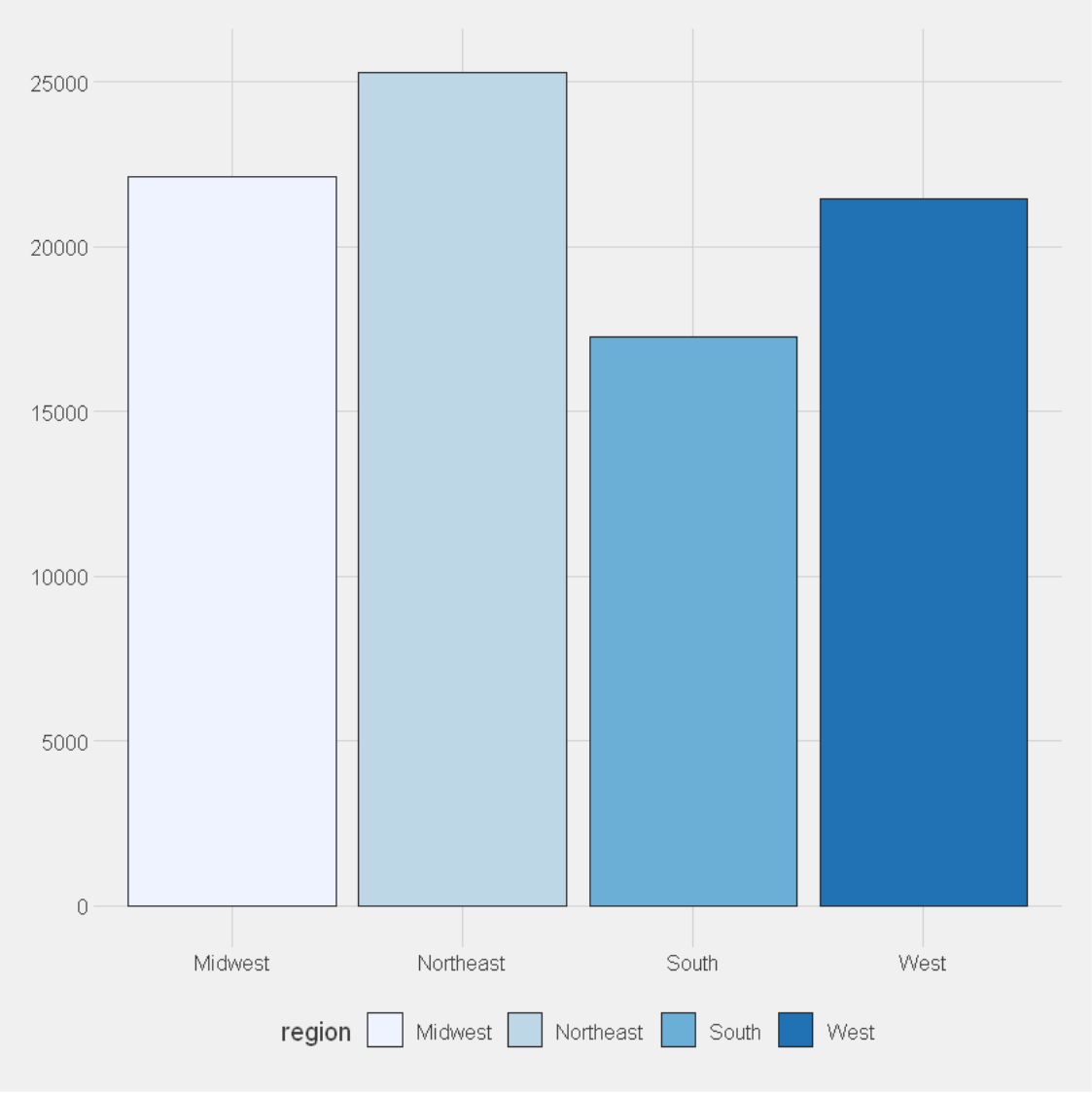


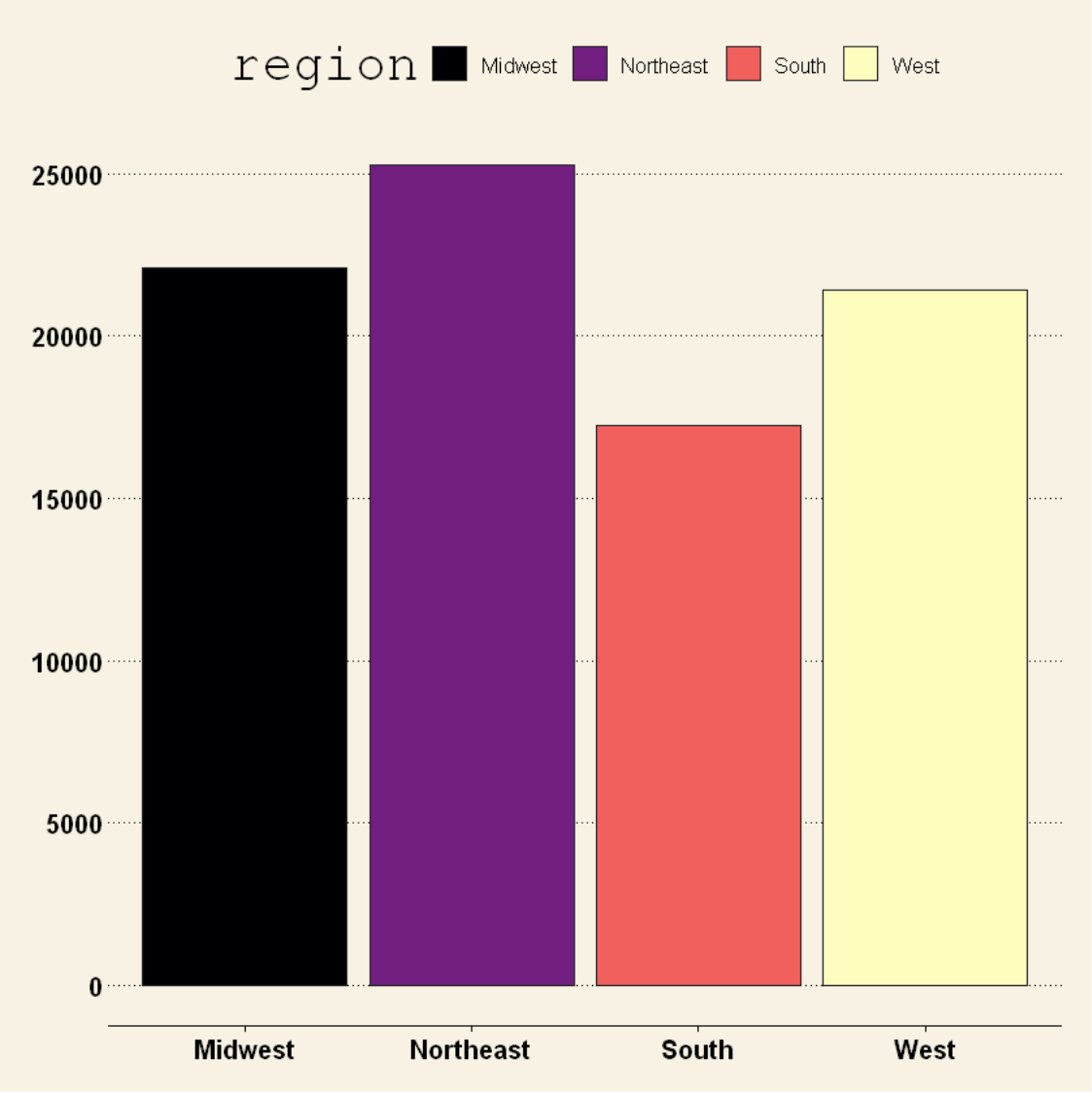
[80]:
```
Plot2 + theme_excel() + scale_fill_brewer(palette='Blues')
Plot2 + theme_excel_new() + scale_fill_brewer(palette='Set3')
```

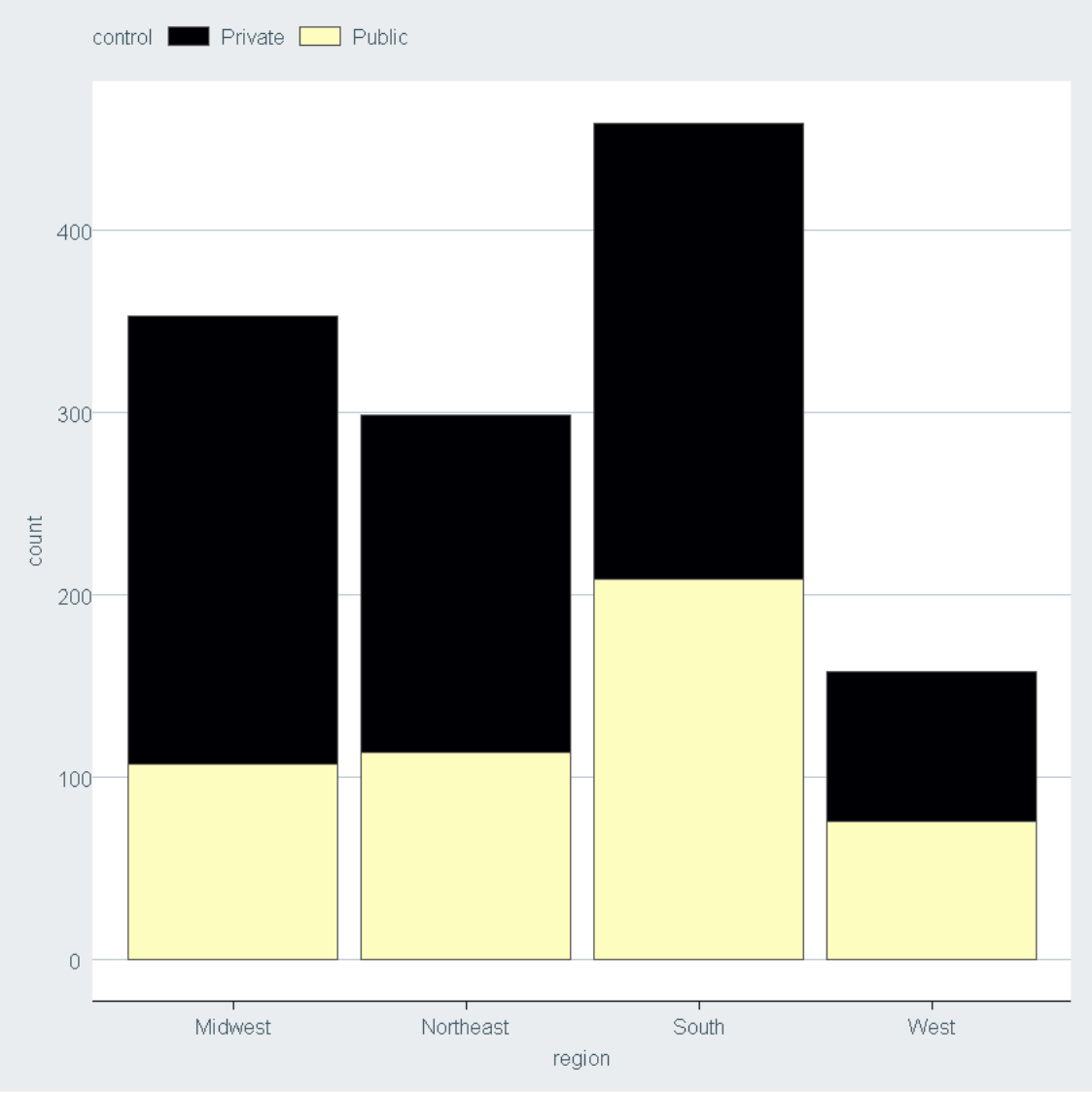

25000
20000
15000
10000
5000
0
average_tuition
Midwest
Northeast
South
West
region
region
Midwest
Northeast
South
West



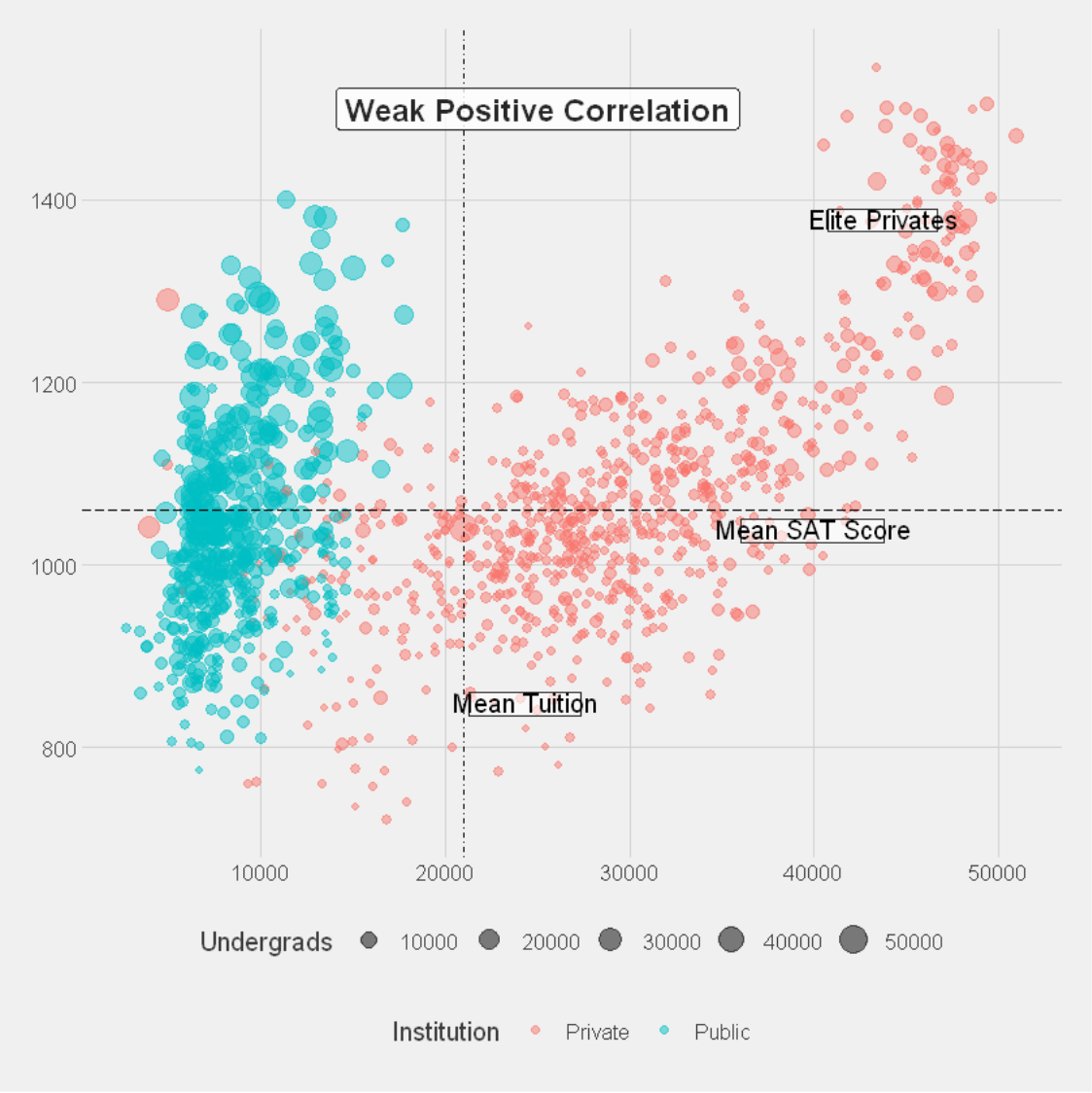
[81]:
```
Plot2 + theme_fivethirtyeight() + scale_fill_brewer(palette='Blues')␣
 ↪#'Spectrals' also cool
Plot3 + theme_wsj() + scale_fill_viridis(option = 'A',discrete = TRUE) #all A-H␣
 ↪are nice but C
Plot4 + theme_economist() + scale_fill_viridis(option = 'A',discrete = TRUE)␣
 ↪#not so good for stacked...
```

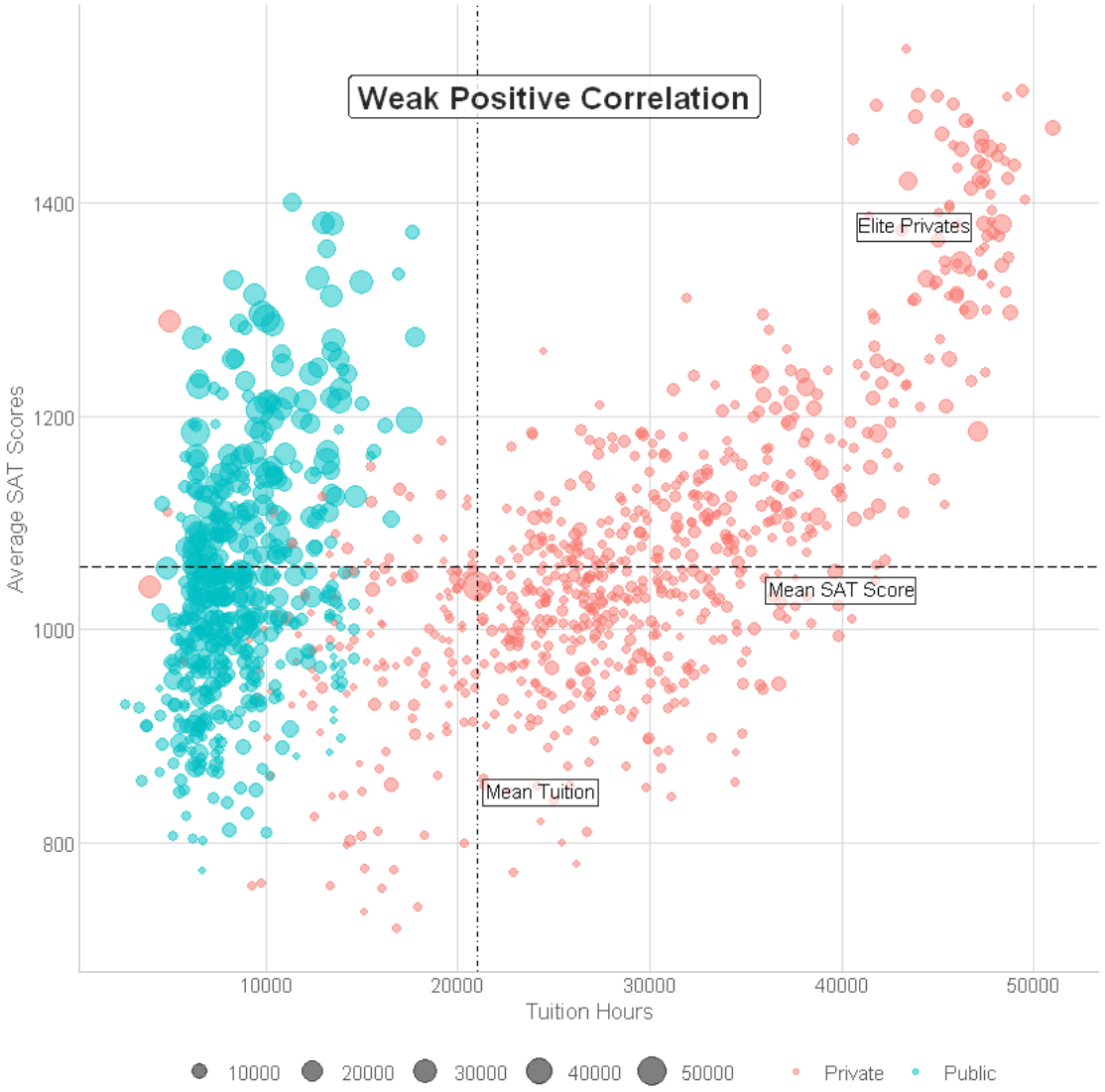

25000
20000
15000
10000
5000
0
Midwest
Northeast
South
West
region
Midwest
Northeast
South
West

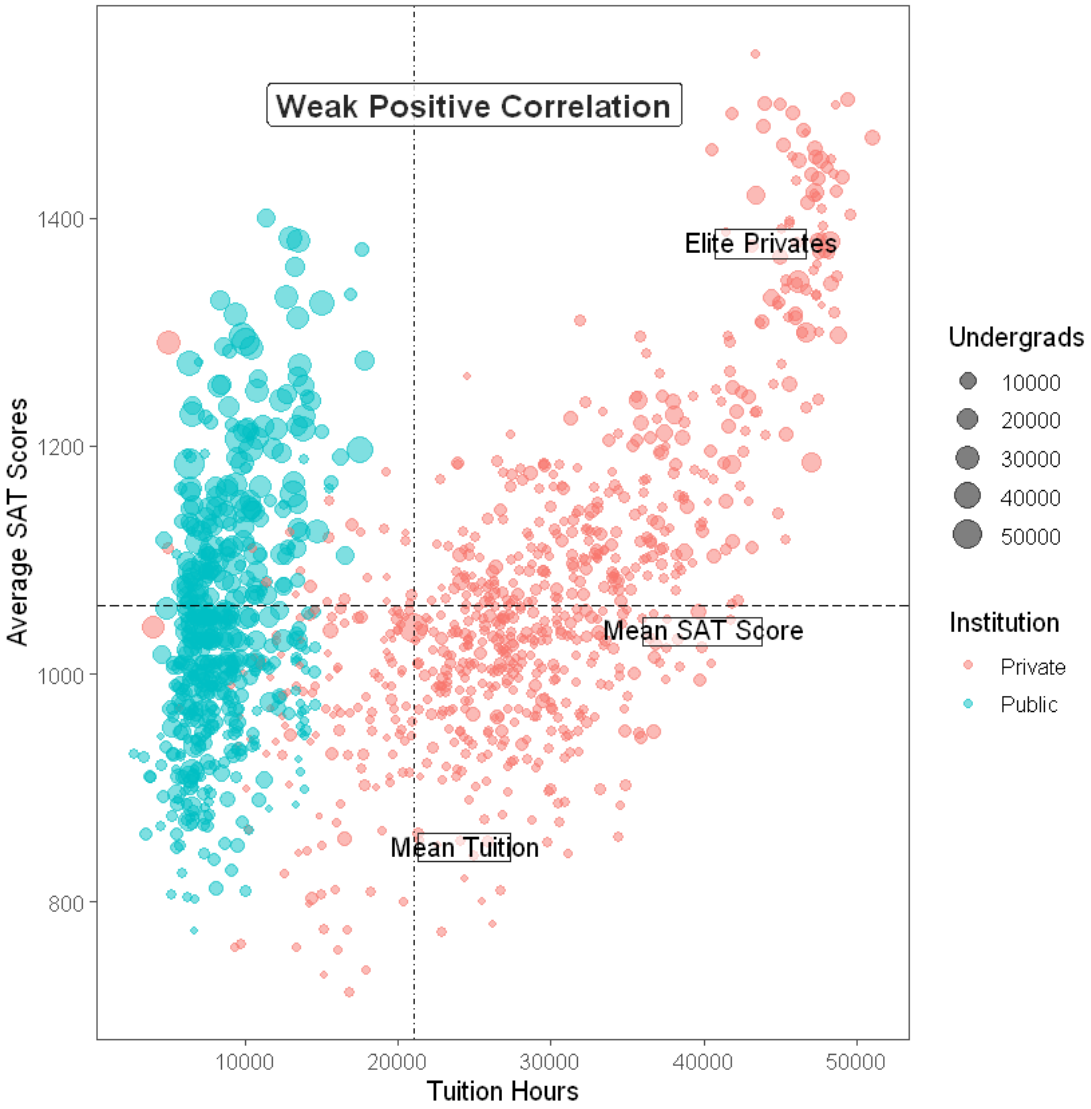

region
Midwest
Northeast
South
West
25000
20000
15000
10000
5000
0
Midwest
Northeast
South
West



[82]:
```
scatter +theme_fivethirtyeight()
scatter + theme_excel_new()
scatter + theme_few()
```


Weak Positive Correlation
Elite Privates
Mean SAT Score
Mean Tuition
1400
1200
1000
800
10000
20000
30000
40000
50000
Undergrads
10000
20000
30000
40000
50000
Institution
Private
Public

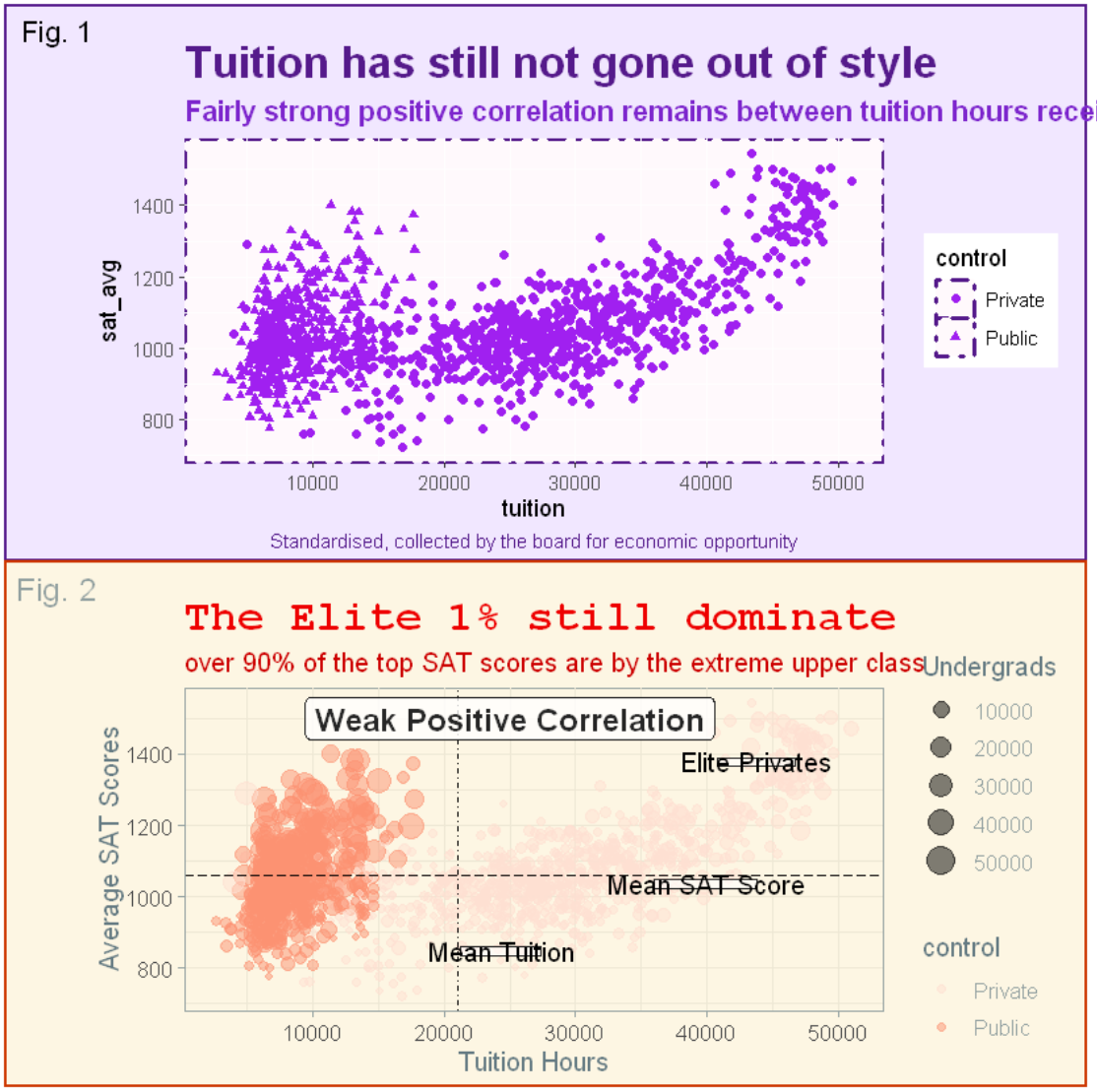

Weak Positive Correlation
Elite Privates
Mean SAT Score
Mean Tuition
Average SAT Scores
1400
1200
1000
800
10000
20000
30000
40000
50000
Tuition Hours
10000
20000
30000
40000
50000
Private
Public



[103]:
```
combineharvester <- scat2 / (scatter +
          theme_solarized() +
          scale_colour_brewer(palette='Reds')+
          labs(title='The Elite 1% still dominate',
               subtitle='over 90% of the top SAT scores are by the extreme␣
 ↪upper class',
               tag='Fig. 2')+
          theme(plot.
 ↪background=element_rect(linetype=1,linewidth=1,colour='orangered3'),
                plot.
 ↪title=element_text(colour='red2',family='mono',face='bold',size=20),
                plot.subtitle=element_text(colour='red3')))
```

```
combineharvester
```

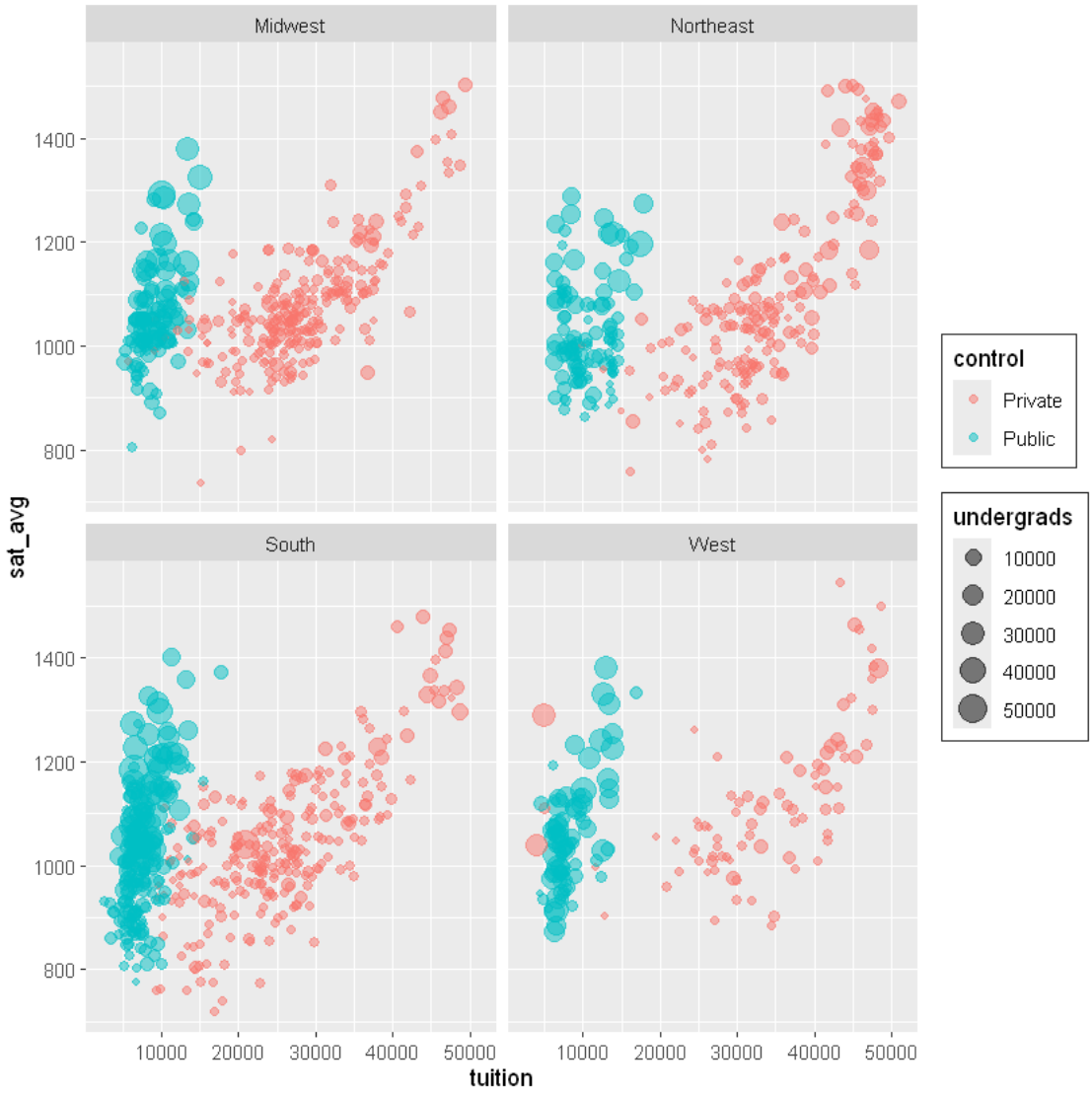
```
Scale for colour is already present.
Adding another scale for colour, which will replace the existing
scale.
```

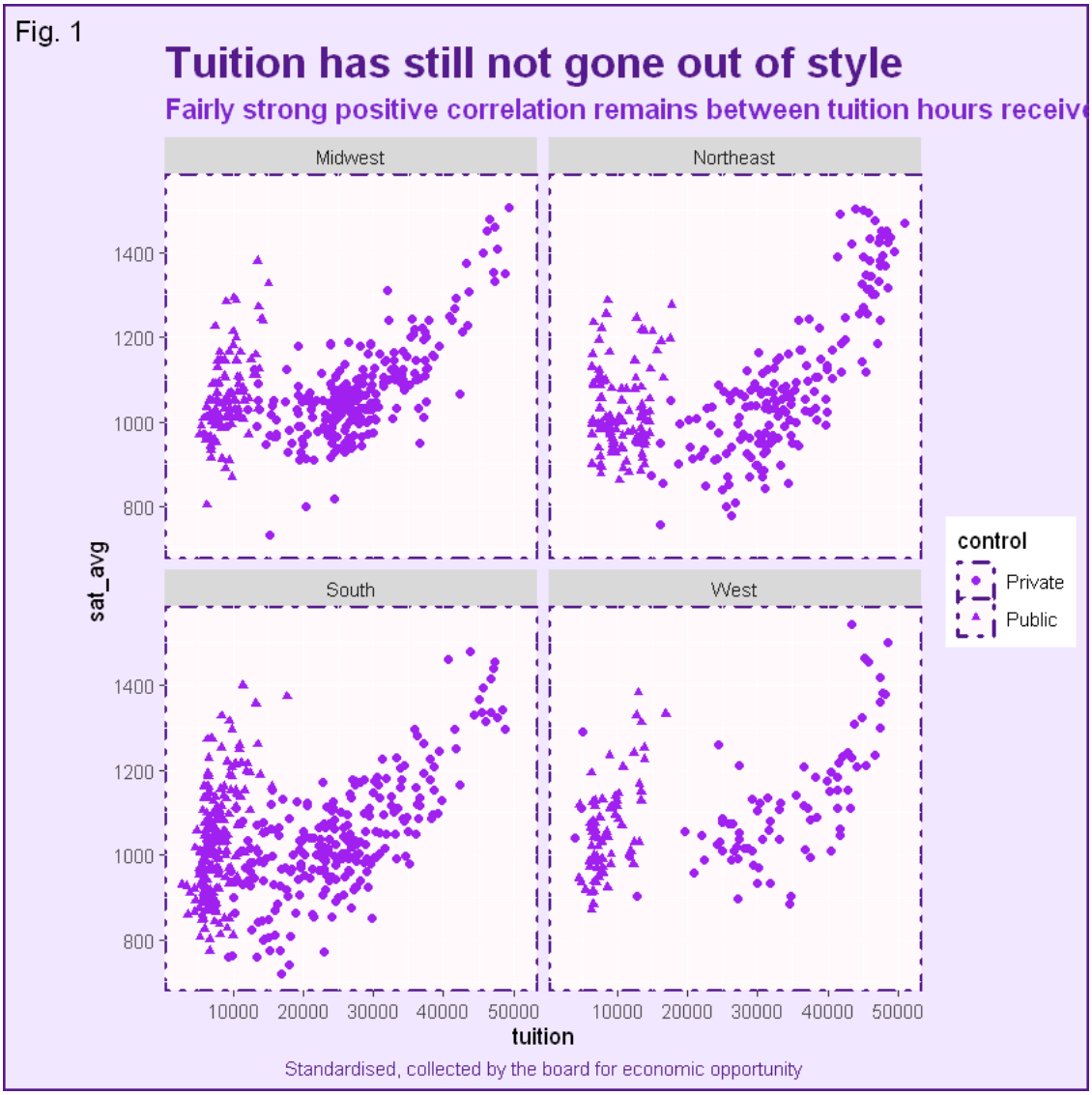


1.2.14 [14] Faceting

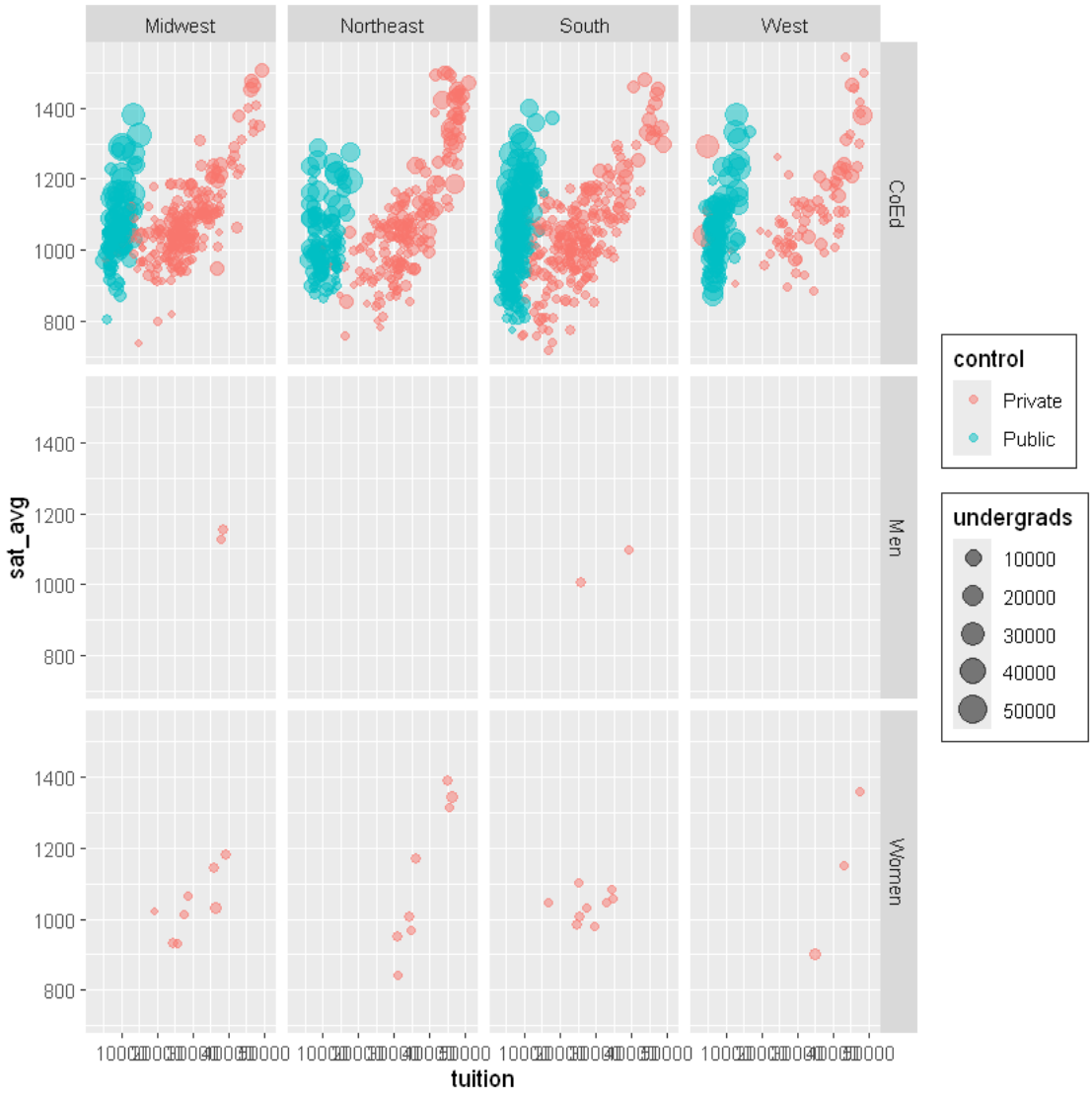
Faceting is a good way to split up data, and more easily compare similar groups (e.g. economic growth by county, or goals scored by football club)

[94]:
```
scat3 +facet_wrap(~region) #splits a graph up into all the diffrent classes
scat2 +facet_wrap(~region) #different visuals
```





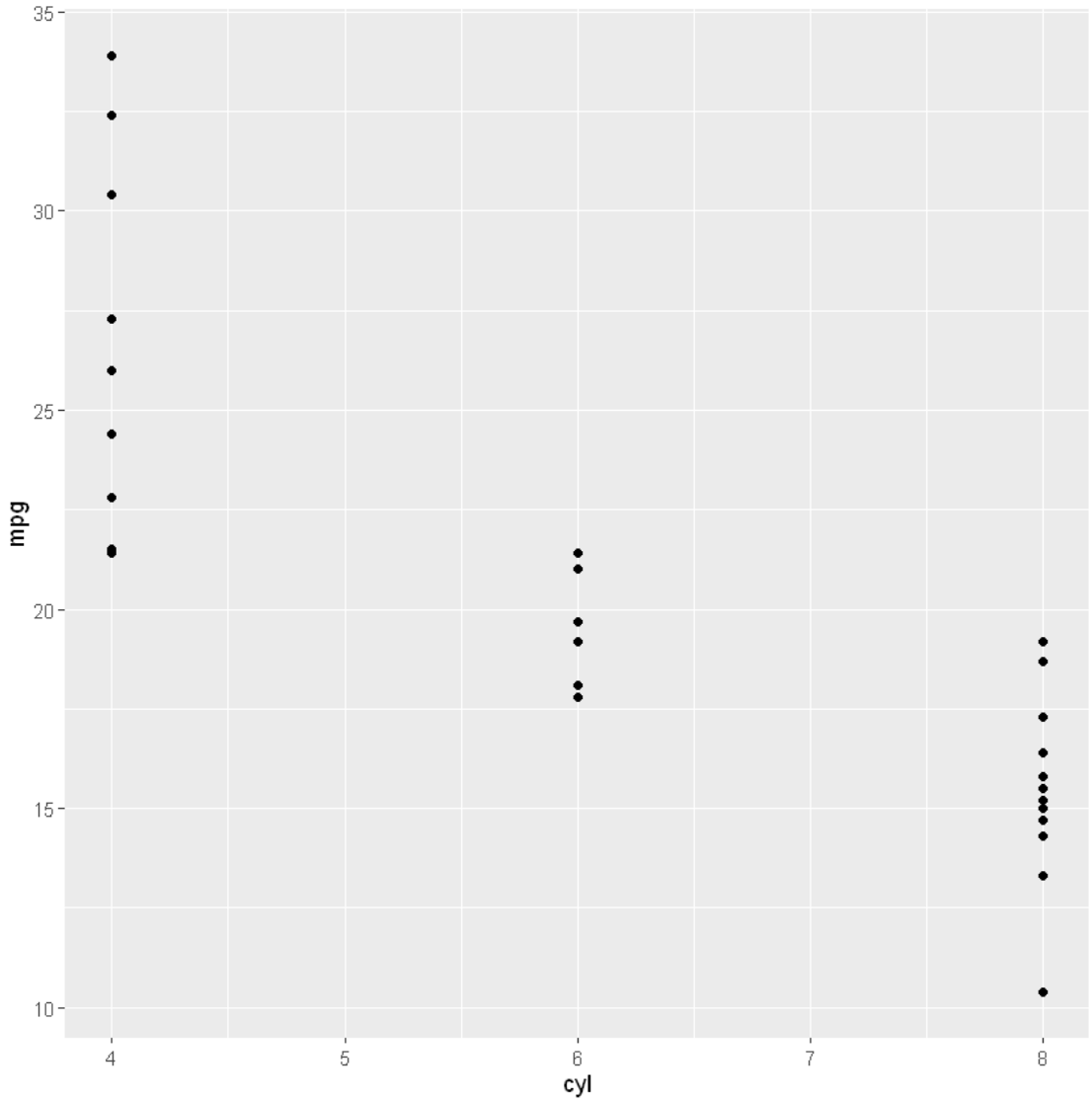
[95]:
```
scat3 + facet_grid(   rows = vars(gender),
                      cols = vars(region))
#as you can see here; the vast majority of schools are unisex (co-education)...
```



1.2.15 [15] Secondary Axis

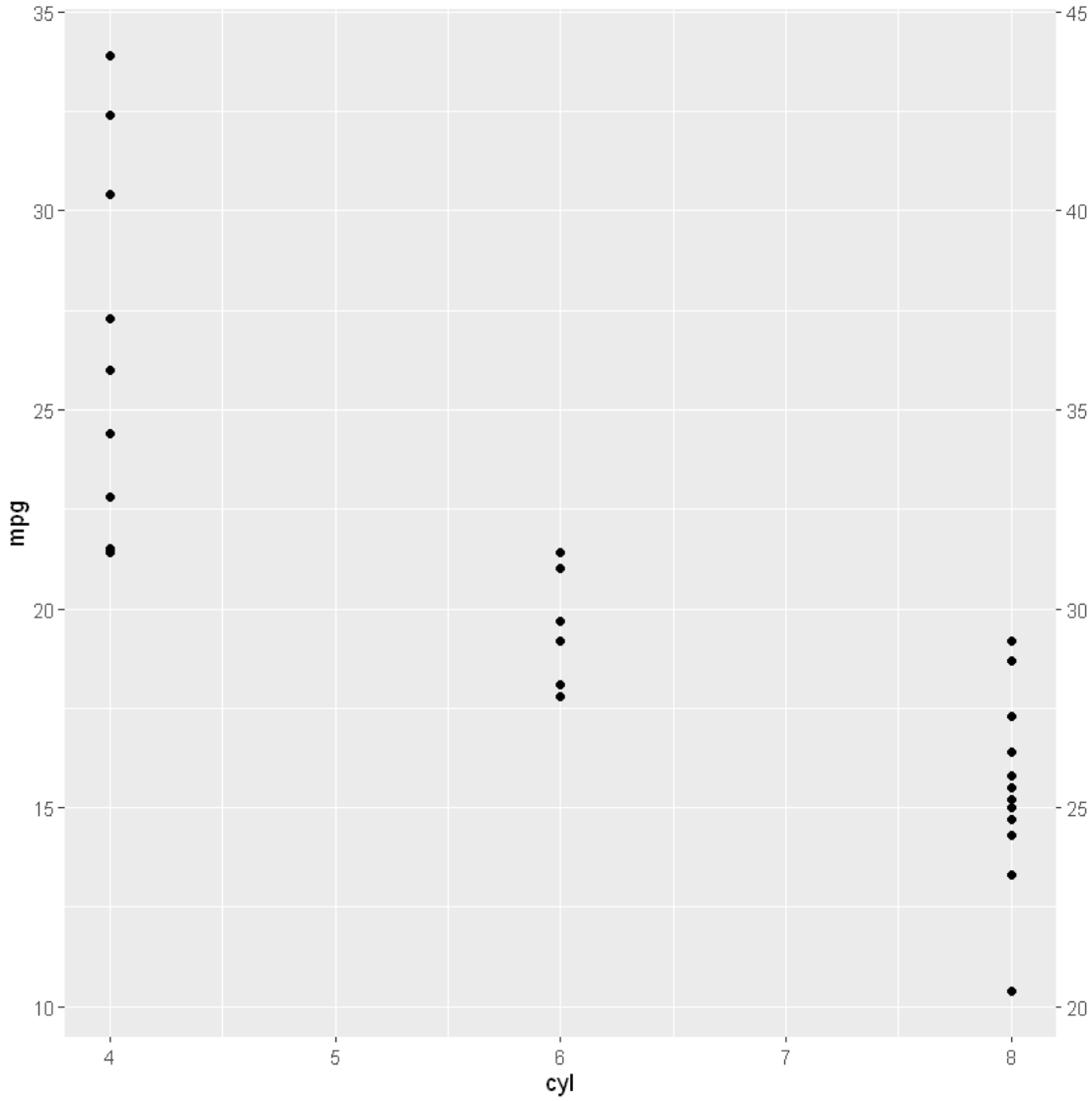
Semi-misleading title, sadly no way to do this....*(have to use workarounds, such as scaling the data down so it visually appears this way - however purely visual secondary axes coupled with this can be effective)*

[99]:
```
p <- ggplot(mtcars, aes(cyl, mpg)) +
  geom_point()
p
```



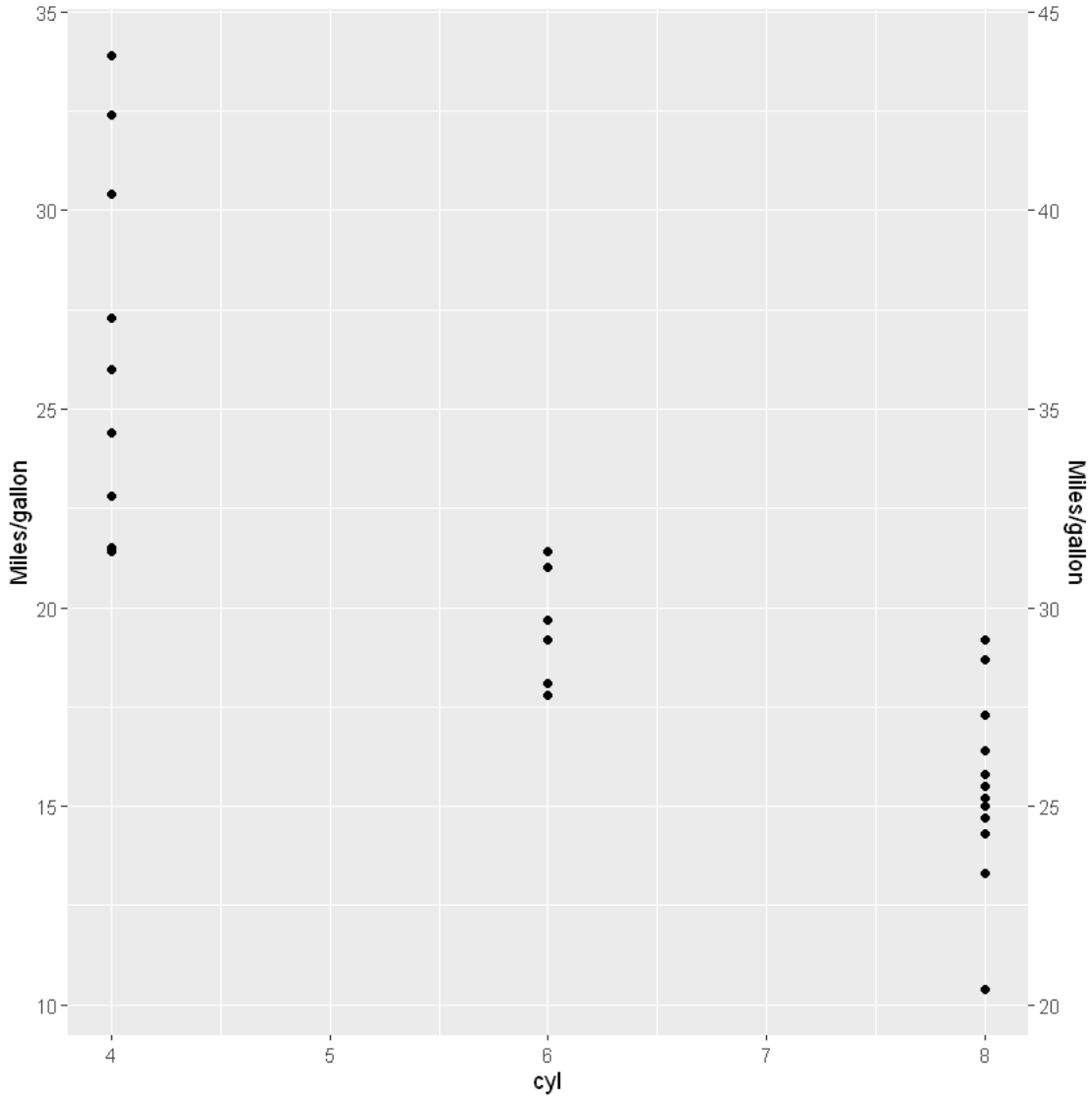
Creating a simple secondary axis, shifted by 10mpg:

[98]:
```
p + scale_y_continuous(sec.axis = sec_axis(~ . + 10))
```



Inheriting the name from the primary axis (can be manually set as you can see):

[100]:
```
p + scale_y_continuous("Miles/gallon", sec.axis = sec_axis(~ . + 10, name =␣
↪derive()))
```



[102]:
```
#sadly... as aforementioned no clean way to scale axes for double graphs like␣
↪in Excel.. have to manually code the data
#e.g. if right scale is 0 to 40,000 and left is -20 to 20 scale factor is 1000X␣
↪temperature*1000
```

[ ]:

[ ]:

[ ]:

[ ]: